



DAWOOD UNIVERSITY OF ENGINEERING AND TECHNOLOGY
GULBERG TOWN KARACHI-74800 (PAKISTAN)

FACULTY OF INFORMATION & COMPUTING SCIENCES
DEPARTMENT OF COMPUTER SCIENCE

Title:

Muhafiz Red Zone Mapping and Community Driven Support

Group members names and ID's

Mazia Sara 21F-BSCS-64

Rabia Imtiaz 21F-BSCS-75

Inshal Atif 21F-BSCS-81

Asva Kamal 21F-BSCS-108

Supervised by:

Dr. Asma Sanam Larik



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CERTIFICATE

This is to certify that the work present in this FYP thesis “Muhafiz Red Zone Mapping and Community Driven Support” has been carried out by group members under the supervision of Dr. Asma Sanam Larik. The work is genuine, original and in our opinion it fulfills the requirements for FYP.

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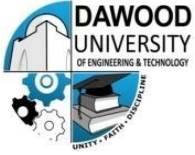
Co-Supervisor:

FYP Coordinator

Chairperson

Dr. Asma Sanam Larik
Assistant Professor

Dr. Fida Hussain Khoso
Associate Professor



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DEDICATION

We wholeheartedly dedicate this project to the vision of building safer and more informed communities through technology. **Muhafiz** represents our commitment to public safety, awareness, and digital empowerment.

This project is a small step toward addressing the challenges faced by citizens in urban environments, where access to real-time safety data can make a critical difference. We hope **Muhafiz** serves as a foundation for future innovations in community protection and responsible data use.

We are also deeply grateful to our families, friends, and mentors whose constant encouragement and support played a vital role throughout this journey.



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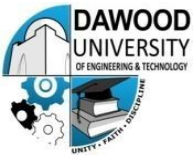
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I would like to express my heartfelt gratitude to Dr. Asma Larik for their invaluable guidance, patience, and insightful feedback throughout this project. Their expertise and encouragement have played a significant role in shaping this research.

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Lastly, I extend my deepest appreciation to my family and friends for their constant support and encouragement. Without their motivation, this achievement would not have been possible.

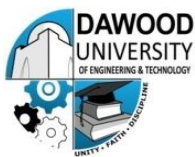


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ABSTRACT

This project presents Muhafiz, a community-driven safety platform designed to help citizens navigate urban environments—specifically Karachi—with increased awareness and security. The system integrates zone-based crime risk mapping, safer route recommendations, and real-time community reporting to support informed decision-making. Key functionalities include dynamic visualization of high-risk zones (Red, Orange, Yellow, Green), risk-aware route suggestions using a custom Dijkstra-based algorithm, and a discussion forum for users to report suspicious activities or incidents in their locality. An admin panel supports content moderation and platform management. Instead of employing complex machine learning models, Muhafiz uses a rule-based, data-driven approach focused on clarity, accessibility, and real-world relevance. **Due to limitations in public crime data availability, a synthetic dataset was generated to simulate realistic patterns and support zone classification and routing.**



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CHAPTER 1: INTRODUCTION

1.1 Introduction

Urban security remains a critical challenge in cities like Karachi, where crime is often underreported and real-time awareness is limited. Existing systems—such as emergency contact apps or police reports—lack integration, user participation, and frequently fail to provide actionable insights for citizens navigating the city.

Muhafiz is a web-based security platform developed using the MERN stack (MongoDB, Express.js, React.js with Vite, and Node.js). It aims to enhance public safety by allowing users to report incidents, visualize red zones (high-risk areas) on a map, and receive safer route recommendations, enhanced with optional AI-based evaluation modules. [1]. The system collects user-generated reports and integrates crime datasets, with the system designed to optionally support machine learning models for zone classification and risk forecasting.

Key Features:

- Red zone visualization using map-based clustering
- Community-driven incident reporting
- AI-powered safer route recommendations [6]
- Role-based access for users and administrators
- Web-first responsive design for broad accessibility

While the initial implementation is rule-based, Muhafiz is architected to allow progressive integration of ML components such as risk prediction and route safety classification.

The platform is divided into two main panels:

- **User Panel:**
For reporting incidents, viewing red zones, reviewing safety updates, and planning safer routes.
- **Admin Panel:**
For managing user reports, moderating discussions, and monitoring overall system health.

Muhafiz is designed to be scalable, lightweight, and easy to deploy. It emphasizes proactive safety by leveraging AI and community data to help users avoid dangerous routes, especially in unfamiliar areas.

1.2 Problem Statement

In densely populated urban areas such as Karachi, the lack of accessible, real-time crime data and intelligent safety tools limits residents' ability to proactively avoid high-risk zones.

[2]. Existing platforms suffer from fragmented and reactive safety measures due to the absence of integration of geospatial analytics, community-driven reporting, and AI-powered risk prediction.

Muhafiz addresses these gaps by providing:

- **Dynamic Red Zone Detection:**
Zone-based identification of crime hotspots, with support for optional ML-based enhancements using severity and frequency analysis. [6]
- **Route Optimization:**
Graph-based pathfinding weighted by real-time risk data to suggest safer routes. [6]
- **Community-Driven Validation:**
Crowdsourced incident reports that enhance and improve dataset accuracy.
- **Web-First Accessibility:**
A browser-based interface ensuring broad compatibility across devices.

1.3 Objectives

The primary objectives of the Muhafiz project are as follows:

- To develop a responsive, web-based platform for urban security, accessible through standard browsers across devices.
- To integrate interactive mapping and geolocation services for real-time visualization of crime-prone areas. [3]
- To design a system capable of integrating AI modules that identify and dynamically update red zones.
- To support safer route recommendations using rule-based logic and optionally enhanced by machine learning techniques. [6]
- To enable community-driven incident reporting, allowing users to contribute local safety insights and validate real-time red zone predictions.
- To ensure a secure backend for managing user reports and sensitive data using role-based access control and modern encryption standards.
- To design an intuitive user interface with separate panels for administrators and regular users, supporting features like heatmaps, community posts, and data visualization dashboards.

These objectives aim to deliver an intelligent, scalable, and user-focused security platform that combines real-world data, machine learning, and community collaboration to support informed decision-making and promote urban safety.

1.4 Challenges

The development of the Muhafiz platform involved several critical challenges, particularly in the areas of data acquisition and system scalability:

- **Lack of Public Crime Data:**
A major challenge was the absence of structured, publicly available APIs or datasets for crime incidents in Karachi. Outreach to local NGOs and police departments did not yield reliable or structured data suitable for analytical use.
- **Complexity of Data Collection:**
With no centralized data source, web scraping became the only viable option. However, scraping from local news websites was ineffective due to inconsistent reporting of specific locations, often missing subdivision-level details essential for geospatial mapping. [3]
- **Geolocation Granularity:**
Even with curated data, accurately mapping incidents to specific blocks or towns in Karachi was difficult. Many locations in news articles lacked geospatial accuracy, requiring extensive preprocessing and standardization. [3]
- **User-Generated Data Validation:**
Ensuring the accuracy and integrity of user-submitted reports is a challenge. Misreported or duplicated entries could distort the platform's red zone forecasts and affect reliability.
- **Privacy and Data Security:**
Given the sensitivity of location-based data, protecting user privacy through proper encryption, secure APIs, and access control was critical. Balancing transparency and data security remains an ongoing concern. [4]
- **Infrastructure and Accessibility:**
As a web-first platform, performance may be constrained in areas with unstable internet connectivity. Offline access or data caching features were considered but add significant complexity.

Overcoming these challenges was essential to building a functional, AI-integrated safety platform. The manual effort behind curating a custom dataset of 18 towns in Karachi highlights the project's focus on local relevance and data-driven decision-making.

1.5 Limitations

The Muhafiz platform faces several inherent limitations that may impact its performance and user experience:

- **Reliance on User Engagement:**
The accuracy and coverage of safety information heavily depend on active and consistent user participation. Insufficient community involvement can reduce overall system effectiveness.
- **Continuous Internet Requirement:**
Core functionalities—such as real-time mapping, community data updates, and dynamic identification of safe or risky zones—require continuous internet access. This limits usability in areas with poor or inconsistent connectivity. [5]
- **Potential Network-Induced Latency:**
In regions with weak network infrastructure, there may be delays in sending alerts or receiving responses, potentially impacting the timeliness of emergency support. [5]
- **Data Granularity Limitations:**
Despite efforts to build a detailed dataset, some incidents lack precise geolocation due to limitations in source data, which may affect the accuracy of zone classification and mapping. [3]

Recognizing these limitations is vital for setting realistic user expectations and guiding future improvements to enhance system resilience, offline capability, and data accuracy.

1.6 Project Scope

Muhafiz is a web-based security platform designed to address Karachi’s urban crime challenges through dynamic red zone mapping, community-driven incident reporting, and AI-powered risk forecasting. The system empowers residents to visualize high-risk areas, share local safety insights, and navigate the city safely using data-driven route recommendations.

Core Features:

- **Red Zone Identification:**
 - Maps crime hotspots using machine learning algorithms trained on historical and crowdsourced data. [6]
 - Dynamically updates risk zones based on frequency, severity, and community validation of incidents.
- **Community Collaboration:**
 - Allows users to submit incident reports with geolocation, photos, and descriptions.
 - Hosts a moderated forum for real-time safety discussions and alerts.
- **Safety-Optimized Routing:**
 - Suggests safer routes by analyzing crime density, time of day, and user-reported threats. [6]
- **Admin Oversight:**
 - Provides tools for moderating reports, validating data, and refining AI predictions.

Target Users:

- **Residents:**
Individuals seeking real-time safety insights for daily travel or exploring new areas.
- **Community Groups:**
Neighborhood organizations aim to monitor local security trends.
- **City Officials:**
Authorities require verified crime data for resource allocation and policy planning.

1.7 Functional Requirements

The Muhafiz platform is designed to enhance individual and community safety by leveraging geospatial insights, machine learning, and user-generated data.[\[7\]](#) Key functional requirements include:

1. AI-Powered Real-Time Mapping

- Dynamic map integration with real-time location tracking and navigation support via GPS and mapping APIs. [\[6\]](#)
- Intelligent route suggestions that automatically avoid red zones (identified dangerous areas) based on live data and predictive analytics.
- Visual safety overlays and heatmaps based on zone severity, recent reports, and machine learning-based risk evaluations.

2. Community-Driven Reporting

- An interactive platform allowing users to report incidents, share observations, and provide feedback on area safety.
- Aggregation and analysis of reports to classify and visualize zones as red (dangerous), yellow (caution), or green (safe). [\[3\]](#)
- Voting and commenting features to validate or challenge submitted data, improving reliability through community consensus.

3. Notification and Alert System

- Real-time alerts when users enter a red zone or deviate from suggested safe routes.
- Timely updates when new community reports or incident alerts are posted within a user's area.

4. User Centric Interface:

- A clean, intuitive, and responsive interface designed for users of all ages and technical backgrounds. [\[5\]](#)
- Dark/light mode toggle and accessibility support for visually impaired users.
- Multilingual options to ensure usability across different regions and demographics.

Framework Destinations and Vision:

Muhafiz points to enable people and communities by advertising a dependable, data-driven security apparatus. By joining AI-assisted ruddy zone location, live course optimization, and collaborative client engagement, the stage bolsters sure route through urban situations.

In spite of progressing challenges in information approval and security administration, Muhafiz builds up a solid establishment for an adaptable and impactful urban security arrangement.

1.8 Conclusion

The growing safety concerns in urban areas like Karachi highlight the urgent need for more proactive, data-informed, and community-driven safety solutions. Existing tools often fall short in terms of real-time awareness, public participation, and practical usability. This chapter introduced the foundation of **Muhafiz**—a platform designed to address these gaps by combining zone-based risk mapping, safer route suggestions, and community reporting into one integrated system. The next chapters will further explore the literature background, system methodology, data collection strategy, and implementation approach that shaped this project into a locally relevant and socially impactful solution.

CHAPTER 2

Literature Review

Literature Review: Existing Applications for Safety

In this section, a comparative review of various women's safety applications is presented. Each application is evaluated based on its core features, strengths, limitations, and relevance to the context of a community-focused safety platform like Muhafiz.

2.1 Safety Pin(2013)

Safeti Pin (2013): Provides features including location sharing, safe place identification, neighborhood safety scores, and safety audits.[\[8\]](#)

- Assesses area safety using factors like public activity, openness, and lighting.
- Does not have interactive user features or real-time emergency response capabilities; instead, it depends on community-contributed data for more precise safety assessments.[\[9\]](#)

2.2 Be Safe (2011)

- Offers live streaming, automated audio/video recording, voice-activated SOS warnings, and the ability to make a phony call in order to divert or dissuade threats. [\[10\]](#)
- Why Effective real-time data exchange improves readiness for emergencies.
- Does not draw attention to red zone statistics or community involvement.

2.3 Go Fearless (2022)

- Contains real-time GPS tracking and Firebase integration for safe data storage. [\[11\]](#)
- Provides many safety modes (Update, Cautious, and Panic) suited to varying threat levels. [\[12\]](#)
- Features an emergency alarm system in addition to offline and internet functionality.
- Relies on initial user input; little adaptation across international contexts.

2.4 My Safety pin (2016)

- Suggests secure courses based on security scores and natural parameters. [\[13\]](#)
- Assesses ranges utilizing nine criteria such as lighting, perceivability, and swarm nearness.

- Collaborates with open and private divisions to move forward scope. [\[14\]](#)
- Information precision is user-dependent, and the interface needs ease-of-use refinements. [\[15\]](#)

2.5 Raksha

- Features loud emergency alarms and SOS capability without net access. [\[16\]](#)
- Prioritizes offline reliability, making sure usability in terrible community zones. [\[17\]](#)
- Simple and useful interface for emergency use.
- Offers minimum community-based capabilities and constrained real-time updates.

2.6 Womenify (2020)

- Provides GPS tracking, access to nearby crime statistics, and protection recommendations.
- Uses real-time facts to assist on-the-go decision-making. [\[18\]](#)
- Leverages government facts to assess and file threat zones. [\[19\]](#)
- Depends heavily on manual user input and lacks dynamic network interaction.

2.7 Research-Based Applications:

Android Based Personal Safety Application for Women (2018):

- Utilized GPS and SMS technology to ship distress signals during emergencies. [\[20\]](#)
- Focused on delivering location records to predefined contacts.
- Operated with minimum user input for fast response.
- Demonstrated viability of fundamental protection capability in mobile environments.

IoT-Based Women Safety System (2022):

- Integrated wearable devices with mobile apps for better protection monitoring. [\[21\]](#)
- Enabled real-time monitoring using IoT sensors and mobile connectivity. [\[22\]](#)
- Provided immediate SOS alert transmission through synchronized hardware and software.
- Emphasized the application of multi-device ecosystems for personal protection.

Mobile App for Predicting Safe Routes (2019):

- Analyzed crime data to suggest safer travel routes to users. [\[23\]](#)
- Focused on algorithmic prediction models for route safety evaluation. [\[24\]](#)
- Included dynamic route recalculations based on risk level inputs.
- Highlighted the need for reliable crime data integration in urban safety tools.

GoFearless: Safety and Security Android-Based Application for Women (2022):

- Implemented progressive emergency triggers, which include gesture-based alerts. [\[25\]](#)
- Offered customizable protection modes for different danger levels.
- Combined ease of use with effective real-time response features.
- Showcased cutting-edge mobile protection utility traits relevant to Muhafiz.

2.8 Muhafiz: Proposed Features

The Muhafiz-Web-based system is designed to include advanced user-oriented features that reflect both practical needs and research strategies. [\[27\]](#)

- Interactive community-controlled security card with red and safe zone dynamic updates. [\[28\]](#)
- AI-based prediction tool for assessing location or route safety.
- Secure user authentication and encrypted personal data management.

2.9 Unique Selling Points

To distinguish itself from existing solutions, Muhafiz introduces a set of unique and innovative features:

Community Discussion Forum:

- Users can discuss safety concerns and exchange local experiences.
- Builds trust, cooperation, and transparency within the user base.
- Promotes users' continued participation to improve data quality.
- Serves as a platform for reporting incidents and learning from others.

Red Zone Mapping:

- Offers a safer alternative route based on live community reports.
- Expects warnings when users enter or approach dangerous locations.
- Continuously updates the security map based on user input and AI knowledge.

Additional Features:

- Includes scheduled alert settings for routine safety checks.
- Provides instructional video tutorials on safety practices.

2.10 Enhancing Muhafiz with Research Insights

Incorporating proven concepts from existing research strengthens Muhafiz as a next-generation safety platform:

Enhanced Safety Modes:

- Inspired by GoFearless, introduces customizable modes (e.g., travel, cautious, emergency).
- Allows users to define settings tailored to various danger levels.
- Enhances readiness and usefulness during sudden events.
- Encourages proactive protection management.

Community Rating System:

- Encourages proactive protection management.
- Modeled according to the user's safety assessment method.
- Users can assess the area based on personal experience and perceived risk.
- Promotes recognition of localized real-time data in safe and uncertain zones.
- Builds crowdsourced security intelligence networks.

Data Optimization:

- Utilizes MangoDb or comparable cloud technology for scalable data handling.
- Ensures rapid data retrieval and synchronization across devices.
- Emphasizes robust protection protocols for personal and location data.
- Supports modular architecture for future feature expansion.

2.11 Comparative Analysis of Muhafiz and Existing Safety Applications

Application	Release Year	Key Features	Strengths	Weaknesses	Muhafiz Differentiation
Safetipin	2013	Safety audits, neighborhood scores	Detailed safety assessments	No real-time zone prediction	Introduces AI-classified color zones (Red/Orange/Yellow/Green) using synthetic + real-time reports
bSafe	2011	Voice SOS, live streaming [31]	Real-time emergency sharing	No location-based community discussion	Adds zone-linked community forums and web-based feedback system
GoFearless	2022	Multi-mode safety, offline support	Works without internet	Limited visualization and integration	Web-first platform with safety overlays and zone-weighted routing
My Safetipin	2016	NGO audits, route suggestions	Verified data sources	Outdated and static interface	Modern web UI, dynamic route risk scoring based on preassigned zone weights

Raksha	2015	Loud alarms, offline SOS	Offline functionality	No map or route-based safety	Incorporates zone-aware safety routing using Dijkstra's algorithm with risk-weighted paths
Womenify	2020	Crime maps, GPS tracking	Good visuals	Static government data	Synthetic data generation with ongoing updates and zone-wise color coding
AASafety	2023	AI crime prediction	Fast alerts, smart scoring	Less accessible for non-tech users	Simple zone-based UI with downloadable safety resources (e.g., self-defense PDFs)
SheGuard	2023	Group safety, predictive alerts	Predictive risk management	Unverified user-only data	Combines structured scraped data + verified incident reports for zone scoring
TrustCall	2019	Trusted contacts, emergency logging	Simple and effective contact system	No predictive zone or route safety	Adds route-level safety navigation based on preclassified zone risks
RescueMe	2021	Scheduled SOS, auto-recording	Good for planned scenarios	Doesn't adapt to dynamic crime risk	Zone-based incident map + path scoring with Dijkstra + zone weights
Guardians (Truecaller)	2021	Guardian tracking, emergency requests [32]	Trusted contact framework	Lacks public crime data integration	Provides zone color overlays and public safety reports for navigational awareness
Citizen	2016	Police scanner alerts, user reporting	Real-time awareness	City-limited availability	Custom synthetic zones + real-time user reports applicable city-wide
Noonlight	2013	Smart device integration [34]	Works with IoT/wearables	Region/device dependency	Accessible web interface without app or hardware dependency
Safe365	2019	Elder GPS tracking [33]	Personal/family safety	Not designed for public area safety	Built for urban public awareness with incident reporting + resource uploads
Red Panic Button	2012	One-click emergency alert	Very easy to use	Lacks contextual awareness	Context-aware routing and visual safety zones through deterministic logic

Muhafiz (Proposed)	2025 (Planned)	Red/Orange/Yellow/Green Zones, Route Prediction (Dijkstra), Community Forum, Incident Reporting, Resource Upload [29]	Risk-aware zone overlays, public discussion, web-first navigation	No mobile app, no offline mode	Uses zone-weighted Dijkstra algorithm for route safety + community discussion, resource sharing, and incident feedback [30]
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2.12 Critique of Muhafiz and Existing Applications

This phase severely analyzes each Muhafiz and similar programs with the aid of figuring out their respective strengths and weaknesses. While many current apps cope with fragments of the protection challenge, they fall quickly in turning into an integrated, user-targeted solution. Muhafiz tries to bridge this hole with the aid of combining more than one technology and network engagement equipment right into an unmarried platform.

2.13 Strengths of Muhafiz

- **Proactive and Reactive Security:** Combines real-time recognition of the situation and an incident reporting feature.
- **Community Engagement:** Enables user discussions through the integrated forum. This improves awareness and information exchange.

2.14 Limitations of Muhafiz

- **Complex Interface:** A big range of capabilities may also cause confusion for less tech-savvy users without a highly intuitive design.
- **Scalability Constraints:** Relies heavily on user-generated content, which may reduce effectiveness in areas with low adoption.
- **Privacy Risks:** Managing sensitive user data requires rigorous encryption, anonymization, and legal compliance.

2.15 Gaps in Existing Applications

- **Lack of Integration:** Most apps fail to supply a complete approach combining protection features, user education, and community feedback.
- **Offline Limitations:** Few systems offer useful functionality in low-network environments.
- **User Input Dependency:** Many programs depend too much on manual inputs, limiting their

usefulness during actual emergencies.

2.16 Recommendations for Muhafiz

- **Simplify the user interface:** Drop the complexity in design and make it easier to access without affecting the required functions.
- **Implement strong security measures:** Protect user data using end-to-end encryption and anonymization techniques.
- **Work with stakeholders:** Partner with government agencies and NGOs to provide reliable safety data early in the platform.

2.17 Technological Depth

- **Web Integration:**
 - Web model supports discussion forums and detailed incident reporting.
- **Shared Database Architecture:**
 - MongoDB ensures consistent data flow and real-time synchronization between web and mobile users.
- **Technology Stack:**
 - **Frontend:** React.js + Vite
 - **Backend:** Node.js for scalable services
 - **Database:** MongoDB Atlas
 - **Auth:** JWT

2.18 Conclusion

This chapter reviewed existing literature and related systems in the domains of crime mapping, urban safety applications, and community reporting platforms. While several international solutions incorporate advanced technologies like machine learning and real-time alerts, many lack contextual relevance to cities like Karachi, where data availability and user engagement remain significant challenges. Furthermore, local applications often fall short in offering integrated safety tools that combine risk-based routing, zone visualization, and participatory features. These gaps in existing solutions justify the development of **Muhafiz**, a focused and locally adapted platform that emphasizes rule-based route safety, user involvement, and dynamic zone awareness. The insights gathered here provide a strong foundation for the system design and methodology discussed in the next chapter.

Chapter 3: Methodology

3.1 Research and Requirement Analysis

The development of **Muhafiz - Red Zone Mapping & Community Safety Platform** began with a data-informed strategy focused on enhancing urban safety in Karachi. The preliminary research and requirement gathering phase included:

- **Crime Data Acquisition:** Local crime data was gathered by scraping online reports and news articles using tools like **Outscraper** and the **Google Custom Search API**. This enabled the collection of hyper-local incident records across Karachi's 18 towns and their subdivisions. [\[34\]](#)
- **Data Augmentation:** To address gaps in real-world data, **Faker.js** was used to generate synthetic crime data. This augmented dataset supported better **risk zone visualization** but was **not used for any machine learning model training**. [\[35\]](#)
- **User Research:** Informal surveys and feedback from Karachi residents guided feature prioritization. The most demanded functionalities were:
 - Visualizing crime-prone "red zones"
 - Suggesting safer routes
 - Enabling real-time community reporting

3.2 System Design and Architecture

Muhafiz follows a modular, service-based web architecture that separates concerns for frontend, backend, data, and API integration.

3.2.1 Functional Overview

The platform is divided into key functional modules:

- **Zone Visualization:** Displays pre-classified crime zones (Red, Orange, Yellow, Green) on an interactive map using colored markers.
- **Route Safety Engine:** Implements **Dijkstra's algorithm** to find the safest path between two locations by reducing exposure to high-risk areas. [\[36\]](#)

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- **Community Forum:** Enables users to post and discuss safety-related updates. All posts are moderated to ensure quality and relevance.

3.2.2 Design Artifacts

- **ER Diagram:** Designed to model core entities like users, incidents, zones, and forum posts within **MongoDB Atlas**.
- **Data Flow Diagram (DFD):** Illustrates how data flows between the **frontend (React)**, **backend (Express.js)**, and **Flask APIs**. It also shows MongoDB handling storage and retrieval.

3.3 Application Development

Development followed **Agile methodology** using weekly sprints, with regular testing and feedback cycles.

3.3.1 Frontend

- Built with **React.js** and **Vite** for a fast development experience.
- **Leaflet.js** was used to visualize geographic zones using interactive-colored markers.
- Admin login and dashboard access are protected using **JWT-based authentication**.
- The UI is responsive and mobile-friendly.

3.3.2 Backend

- Created using **Node.js** and **Express.js**, offering RESTful APIs for incident reporting, user accounts, and forum management.
- Connects securely with **MongoDB Atlas** for data storage.
- Performs CRUD operations for incidents, comments, zone data, and user profiles.

3.3.3 Python Services (Flask API)

- A **Flask-based microservice** handles two main features:
 1. **Zone Mapping API:** Serves zone classification data from a preprocessed CSV file.
 2. **Safer Route Suggestion:** Uses a **custom Dijkstra-based algorithm** to

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compute safest paths, factoring in crime severity and zone classification.

[\[36\]](#)

- Unlike traditional ML-based systems, **no real-time training or prediction occurs**. All logic is deterministic and rule based.
- **CORS support** is enabled to allow smooth integration with the frontend.

3.4 System Implementation

3.4.1 Flask-Based Python Services Red Zone Detection

A minimal Flask API exposes Karachi's risk zones to the front end. It works as follows:

- Loads a **preprocessed CSV file (karachi_crime_dataset_ranked.csv)** containing subdivision, town, coordinates, and their pre-assigned risk zones.
- Extracts only required fields: subdivision, town, latitude, longitude, and risk zone.
- Responds via the endpoint /get-zone-data with a JSON array of zones.
- No ML clustering is performed; zone classification is already embedded in the CSV based on crime frequency and severity.

Safer Route Suggestion

- A **custom Dijkstra's algorithm** implementation calculates safer routes between locations.[\[36\]](#)
- Routes are weighted based on crime severity and zone risk levels:
 - Red zones → higher weight (discouraged paths)
 - Green zones → lower weight (preferred paths)
- The algorithm avoids areas with high crime scores, optimizing not just for shortest distance but for **user safety**.

3.4.2 Deployment of Python Services

- Both the zone and route APIs are deployed as standalone Flask services.
- They are containerized and integrated via internal endpoints for ease of access from the frontend and backend.

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3.4.3 Deployment of Python Services

- Both the zone and route APIs are deployed as standalone Flask services.
- They are containerized and integrated via internal endpoints for ease of access from the frontend and backend.

3.5 Database Layer

The platform uses **MongoDB Atlas** as its primary database. Key features include:

- **2dsphere Geospatial Indexing:** Enables efficient spatial queries for map-based data like incident locations or zone filtering.
- **Flexible Schema:** MongoDB's document-based storage adapts easily to different types of data (incidents, discussions, zone levels).
- **Cloud Integration:** Offers environment variable support, auto backups, and scaling.

3.6 Security Considerations

Security is a key pillar of the Muhafiz system. The following measures were implemented:

- **JWT Authentication:** Ensures secure sessions for both users and admins using access and refresh tokens.
- **Role-Based Access Control (RBAC):**
 - General Users: Can report and view incidents.
 - Admins: Full access to dashboard, data, and user control.
- **AES-256 Encryption:** Protects sensitive incident data from unauthorized access. [\[37\]](#)
- **Rate Limiting:** Enforced via Express middleware to prevent request spamming or brute-force attacks.

3.7 Deployment Strategy

Each platform component was deployed to a suitable service for maximum reliability and scalability:

Component	Platform	Justification
Frontend (React + Vite)	Vercel	Seamless CI/CD with GitHub, global CDN, ideal for React apps
Backend (Node.js + Express)	Render	Developer-friendly, supports Node.js, easy ENV management, free tier available
Routing Logic Service (Flask/Dijkstra)	Railway	Quick Python API deployment, supports containerized microservices, scalable
Database (MongoDB Atlas)	MongoDB Atlas	Fully managed NoSQL DB, geospatial indexing, role-based access, backup support

Why This Stack?

- **Render** simplified backend hosting for Express.js.
- **Railway** supported fast and modular deployment for Flask APIs.
- **Vercel** handled frontend builds efficiently via GitHub integration.
- **MongoDB Atlas** ensured secure, flexible, and spatially-aware data handling.

3.8 Requirements Specification

3.8.1 Functional Requirements

The Muhafiz platform supports multiple features aimed at ensuring public safety, spatial awareness, and user engagement. Key functional requirements are outlined below:

1. Incident Reporting

- Users can submit crime or suspicious activity reports using structured fields such as:
 - **Crime type**
 - **Location (coordinates or area name)**
 - **Severity level**
 - **Detailed description**
- Users can attach media (images or videos) to their reports.
- All uploaded media undergo **automated EXIF metadata removal** to protect user privacy.

2. Red Zone Detection

- A **Flask-based API** serves pre-classified risk zone data to the frontend application.
- The dataset (karachi_crime_dataset_ranked.csv) includes:
 - Subdivision
 - Town
 - Latitude and Longitude
 - Risk Zone (Red, Orange, Yellow, Green)
- **Zone classification is performed offline** using severity and crime frequency metrics during data preparation.
- The API provides this data via the /get-zone-data endpoint in **JSON format**.
- **CORS (Cross-Origin Resource Sharing)** is enabled to allow access from frontend clients.

3. Safe Route Suggestion

- A custom route engine computes **safest travel paths** between two locations using:
 - **Dijkstra's algorithm** [\[36\]](#)
 - **Crime severity-based edge weights**
- Paths are optimized to avoid zones with high risk or incident density.

- This allows safer commuting decisions in urban areas.

4. Community Forum

- An in-app moderated discussion platform allows users to:
 - Share local safety tips
 - Report suspicious individuals or vehicles
 - Post real-time neighborhood alerts
- **Toxic content filtering** is handled on the frontend using **TensorFlow.js** to ensure constructive discussions.

5. Admin Panel

- A secure administrative interface is available for platform management.
 - Verify submitted incident reports.
 - Moderate forum content
 - Monitor analytics and flagged data in real-time

3.8.2 Non-Functional Requirements

1. Security

- Secure authentication via **JSON Web Tokens (JWT)** with support for: [\[37\]](#)
 - Access tokens
 - Refresh tokens
- **Role-Based Access Control (RBAC)** ensures proper authorization:
 - General users
 - Administrators
- API routes are protected using **rate limiting** and token validation middleware.

2. Performance

- Target response times:
 - **< 500 ms** for zone rendering and route suggestions
- Services are horizontally scalable to accommodate high traffic and data volume.

3. Scalability

- System designed to handle:
 - **100,000+ crime records**
 - Simultaneous requests from multiple users without performance degradation

4. Compatibility

- Fully **responsive layout** supporting:
 - Desktops
 - Mobile devices
- Tested on all major browsers including:
 - **Chrome**
 - **Firefox**
 - **Safari**

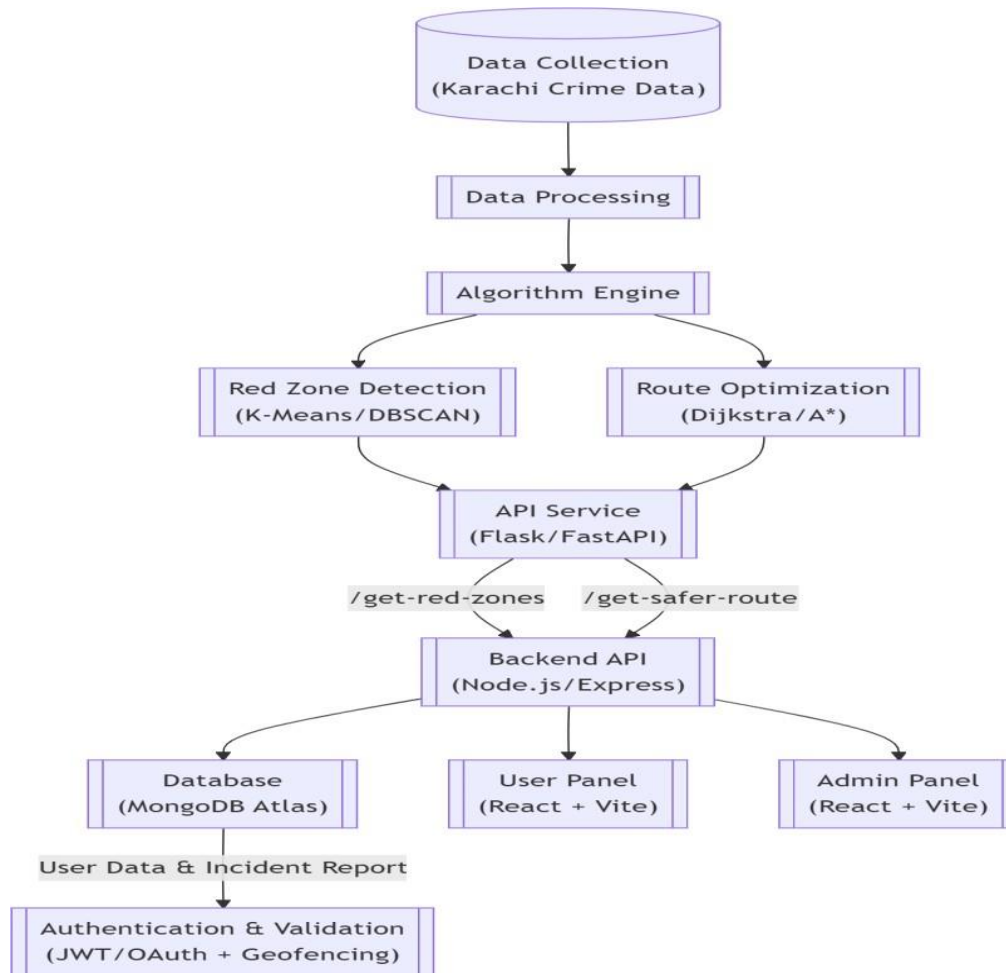
5. Maintainability

- Codebase follows **modular structure** for easy updates and testing.
- Project includes:
 - **CI/CD pipelines**
 - **Git-based version control**
 - Automated test hooks for backend and frontend components

3.8.3 System Constraints

- **Web-first architecture:** Muhafiz is designed as a browser-accessible web application with no native mobile dependencies.
- **No real-time ML predictions:** All machine learning computations (e.g., zone classification) are performed offline and exposed through Flask APIs.
- **Geospatial capabilities** rely on **Leaflet.js** for map visualization, with optional support for Google Maps APIs if required.

System Design



Module 1: **Karachi Crime** Dataset Collection and Synthesis

3.1 Karachi Crime Dataset Collection and Synthesis

This section documents the complete process followed to collect, clean, and enrich Karachi's crime data for use in the Muhafiz platform. The resulting dataset became the foundation for the system's two key features: **Red Zone Mapping** and **Safer Route Suggestion**. Due to limited access to structured public data sets, a hybrid data acquisition strategy was adopted, involving NGO outreach, police petitions, government dashboards, and synthetic data generation. [\[39\]](#)

3.1.1 Data Sourcing from NGOs and Police

NGO Outreach

- **Process & Hurdles:** Obtain detailed crime incident datasets from safety-focused NGOs operating in Karachi.
- **Method:** Contacted over **10 NGOs** via email, phone calls, and in-person visits.
- **Response:** Zero cooperation. Most cited "data sensitivity" or lacked structured datasets.
- **Challenges:**
 - Most organizations declined to share due to **data sensitivity**.
 - Others lacked structured formats, relying only on anecdotal reporting.
- **Key Insight:** NGOs primarily aggregate statistics (e.g., annual crime reports), not granular incident-level data.

NGO	Response
Aurat Foundation	No Response
CPLC	No Response
WAR Against Rape	No Response
Alkhidmat Foundation	No Response

Official Data from CPLC and Police

- **Police Records:**

Obtained 886 petitions via court access (Malir District) – mostly useless due to formatting issues, no subdivisions, and repeated formats.
- **Methodology:**
 - Contacted 15+ NGOs via email, phone, and in-person visits.
 - Most organizations either declined due to **data sensitivity** or maintained only aggregated annual reports with no location-based details.
- **Conclusion:** NGO-sourced data was not usable for the platform's zone or route modeling needs.[\[39\]](#)[\[40\]](#)

Official Data from Police and CPLC

1. Malir District Petition Data

Obtained 886 petitions via court access (Malir District) – mostly useless due to formatting issues, no subdivisions, and repeated formats.

Police Excel Sheet: Obtained via formal contact.

- Unstructured format: no subdivisions, inconsistent date and crime fields

Raw Police Petition Data (Malir

District) Raw Police Petition Data

(Malir District)

S.No	Court Name	Case No	Year	Parties	Category	Under Section	Sentup Court	Status
1	District & Sessions Judge, Malir	12	2025	Saad Gul V/S SHO Sachal	Criminal Petition U/S 22-A	Cr.PC- 22-A	Additional District & Sessions Judge VI	Disposed (Sentup to Court)
2	District & Sessions Judge, Malir	15	2025	Mst Mumtaz Ali V/S SHO SSHIA	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge VIII	Disposed (Sentup to Court)
3	District & Sessions Judge, Malir	14	2025	Rafaqat Ali V/S SHO Airport	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge VIII	Disposed (Sentup to Court)
4	District & Sessions Judge, Malir	11	2025	Abdul Rasheed V/S SHO Malir City	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge VI	Disposed (Sentup to Court)
5	District & Sessions Judge, Malir	8	2025	Muhammad Abid Ali Khan V/S SHO Sachal	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge II	Disposed (Sentup to Court)

6	District & Sessions Judge, Malir	7	2025	Mst Nusrat Naz V/S SHO Quaidabad	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge VIII	Disposed (Sentup to Court)
7	District & Sessions Judge, Malir	2	2025	Muhammad Khamiso V/S SHO SSHIA	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge II	Disposed (Sentup to Court)
8	District & Sessions Judge, Malir	3	2025	Saifullah Kalhoro V/S SHO Shah Latif	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge III	Disposed (Sentup to Court)
9	District & Sessions Judge, Malir	4	2025	Zulfiqar Ali V/S SHO Malir Cantt	Criminal Petition U/S 22-A	Cr.PC - 22-A	Additional District & Sessions Judge V	Disposed (Sentup to Court)

Note: The complete police petition dataset contains **886 rows** and is available in digital form. Common issues included lack of subdivision details, repeated case formats, and inconsistent formatting. so this data was not useful for us.

Despite the quantity, this data lacked **spatial granularity** and was not suited for GIS-based applications.

2. Police Excel Dataset

- Obtained through formal outreach.
- Challenges:
 - Missing subdivision fields
 - Inconsistent formatting (date, crime type columns)

3.1.2 Limitations of Public Crime

Dashboards CPLC Monthly Reports

The Citizens-Police Liaison Committee (CPLC) provides **monthly crime statistics** on their official platform:

https://web.cplc.app/khi_statistic/

However, this data is **aggregated city-wide**, offering **no breakdown by area, town, or subdivision**, and thus is unsuitable for red zone mapping or localized ML training.

- + No incident-level details
- + No location granularity
- + No geocoordinates or timestamps

Sample excerpt from January 2025 (01–31 January 2025):

Sample excerpt from January 2025 (01–31 January 2025):

S.No	Crime Type	Count
1	4-Wheelers Snatched	31
2	4-Wheelers Theft	151
3	2-Wheelers Snatched	617
4	2-Wheelers Theft	3,413
5	Mobile Phone Snatched	1,585
6	Kidnapping for Ransom	3
7	Extortion	11
8	Homicide	54
9	Bank Robbery / Dacoity	0

As shown above, CPLC only publishes monthly totals without any spatial or incident-level breakdown. This made their statistics **informative but unusable** for red zone classification or location-based analysis in our system.

OpenData Pakistan (ODP)

Analyzed a third-party source collecting national-level statistics.

This portal compiles crime reports across Pakistan by scraping and aggregating public records, news articles, and government announcements. However, the limitations were consistent:

Feature	Available ?	Comments
Granular Incident Data	+	No individual incident entries
Subdivision/Geotags	+	Only district or province-level labels
Karachi Focus	+	Covers all of Pakistan; no Karachi-specific breakdown
Timestamps	+	Most data is historical/yearly summaries

3.1.3 Key Insight: Why External Data Sources Were Insufficient

Across all sources—NGOs, police records, CPLC statistics, and open datasets—the following limitations were consistently found:

- Lack of **subdivision-level crime reporting**
- Absence of **geocoordinates or GIS tagging**
- No **structured incident-level datasets**
- Minimal **timestamped data**

This forced the Muhafiz team to pivot toward **web scraping of local news portals** and **synthetic data generation** to simulate localized crime incidents with appropriate metadata.[\[39\]](#)

3.2 Web Scraping Strategy for Crime Data

Due to the limitations of official and NGO-sourced datasets, a web scraping strategy was designed to extract localized crime incident information from online news portals and search engines. The scraping was conducted in two phases. [\[39\]](#)

3.2.1 Phase 1 – News Media Scraping

- **Tools:** BeautifulSoup and Selenium
- **Sources:** *Dawn*, *Geo News*, *ARY*, *The News*
- **Coverage:** Scraped ~10 years of articles using keyword and location filters.
- **Limitations:**
 - ~90% articles lacked **exact location granularity**.
 - Frequent **anti-bot defenses** and vague terms like “*near Gulshan*” complicated parsing.

3.2.2 Phase 2 – Outscraper & Google CSE API Integration

To increase precision and automation in web search and scraping, the following tools were employed:

A. Outscraper

- Karachi divided into 18 towns; searched terms like:
 - “*robbery in Ittehad Town, Baldia*”
 - “*snatching in Baghdadi, Lyari*”
- Overcame API limits using **multiple accounts**.
- Output required **manual filtering** and formatting.

B. Google Custom Search API (CSE)

- Over **30+ customized keyword phrases** were generated for each subdivision.
- Python scripts were used to automate queries and store results.

```
response = requests.get(
    "https://www.googleapis.com/customsearch/v1",
    params={"q": keyword, "key": API_KEY, "cx": CSE_ID, "num": 5}
)
```

- Resultant JSON responses were parsed and saved in CSV format for later filtering.

This two-layered approach allowed for more precise, subdivision-level tagging of crime incidents for Karachi.

3.3 Data Cleaning, Structuring, and Mapping

Once raw crime data was collected, it underwent a structured multi-step cleaning and mapping process to convert it into a usable format.

3.3.1 Cleaning & Preprocessing

Task	Description
Crime Type Normalization	Unified terms like “robbery”, “snatching”, “firing”, etc.
Location Extraction	Extracted locality and landmark names using a mix of manual and NLP techniques
NLP Tools Used	spaCy for named entity recognition (NER) and custom regex filters
Subdivision Tagging	Manually verified and matched vague place names to Karachi’s official map

Example: "near NIPA" was mapped to **Gulshan-e-Iqbal**, while "UP Mor" was mapped to **North Karachi**.

3.3.2 Town and Subdivision Mapping

Karachi was officially broken down into:

- **18 towns**
- **170+ subdivisions**

To ensure consistency and correctness:

Data Source	Purpose
Local government records	Verified official subdivision boundaries
Community-sourced maps	Cross-validated town-subdivision structure
Ground-level knowledge	Used for ambiguous or informal area names

Outcome: A clean, location-tagged dataset ready for zone classification and safe route modeling.

- **Normalized** crime type names (e.g., *snatching*, *firing*).
- **Extracted locations, towns and sub divisions** using a mix of **manual tagging** and **spaCy NLP**.
- Mapped vague descriptions to **official subdivisions**.

Town and Subdivision Mapping

- Karachi divided into **18 towns** and **170+ subdivisions**.
- Subdivisions verified using:
 - Local government records
 - Community maps
 - Ground-level knowledge

Town Name	Subdivisions
1. Baldia Town	Saeedabad, Ittehad Town, Rasheedabad, Nai Abadi, Gulshan-e-Ghazi, Islam Nagar, Muhajir Camp, Afridi Colony
2. Bin Qasim Town	Ibrahim Hyderi, Landhi Industrial Area, Bhains Colony, Rehri Goth, Gulshan-e-Hadeed, Steel Town
3. Gadap Town	Surjani Town, Taiser Town, Gulshan-e-Maymar, Manghopir, Murad Memon Goth, Darsano Chhanno, Kathore
4. Gulberg Town	Aisha Manzil, Ancholi, Samanabad, Water Pump, Shafiq Mill Colony, Yousuf Plaza, Gulberg Block 5
5. Gulshan-e-Iqbal	Block 1–13D, Dhoraji Colony, Civic Center, PIB Colony, Aziz Bhatti Park Area, Federal B Area (adjacent blocks)
6. Jamshed Town	PECHS Block 1–6, SMCHS, Soldier Bazaar, Guru Mandir, Jacob Lines, Garden East, Garden West
7. Keamari Town	Keamari No. 1, Sultanabad, Machhar Colony, Bhutta Village, Manora, Shireen Jinnah Colony
8. Korangi Town	Korangi No. 1–6, Korangi Industrial Area, Allah Wala Town, Zaman Town, Nasir Colony, Bilal Colony
9. Landhi Town	Landhi No. 1–6, Sherpao Colony, Bhutto Nagar, Khokhrapar, Future Colony, Dawood Chowrangi
10. Liaquatabad Town	Dak-khana, Sharifabad, Bandhani Colony, Super Market, Firdous Colony, Al-Karam Square, Gharibabad
11. Lyari Town	Baghdadi, Chakiwara, Bihar Colony, Kalakot, Nayabad, Singo Lane, Shah Baig Line, Daryabad
12. Malir Town	Malir Halt, Malir City, Saudabad, Model Colony, Kala Board, Jafar Tayyar Society, Jinnah Square
13. New Karachi Town	Sector 5-A to 5-E, Sector 11-G, Godhra Camp, Shah Nawaz Bhutto Colony, Fatima Jinnah Colony, Khamiso Goth
14. North Nazimabad	Block A to L, Hyderi, Shadman Town, Buffer Zone, Paposh Nagar, KDA Chorangi
15. Orangi Town	Sector 1 to 16, Mujahid Colony, Madina Colony, Ghaziabad, Islam Chowk, Aligarh Colony
16. Saddar Town	Saddar, Burns Road, Empress Market, I.I. Chundrigar Road, Civil Lines, Clifton Block 1–9, Cantt Station
17. Shah Faisal Town	Shah Faisal Colony No. 1–5, Natha Khan Goth, Al-Falah Society, Rafah-e-Aam Society, Green Town
18. SITE Town	Metroville, Haroonabad, Pak Colony, Frontier Colony, Qasba Colony, SITE Industrial Area

3.1 Zone Risk Classification and Ranking

Each subdivision was labeled and ranked based on:

- **Crime frequency**
- **Incident severity**

Risk Categories:

-  High Risk
-  Medium Risk
-  Low–Medium Risk
-  Low Risk / Safe
-  Not Listed / Unknown

3.2 High-Risk Subdivision Assessment

The following section presents an in-depth risk evaluation of Karachi's subdivisions based on reported incidents of mobile snatching, theft, robbery, and gang-related activity. The risk levels are derived from scraped news data, CPLC trends, and local analysis.

High Risk Zones :

Subdivisions consistently reported street crimes, gang activity, and low police visibility.

1. Lyari Town

Overall Risk: High (Crime-prone, under-policed)

Key Subdivisions:

- **Baghdadi:** Street crime, mugging
- **Kalakot, Shah Baig Line, Singo Lane:** Gang activity, theft
- **Chakiwara:** Mugging, theft
- **Agra Taj Colony:** Theft, mugging

2. Korangi Town

Overall Risk: High (Frequent snatching and robbery)

Key Subdivisions:

- **Bilal Colony, Nasir Colony:** Theft, snatching
- **Zaman Town, Mehran Town:** Mugging, robbery
- **Korangi Industrial Area:** Street crime

3. Landhi Town

Overall Risk: High (Gang presence, red zones)

Key Subdivisions:

- **Sherpao Colony, Bhutto Nagar:** Street crime
- **Quaidabad, Khokhrapar:** Mugging, robbery

4. **Orangi Town**

Overall Risk: High (Theft-prone, limited policing)

Key Subdivisions:

- **Ghaziabad, Islamia Colony:** Mugging, theft
- **Pirabad, Mominabad:** Robbery, snatching

5. **Baldia Town**

Overall Risk: High (Multiple vulnerable clusters)

Key Subdivisions:

- **Ittehad Town, Saeedabad:** Mugging, robbery
- **Rasheedabad, Nai Abadi:** Street crime

6. **SITE Town**

Overall Risk: High (Industrial zones with poor safety)

Key Subdivisions:

- **Metroville, Qasba Colony:** Theft, snatching
- **Pak Colony, Haroonabad:** Robbery,

street crime Medium–High Risk Zones

7. **Keamari Town**

Overall Risk: Medium (Mixed safety zones)

Key Subdivisions:

- **Machhar Colony, Bhutta Village:** Theft, robbery (Medium–High)
- **Keamari No. 1:** Moderate risk

8. **Shah Faisal Town**

Overall Risk: Medium (Varied safety across blocks)

Key Subdivisions:

- **Shah Faisal Colony No. 5:** Violence, street crime (Medium–High)
- **Natha Khan Goth:** Mugging, theft

9. **Malir Town**

Overall Risk: Medium (Semi-urban, partial red zones)

Key Subdivisions:

- **Model Colony, Saudabad:** Theft, mugging (Medium–High)
- **Malir City, Jinnah Square:** Moderate crime

10. **Bin Qasim Town**

Overall Risk: Medium (Mixed rural-industrial profile)

Key Subdivisions:

- **Ibrahim Hyderi, Gulshan-e-Hadeed:** Mugging, theft (Medium–High)
- **Rehri Goth, Steel Town:** Moderate activity

11. **Gadap Town**

Overall Risk: Medium (Large zones with high-risk pockets)

Key Subdivisions:

- **Surjani Town (Sector 7-A/B):** Moderate risk
- **Ahsanabad:** Land grabbing concerns (Low–Medium)

12. **Jamshed Town**

Overall Risk: Medium (Diverse urban clusters)

Key Subdivisions:

- **Baloch Colony, Tariq Road:** Mobile theft, snatching (Medium–High)
- **PECHS, Soldier Bazaar:** Moderate incidents

13. **Gulberg Town**

Overall Risk: Medium (Mostly safe, with a few red spots)

Key Subdivisions:

- **Ancholi, Water Pump:** Robbery, snatching (Medium–High)
- **Shafiq Mill Colony:** Theft, mugging

14. **New Karachi Town**

Overall Risk: Medium (Demographic diversity and underreporting)

Key Subdivisions:

- **Godhra Camp, Khamiso Goth:** Mugging, robbery (Medium–High)
- **Sector 5-D/E:** Moderate

safety Low-Risk Towns

15. **Gulshan-e-Iqbal Town**

Overall Risk: Low (Well-planned and patrolled)

Safe Subdivisions:

- **Blocks 4–7, 13D/E:** Minimal crime reports
- **Federal B Area Block 1–6:** Low incidents

16. North Nazimabad Town

Overall Risk: Low (Affluent, well-guarded)

Safe Subdivisions:

- **Blocks H–L, Hyderi:** Secure residential areas
- **Shadman Town:** Patrolled and low crime

17. Liaquatabad Town

Overall Risk: Low (No recent alerts or red zones)

Safe Subdivisions:

- **Federal B Area:** Mostly moderate, no recent red zone reports
- **Al-Karam Square:** Low crime

18. Saddar Town

Overall Risk: Low (Safe commercial and residential mix)

Safe Subdivisions:

- **Clifton Blocks 2–5:** Secure zones
- **Civil Lines, Cantt Area:** Low-risk and well-guarded.

Analysis based on real-time scraping, subdivision-level mapping, and verification from past incident archives.

Synthetic Data Generation Need for Synthetic Data

- The scraped and sourced dataset yielded only ~1800 usable records, which was insufficient for comprehensive zone mapping and safer route planning.

Generation Strategy

- Employed **Faker.js** with real-data constraints to generate **100,000 synthetic crime records**.
- Ensured alignment with:
 - Authentic **Karachi subdivision names and coordinates**
 - **Risk-level-based incident frequency**
 - Realistic **crime types**, time distribution, and severity patterns

Bias Correction

- Media underreporting was corrected by **up-weighting** crime incidents in areas like **Lyari**, which are known to be high-risk but often underrepresented in news coverage.

3.3 Town-wise Risk-Level Classification

The table below provides a town-wise risk classification based on crime frequency, intensity, and known gang or law enforcement presence. These zones were defined using both real and synthetic datasets combined with field insights.

Risk Legend:

- **High Risk** – Gang presence, frequent crimes (snatching, assaults, robbery)
- **Medium Risk** – Mixed safety, reported incidents
- **Low-Medium Risk** – Mostly safe with minor incidents
- **Low Risk** – Guarded or well-patrolled zones

Town	Risk Level	Remarks
Lyari Town	High	Gang activity, high crime density
Korangi Town	High	Frequent snatching, thefts
Landhi Town	High	Red zones, gang-prone pockets
Orangi Town	High	Densely populated, under-policed
SITE Town	High	Industrial theft, unsafe regions
Baldia Town	High	Clustered crime-prone subdivisions
Keamari Town	Medium	Mixed zones with varying risk
Shah Faisal Town	Medium	Block variations and moderate risk areas
Malir Town	Medium	Semi-urban zones, some hotspots
Bin Qasim Town	Medium	Rural-industrial landscape
Gadap Town	Medium	Peripheral town, crime pockets
Jamshed Town	Medium	Urbanized; moderate policing
Gulberg Town	Medium	Some subdivisions with safety concerns
New Karachi Town	Medium	Complex demographics, mixed safety
Gulshan-e-Iqbal Town	Low	Planned areas, generally safe
North Nazimabad Town	Low	Affluent and patrolled
Liaquatabad Town	Low	No recent alerts
Saddar Town	Low	Clifton, Cantt; business-safe

3.4 Summary of Challenges

Throughout the pipeline, significant hurdles were encountered. The most common issues stemmed from data privacy restrictions, unstructured formats, vague locations, and inconsistent naming conventions.

Challenges Overview:

Category	Challenge
NGO Data	No cooperation due to privacy; lacked digital, structured formats
Police Records	Unstructured; missing coordinates and timestamps
News Scraping (Phase 1)	Anti-bot systems (e.g., CAPTCHAs), vague geolocation in text
Outscraper / Google CSE	API limits, manual tuning of keywords needed

API	
Data Cleaning	Inconsistent formats and missing fields
Subdivision Mapping	Required manual effort for over 170 areas
Synthetic Generation	Needed careful balancing for realistic, bias-corrected data

3.5 Data Processing Pipeline

This section explains the transformation applied to raw and synthetic crime records to produce a **structured, analysis-ready dataset** that supports zone classification and route safety features.

3.5.1 A. Location Mapping with NLP + Geopy

Process	Description
Text Location Extraction	Used spaCy to detect place names from article text
Geocoding	Geopy Nominatim API to fetch latitude/longitude for each subdivision

```
from geopy.geocoders import Nominatim
geolocator = Nominatim(user_agent="muhafiz_fyp")
location = geolocator.geocode("Ittehad Town, Baldia, Karachi")
latitude = location.latitude
longitude = location.longitude
```

3.5.2 B. Feature Engineering

Column	Description
INCIDENT_ID	Auto-generated unique ID
TOWN	Town name mapped from subdivision
TOWN_RISK_LEVEL	Aggregated risk label (High, Medium, Low)
TOWN_PRIORITY_RANK	Static numeric rank (e.g., Lyari = 1, Gulshan = 18)
SUBDIVISION	Cleaned and mapped from incident location
SUBDIVISION_RISK_LEVEL	From risk scoring logic
SUBDIVISION_PRIORITY_RANK	Derived from frequency + severity
SEVERITY_SCORE	Float (0.1–1.0) representing incident danger level
SEVERITY	Label: Low / Medium / High
DATE	Date of incident (real or synthetically generated)

LATITUDE, LONGITUDE	Coordinates using geopy
CRIME_TYPE	Cleaned and standardized type

IS_RED_ZONE → IS_WHITE_ZONE	One-hot encoded risk zones for frontend use
RISK_ZONE	Final risk color label
SOURCE	Scraped / Synthetic / Police / News
RANK	Final composite rank

3.5.3 C. One-Hot Zone Encoding

Zone	Condition (SEVERITY_SCORE)
RED Zone	≥ 0.85
ORANGE Zone	0.70 – 0.85
YELLOW Zone	0.50 – 0.70
GREEN Zone	0.30 – 0.50
WHITE Zone	< 0.30

- These flags are critical for route filtering and map overlays.

3.5.4 D. Final Output

- Final CSV dataset: **100,000 rows**, each containing geocoded and zone-tagged records
- Supports:
 - Risk-aware route computation
 - Interactive map overlays
 - Subdivision-level trend analysis

3.6 Red Zone Mapping and Safer Route

Algorithm Module Focus

This module focuses on the algorithmic system that powers:

1. **Red Zone Visualization** – dynamic mapping of crime intensity
2. **Safer Route Suggestion** – secure path planning avoiding dangerous areas.

3.6.1. Red Zone Mapping Data Integration

Component	Details
Road Network (OSMnx)	Extracted from OpenStreetMap (driveable roads only)
Geocoded Incidents	Mapped crime records to subdivisions using coordinates
Zone Encoding	Assigned RISK_ZONE to subdivisions, then tagged nearby road segments
Spatial Join Tool	Used GeoPandas to attach zone data to streets within ~300m proximity

Zone Color Legend

Risk Level	Criteria	Zone Code	Color
High	Gang/criminal hotspots	RED	Red
Medium	Moderate risk	ORANGE	Orange
Low–Medium	Occasionally unsafe	YELLOW	Yellow
Low / Safe	Secure or affluent	GREEN	Green
Unknown	No verified data	WHITE	White

Visualization

- Rendered in **Leaflet.js**
- Pulled from Flask API /red-zone-map
- Subdivisions shown as color-coded overlays.

3.6.2 Safer Route

Suggestion Graph

Construction

- **OSMnx** used to build a `networkx.MultiDiGraph`
- Each road segment (edge)

weighted: `edge_weight = base_distance + zone_penalty`

Penalty Table

Zone	Penalty Weight
Red	+1000
Orange	+400
Yellow	+200
Green	0
White	+50

Routing Modes

Mode	Description
Safest Route	Excludes Red/Orange/Yellow zones
Medium-Safe Route	Avoids Red; allows Orange/Yellow with penalty

Fastest route is not active yet (planned feature)

Algorithm & Flow

```
path = nx.shortest_path(graph, source, target, weight='custom_weight')
```

- Returns: route path, zone summary, and total distance

3.6.3 Visualization and Interaction Frontend (React + Leaflet)

- **Inputs:** Source & destination
- **Displays:**
 - Safer path overlay
 - Zone colors
 - Crime heat marker

Flask API

- Endpoint: /get_route
- Input:

```
{  
  "source": [lat, lon],  
  "destination": [lat, lon],  
  "mode": "safe" |  
  "medium"  
}
```

- Output:

```
{  
  "route_polyline":  
  [...],  
  "total_distance":  
  4200,  
  "zone_summary":  
  { "red": 0,  
    "orange": 2,  
    "yellow": 5  
  }  
}
```

Express ↔ Flask Communication

- Node.js sends request:

```
const response = await  
  axios.post("http://localhost:5000/get_route", { source,  
    destination, mode  
  });
```

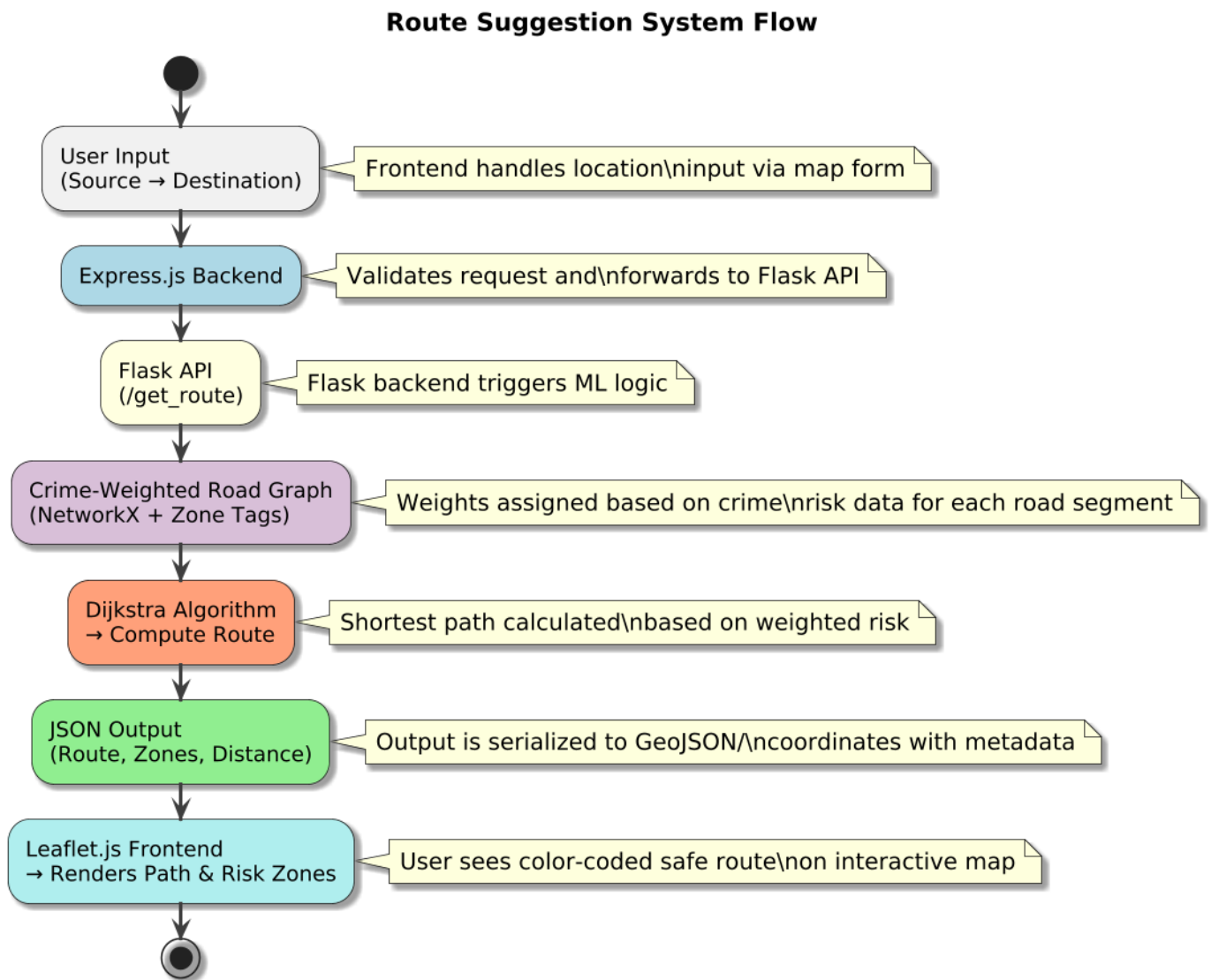
- Flask handles routing → responds → frontend renders map

3.6.4 Route Engine Benefits

Feature	Description
Real-time Risk Awareness	Adjusts to zone-tagged datasets
Scalable Design	Port to any city with risk-encoded maps
ML-Free Engine	Rule-based Dijkstra logic; no model training

3.6.5 System Flow Diagram

Figure 3.18.1 – Route Suggestion System Flow



3.7 Integration of Flask APIs with Express Backend

This section explains the architecture and API-level integration between the **Flask-based microservices** (used for Red Zone Mapping and Route Suggestion) and the **Node.js (Express)** backend, which powers both the user and admin panels of the Muhafiz platform.

3.7.1 Overview of Integration Architecture

The system follows a **modular microservices architecture**:

Component	Responsibilities
Express.js	API gateway, user routes, admin routes, JWT auth, frontend API routing
Flask (Python)	Crime zone generation, safer route algorithm, crime-weighted logic, JSON responses

This decoupled structure ensures independent scaling, isolated logic, and cross-language flexibility

3.7.2 Key APIs and Responsibilities

API Endpoint	Hosted In	Function
/get_route	Flask	Computes safest or medium-safe route between two coordinates
/red-zone-map	Flask	Returns zone boundary data with color-coding (Leaflet-ready)
/api/route	Express	Accepts user request, sends to Flask /get_route
/api/red-zones	Express	Forwards request to Flask to retrieve zone overlays

3.7.3 Communication

Flow Step-by-step API

Workflow:

1. **User Input (Frontend)**
 - Selects source, destination, and route type (safe or medium).
2. **Express.js Receives Request**
 - Hits either /api/route or /api/red-zones.
3. **Proxy to Flask**
 - Using axios in Node:

```
const axios = require("axios");
const res = await
  axios.post("http://localhost:5000/get_route", { source:
    [lat1, lon1],
    destination: [lat2, lon2],
    mode: "safe"
  });
```

4. **Flask Engine Processes Request**
 - Runs Dijkstra on a zone-tagged road graph.[\[38\]](#)
 - Returns JSON with route polyline, zone summary, total distance.
5. **Express Responds to Frontend**
 - Sends data back to frontend for rendering in Leaflet.js.

3.7.4 Error Handling and Optimization

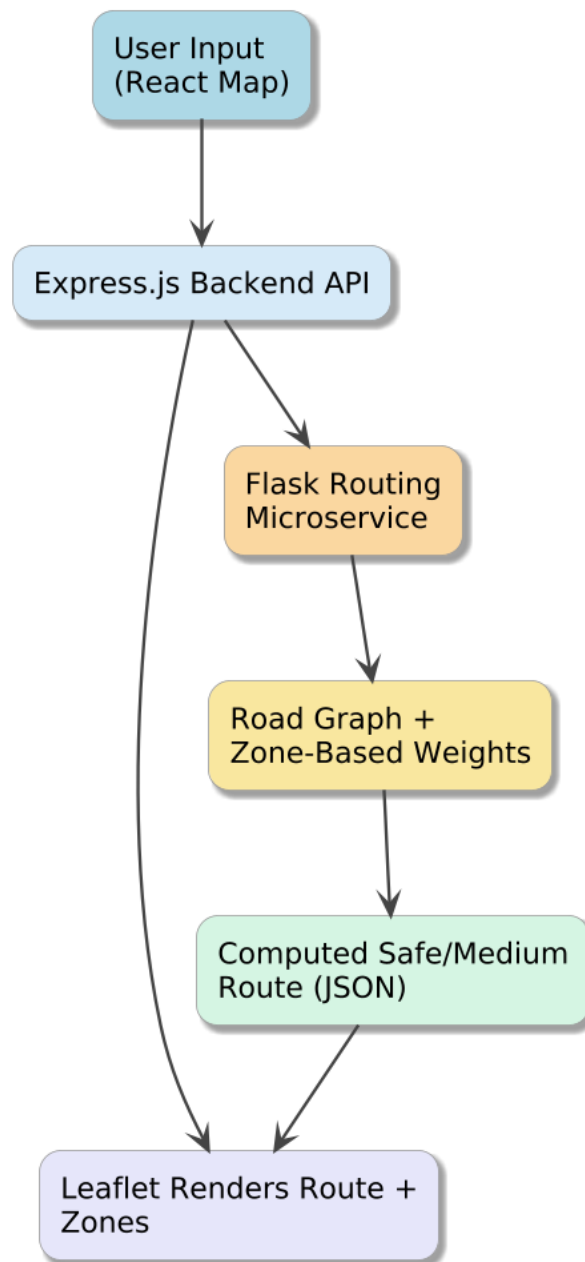
Feature	Implementation
Timeout Handling	Express enforces timeout for Flask response
Retry Logic	Optional retry attempts on Flask failure
Response Validation	Checks if returned data contains route, zones, and distance fields
Caching (Optional)	Redis/memory cache for popular routes or zone overlays
Load Management	Flask service runs independently and can be scaled separately

3.7.5 Deployment Considerations

Component	Platform	Notes
Express.js Backend	Render / Railway	Central backend, API gateway
Flask Service	Render / EC2	Independent microservice for zone + route logic
Database	MongoDB Atlas	Shared among services, 2dsphere for geoqueries
Frontend	Vercel	Sends route/zone requests via Express API

3.7.6 Integration Flow Diagram

Integration Flow Diagram - Route Engine



3.7.7 Summary

This integration enables:

- **Seamless frontend-backend interaction**
- **Language independence** (Node ↔ Python)
- **Modular ML logic** separate from app operations
- **Scalable backend**, allowing Flask to scale on-demand

3.8 Muhafiz Admin & User Panel – Frontend Architecture and Features

This section documents the design of the **Admin and User Panels**, built with **React.js (Vite)** and integrated with Express and Flask microservices. Each panel serves distinct user roles, enabling reporting, map interaction, and platform management.

3.8.1 Frontend Architecture (React + Vite)

Layer	Technology
UI Framework	React.js (Vite)
Styling	Tailwind CSS
Maps	Leaflet.js + GeoJSON
State Mgmt	Context API + Axios
Routing	React Router
API Calls	Express.js REST API

Vite ensures faster HMR (Hot Module Reload), optimized production builds, and modular design.

3.8.2 Backend Integration

- React frontend connects to:
 - **Express API** for all database and auth operations.
 - **Flask API** for real-time ML services (routes/zones).

JWT-based authentication protects API routes.

3.8.3 MongoDB Atlas – Central Database

Collections	Purpose
Users	Auth, roles
reports	Incident records
Zones	Red/orange/yellow zone metadata
community_posts	Discussion threads
admin_logs, feedbacks	Moderation, analytics, suggestions

Supports **geospatial indexing** with 2dsphere for fast queries.

3.9 Admin Panel Features

Module	Functionality
Dashboard	Stats overview, weekly trends, zone map snippet
Incident Manager	Review, filter, approve/reject reports
Approval Queue	Pending reports needing admin action
Red Zone Manager	Override ML zones, see last update time and incident counts
Analytics Dashboard	View heatmaps, charts, distribution graphs
User Management	Suspend/delete users, assign roles
Post Moderation	Moderate user discussions and comments
Safety Resources	Upload PDFs or links (e.g., police hotlines)
Settings	Theme switch, password change, logout

Optional Add-ons:

Feature	Purpose
Alerts & Notifications	Push messages to users
Audit Logs	Admin activity tracking
Export Tools	Export reports or users as CSV
Feedback Center	View platform suggestions

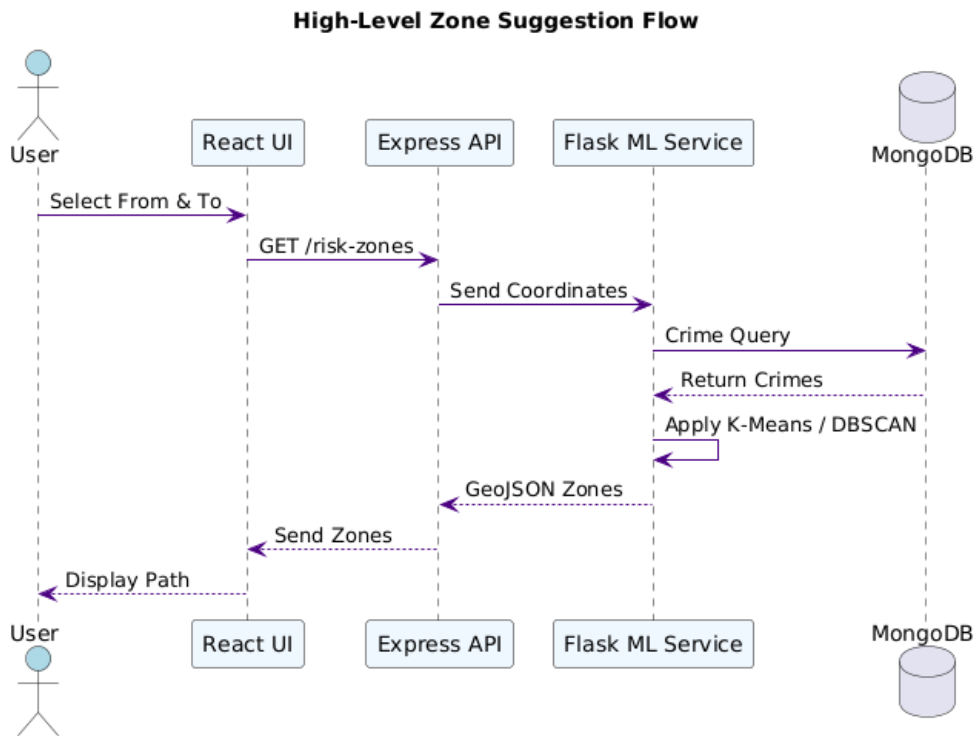
3.10 User Panel Features

Feature	Description
Profile Page	User info, report history, bookmarked threads
AI Chatbot Assistant	Tips, route suggestions, emergency help (voice + text) (in future)
My Reports	View report status, timestamps, location
Live Alerts	Real-time incident notifications (zone-based)
Travel History	Log of past suggested routes (in future)
Feedback Form	Submit issues or improvement ideas
Location Sharing	Share real-time location with trusted contacts (optional)
Multilingual UI	Support for English and Urdu (in future)
Gamification	Badges for active reporters and helpers (in future)

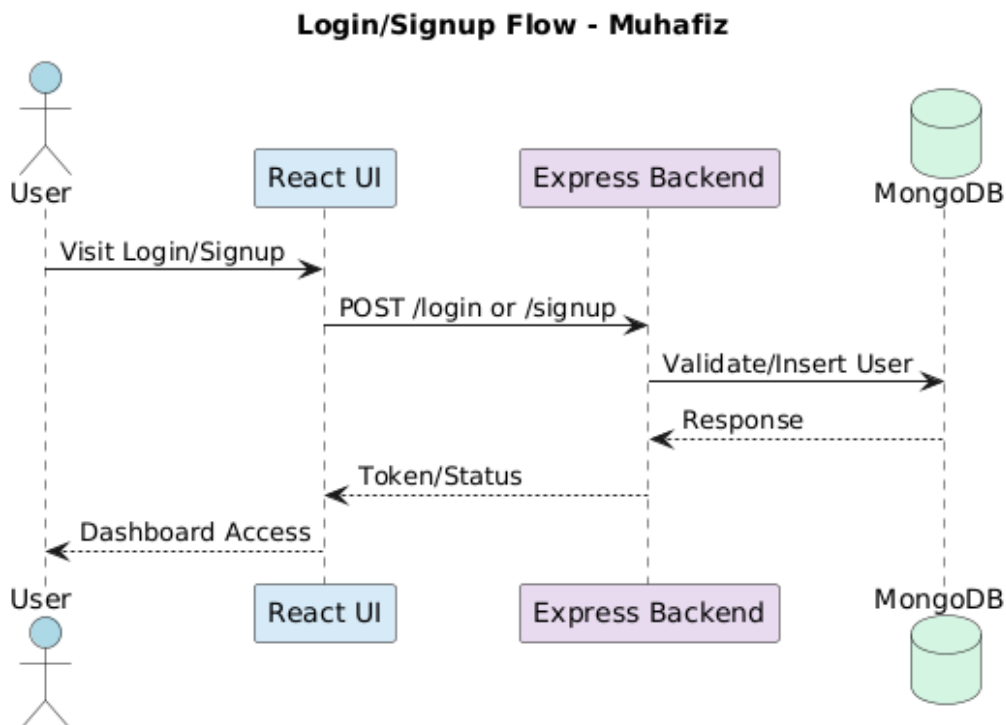
3.11 Summary: Admin vs User Panel

Feature Area	User Panel	Admin Panel
Incident Handling	Submit & track reports	Moderate, approve, archive reports
Routing	Request safe route	View routes, zone updates
Red Zone Awareness	View zone overlays	Override zones, edit risk tags
Community Features	Post, bookmark, react	Moderate posts and comments
Security & Auth	JWT tokens, session control	Role-based admin access
Analytics	Basic stats or none	Full visual analytics + heatmaps
Data Tools	View only	Export, log access, manual override

1.High-Level Overview Sequence Diagram

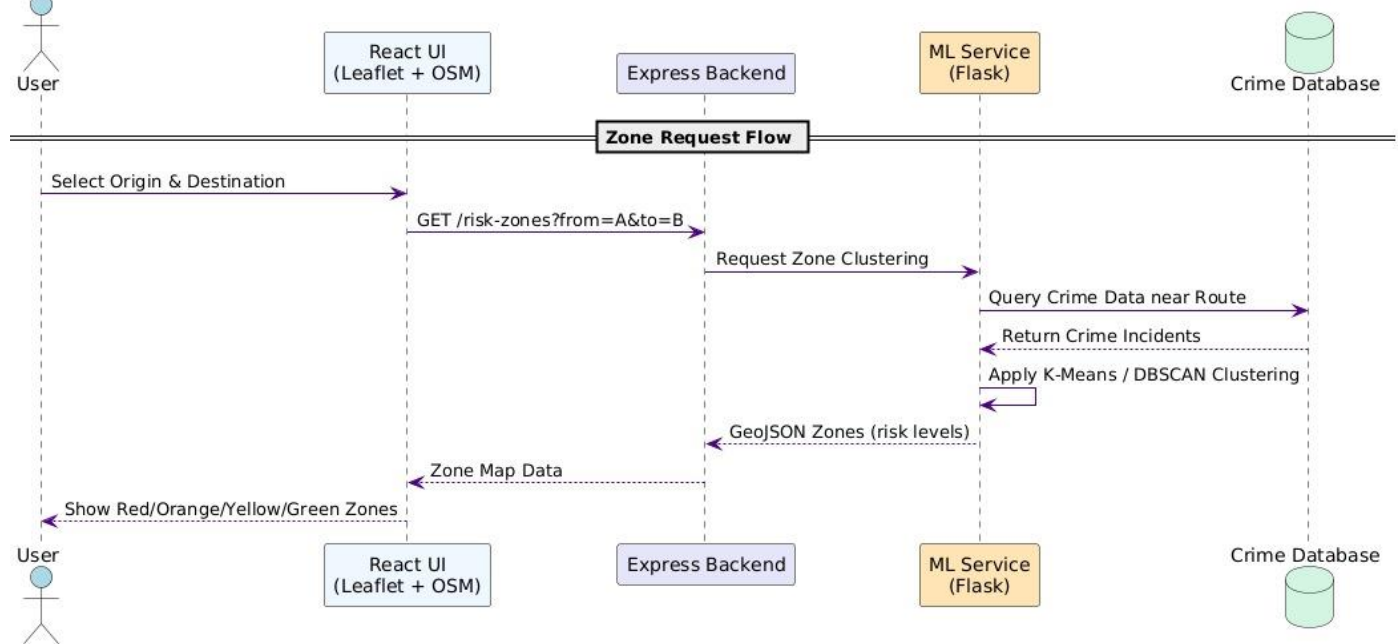


a. Feature-Wise Detailed Diagrams



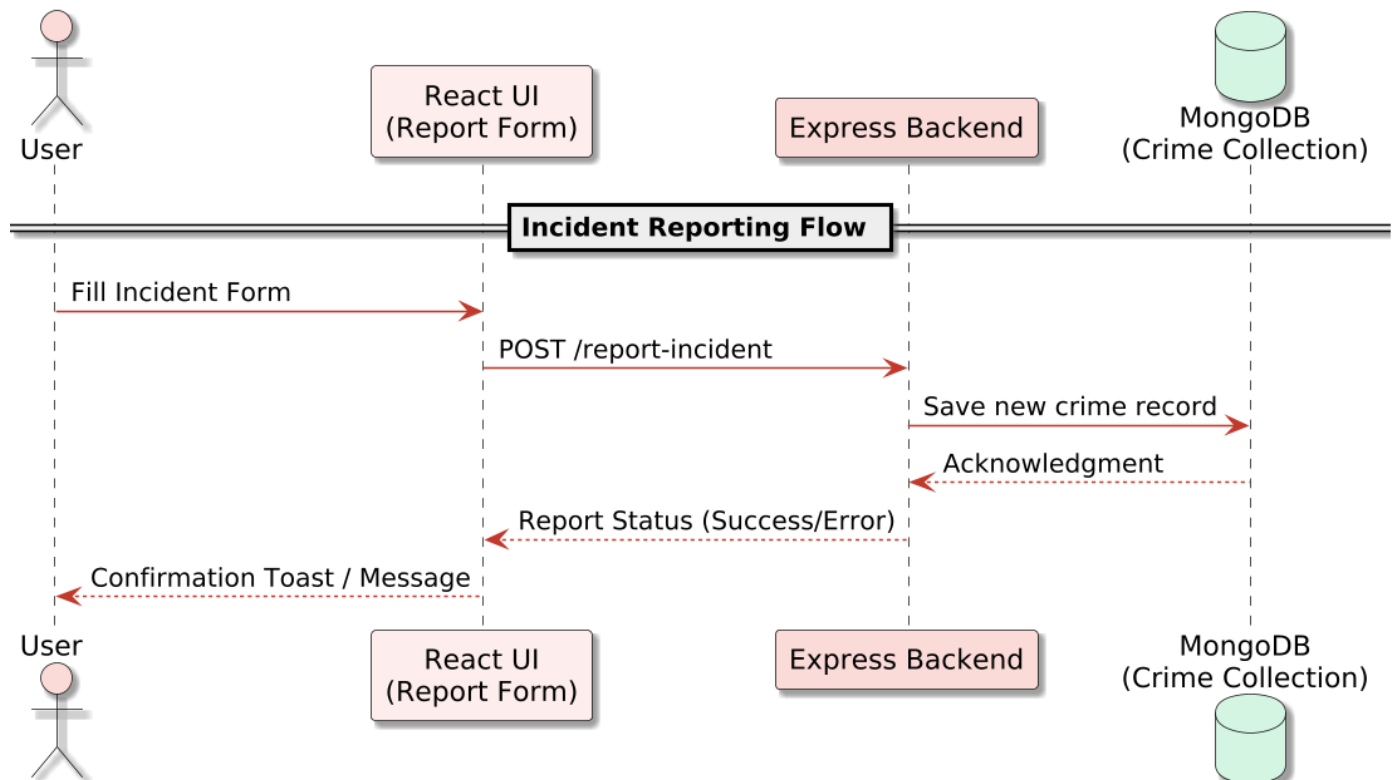
b. Red Zone Map Viewing

Muhafiz - Red Zone Mapping Flow (High-Level Sequence Diagram)



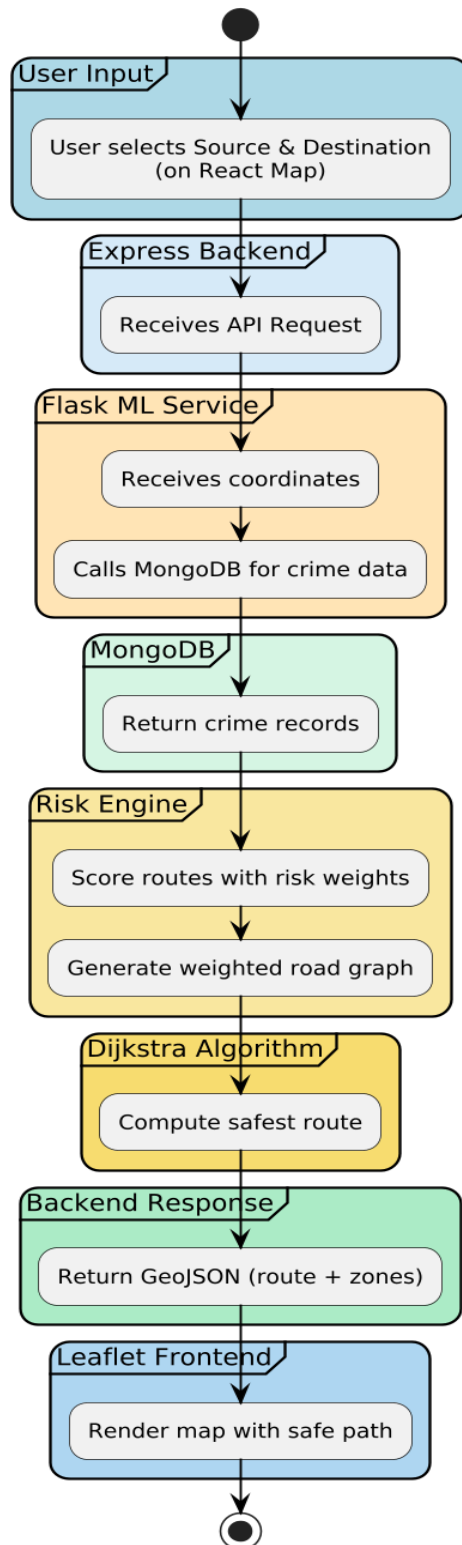
c. Incident Reporting

Incident Reporting Flow - Muhafiz



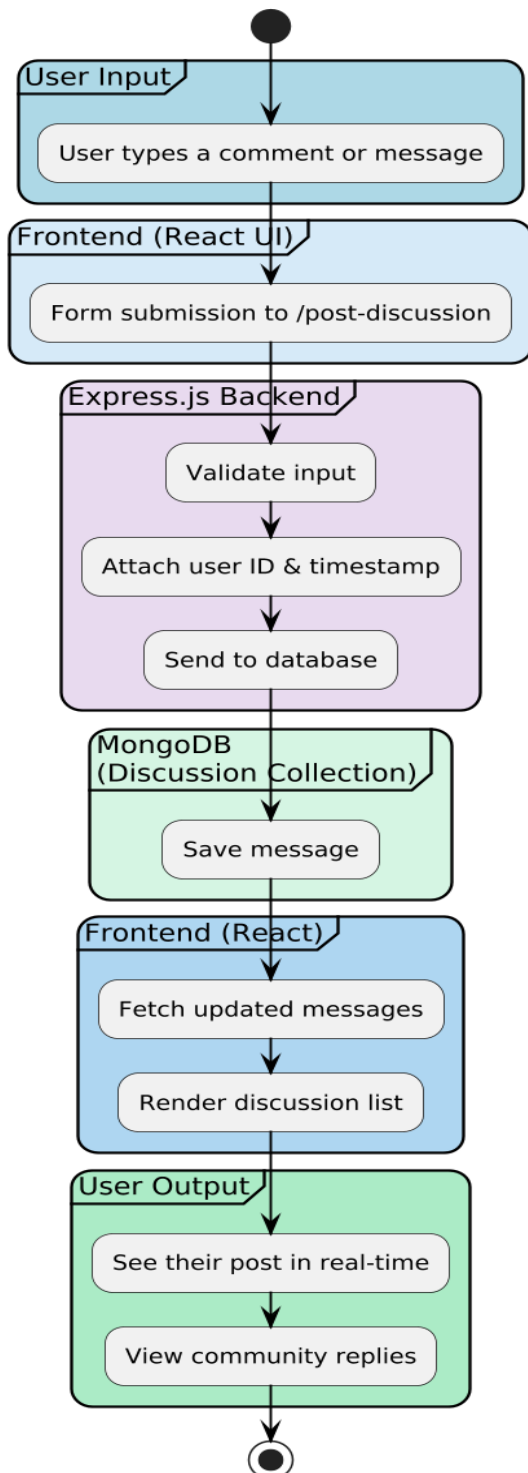
d. Safer Route Suggestion

Muhafiz - Safer Route Suggestion Flow (Clean & Colored)



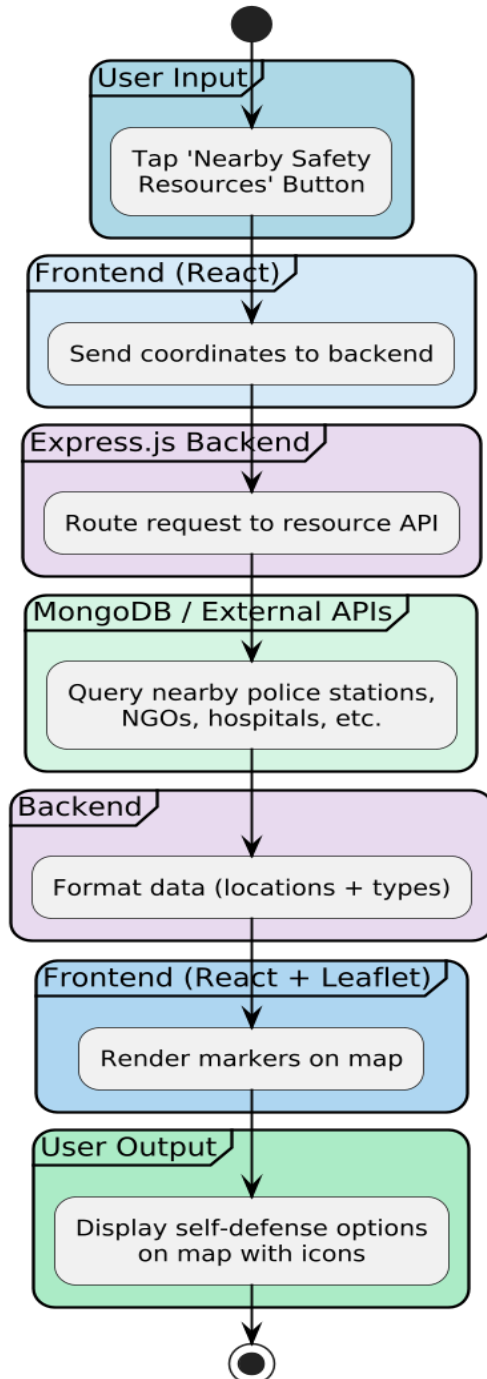
e. Community Discussion

Muhafiz - Community Discussion Flow



g. Self-Defense Resources Download

Muhafiz - Self Defense Resources Flow



2.Data Flow Diagram (DFD – Level 1)

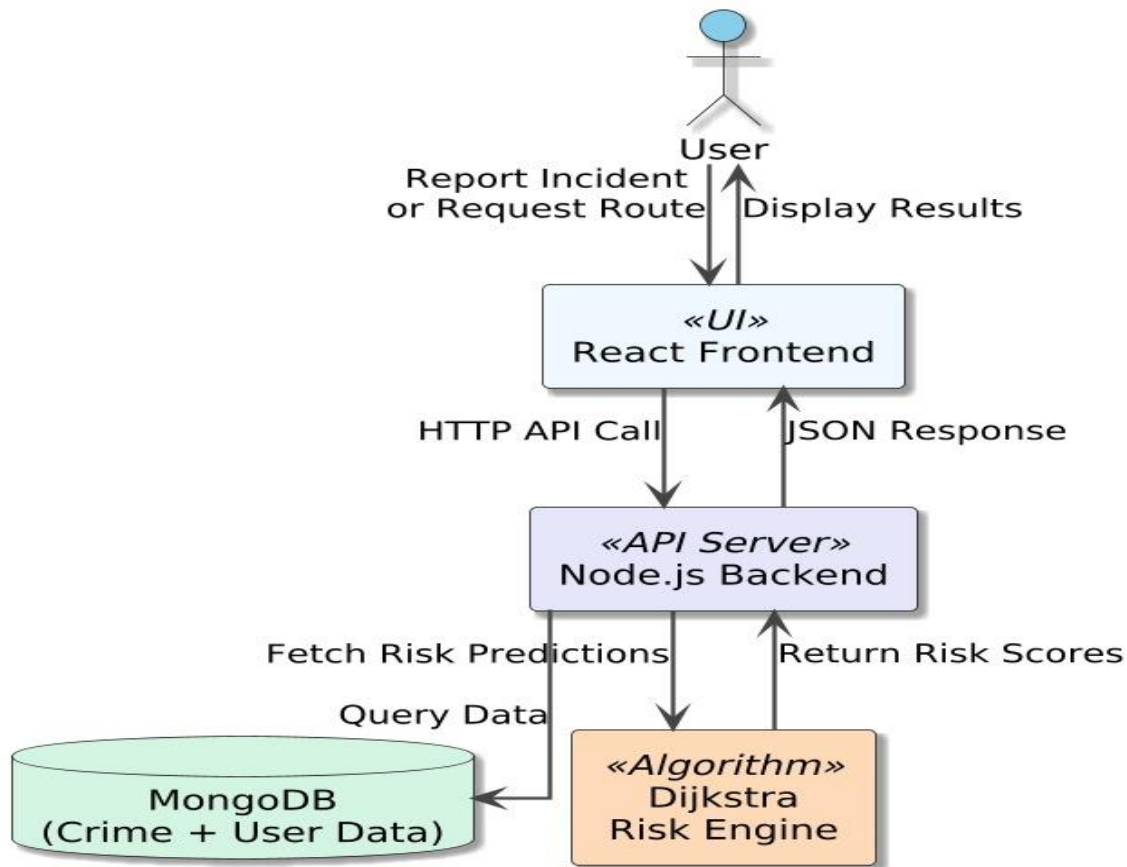


Figure: System Interaction Flow of Muhafiz Platform

This diagram illustrates the technical interaction flow between the user interface and backend systems in the Muhafiz application. Users report incidents or request safe routes through the React-based frontend. These actions trigger API calls to the Node.js backend, which interacts with the MongoDB database to fetch data and with the Python-based risk prediction module to compute zone-aware scores. Results are returned in JSON format and displayed to the user in real-time.

3.Data Flow Diagram (DFD – Level 2)

Purpose: Illustrate end-to-end data processing across Muhafiz's core modules and their interactions.

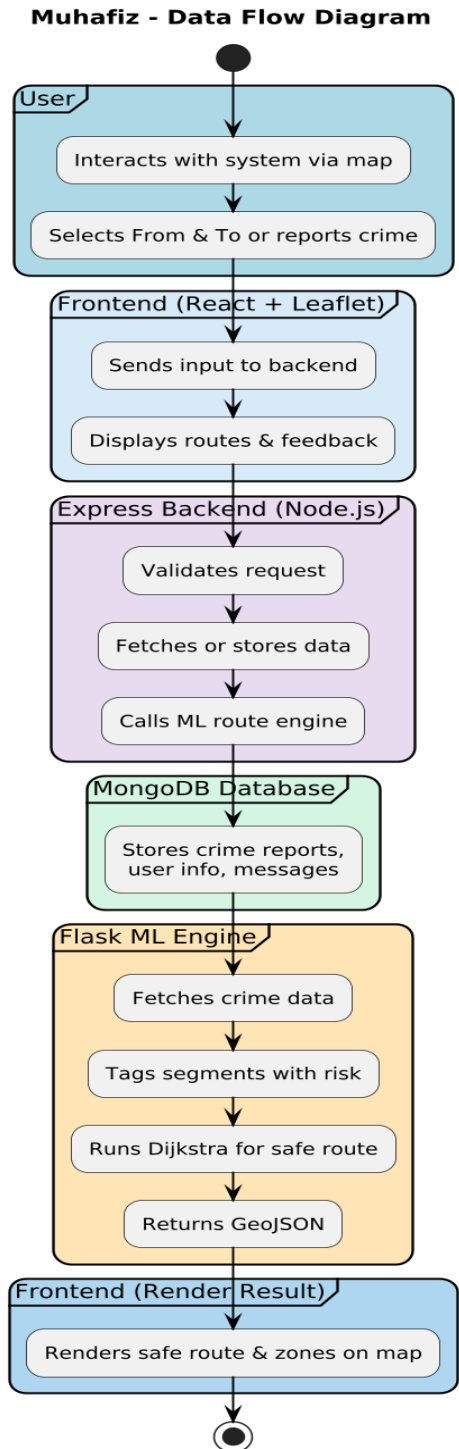
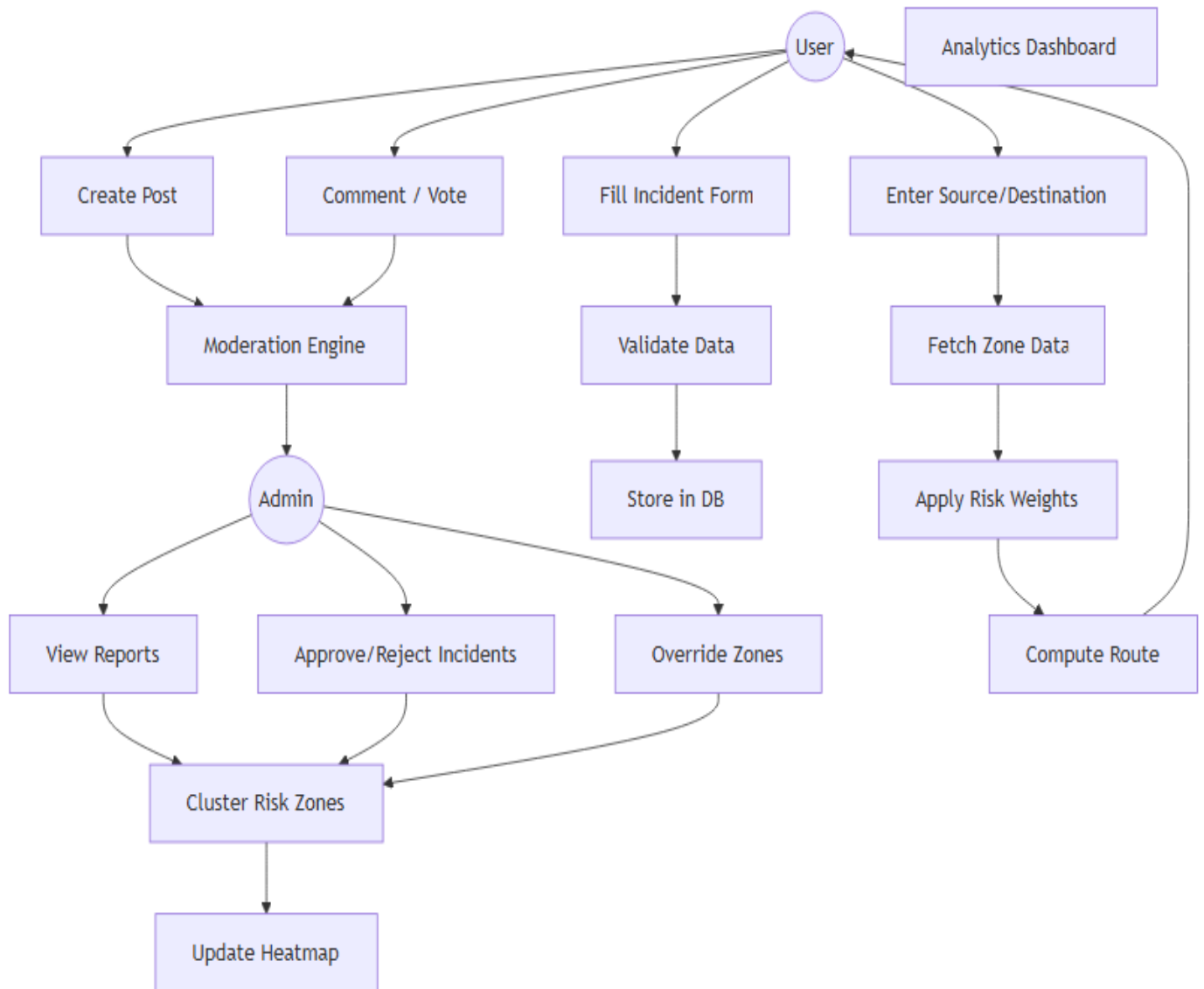


Figure: DFD Level 2 – Muhafiz Platform



This diagram represents the internal data flow of the Muhafiz system. Users interact with modules such as incident reporting, route planning, and the community forum. Submitted incident data is validated and stored in the database. Route planning retrieves zone data, applies predefined risk weights, and computes the safest or medium-risk path using graph-based logic. In the community forum, user-generated content is moderated. Admins can review reports, override zone classifications, and update risk zones through the analytics panel. The Red Zone Engine updates the heatmap based on incident clustering logic and administrative input.

4. Activity Diagram

Purpose: Visualize end-to-end user interaction flow and system responses



Figure: Activity Diagram Overview

The activity diagram illustrates the core workflow of the Muhafiz system. It begins with user authentication (login or registration), followed by key actions such as incident reporting (which triggers ML-based updates), safe or medium route planning, and participation in the AI-moderated discussion forum. Admin-specific activities include reviewing user reports and manually overriding zone classifications. Throughout the process, the system handles database storage, heatmap rendering, and path computation in the background.

5. Agile Life Cycle Diagram

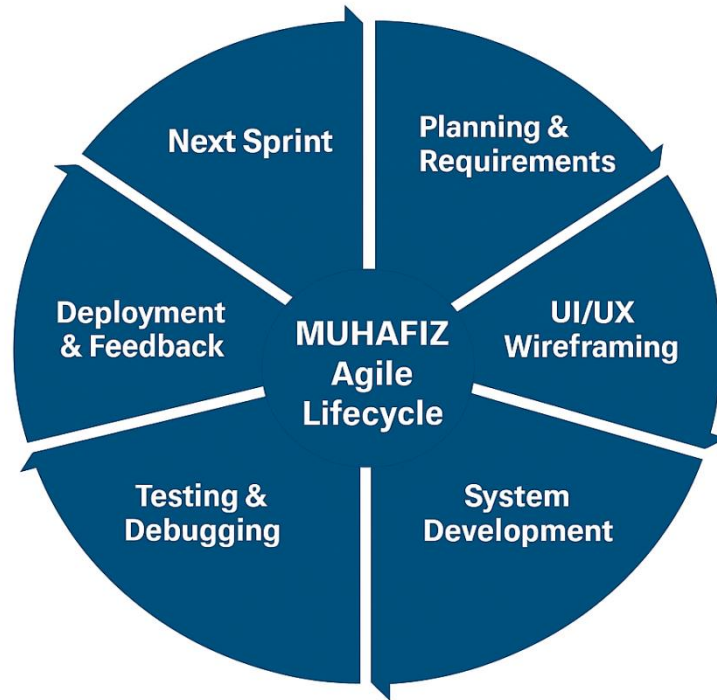


Figure: Muhafiz Agile Lifecycle Diagram

Purpose: The diagram illustrates the Agile development lifecycle followed in the Muhafiz project. It highlights six iterative stages: Planning & Requirements, UI/UX Wireframing, System Development, Intelligent Route Engine Integration, Testing & Debugging, Deployment & Feedback, and Next Sprint. Each stage represents a key development phase aligned with Agile principles, enabling modular delivery, user-centric design, and continuous improvement.

6. Class Diagram

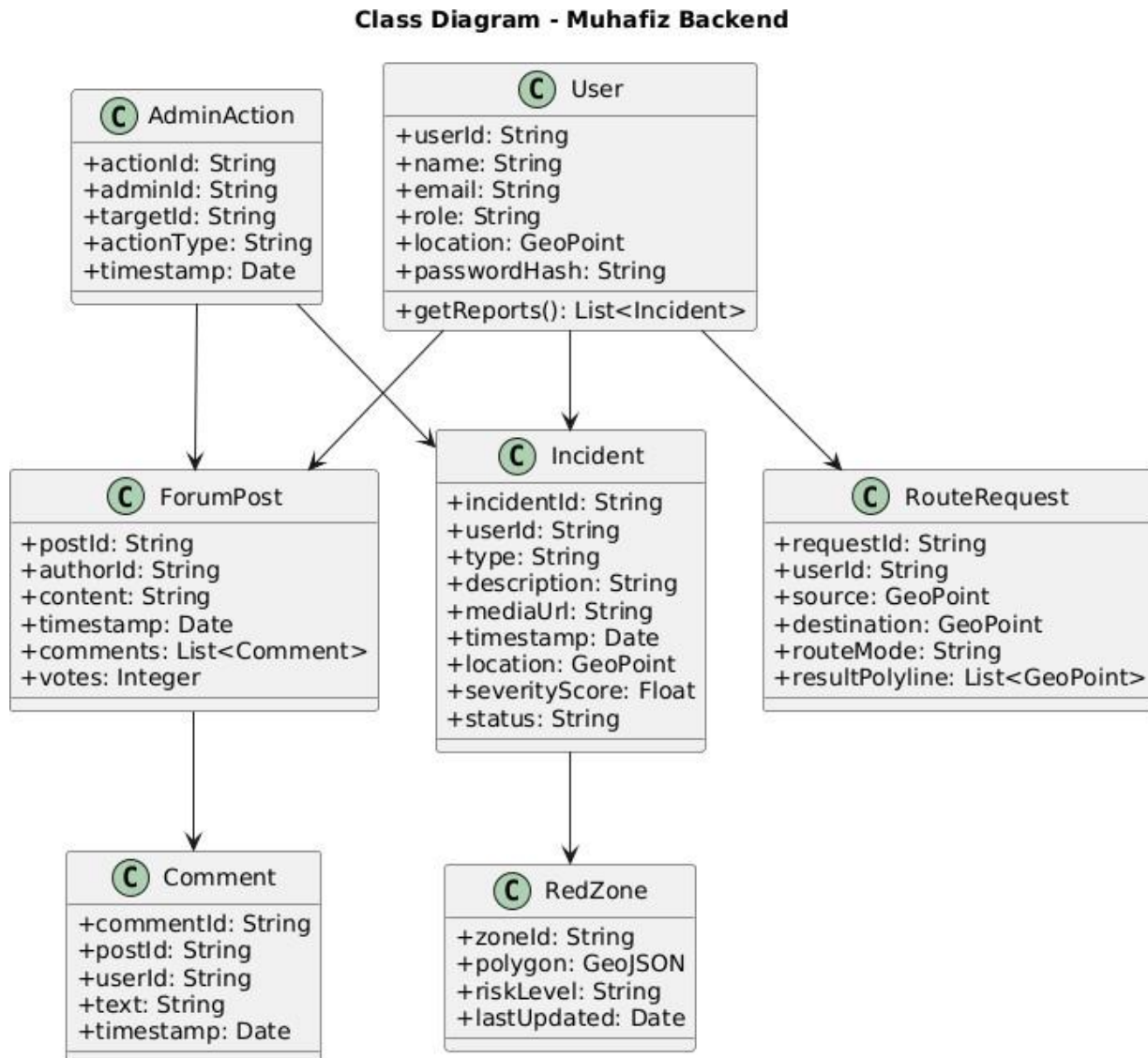


Figure : Class diagram showing 7 key entities:

- *User (profile/auth), Incident (reports), RedZone (risk areas)*
- *Relationships: User creates Incidents, AdminActions modify entities*
- *Critical attributes: GeoPoint locations, severityScore, riskLevel*

7 .ER Diagram (Entity-Relationship Diagram)

Purpose: Illustrate NoSQL data model and collection relationships.

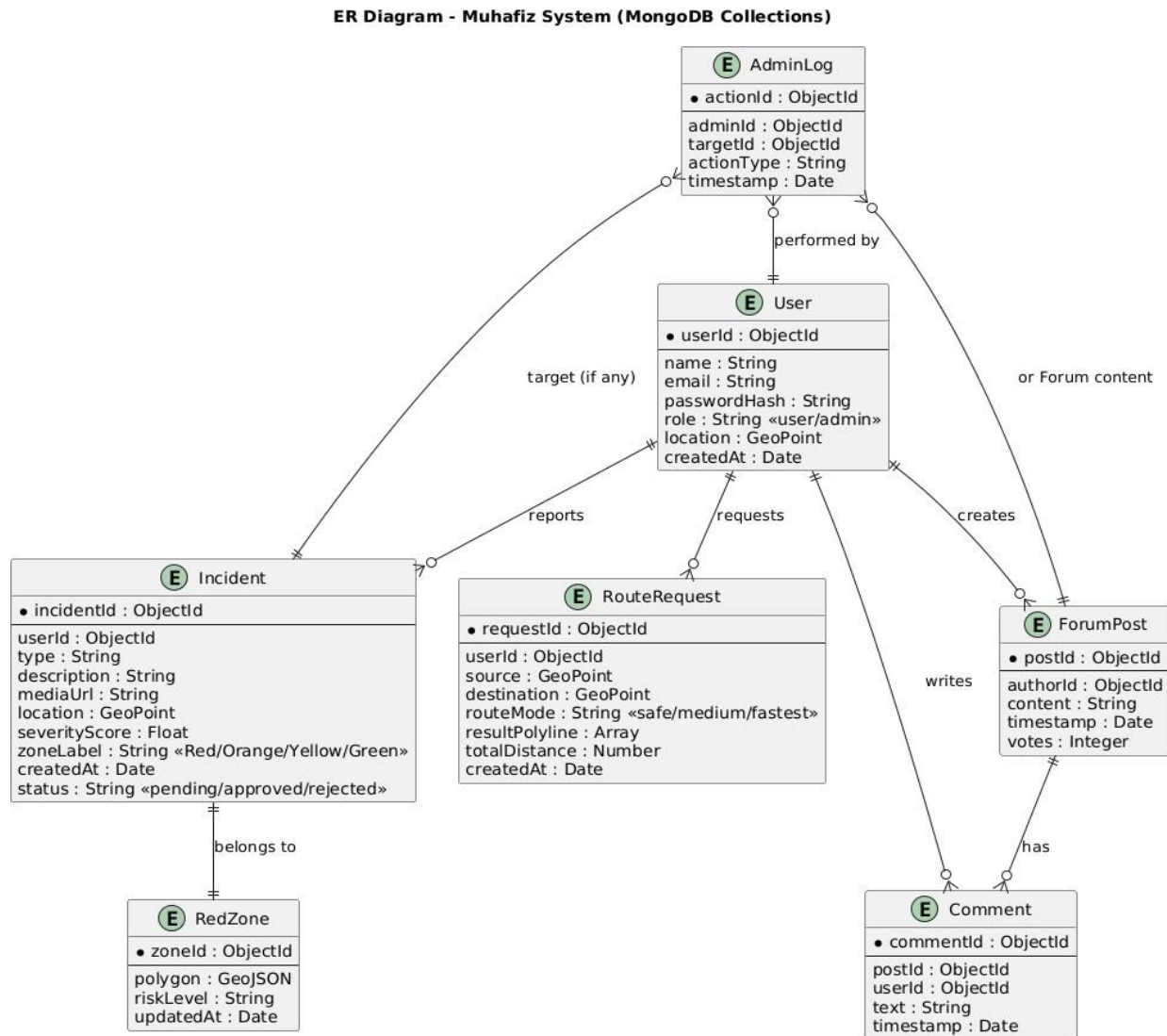


Figure : ERD for MongoDB collections:

- *Core collections: Incident (status: pending/approved), RedZone (GeoJSON polygons)*
- *Relationships: User reports → Incident belongs to → RedZone*
- *Special fields: routeMode («safe/medium/fastest»), GeoPoint coordinates*

8. Gantt Chart

Purpose: Visualize development phases and milestones.

Muhafiz – Red Zone Mapping and Community-Driven Support

Task	Mar 1	8 15	22 29	Apr 5	12 19	Apr 3	May 3	May 1	May 7	24 4
Requirement Analysis										
Literature Review										
Data Collection										
System Design										
Frontend Development										
Backend API Development										
Machine Learning										
Final Report & Documentation										
Finul Report & Documentat										
Community-Driven Support										

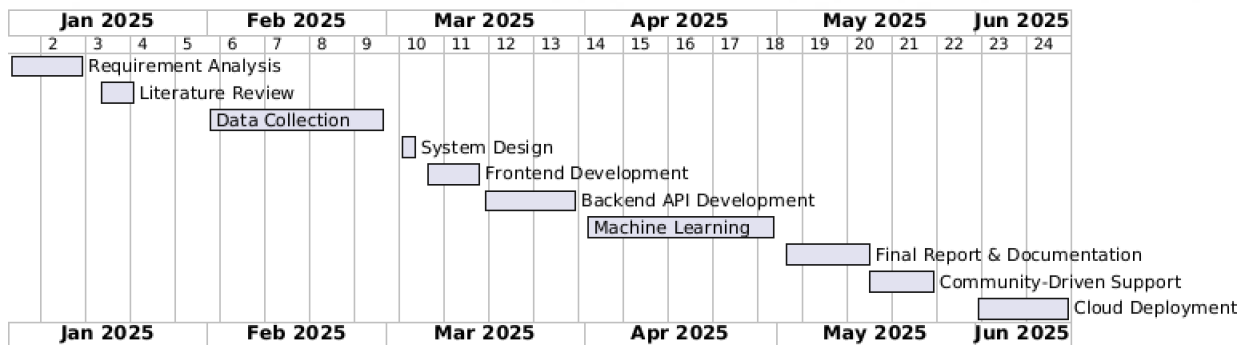


Figure : Gantt chart spanning Jan-Jun 2025:

- *Key phases:* Requirement Analysis (Jan), Geospatial API integration (Mar), ML development (Apr)
- *Critical milestones:* Backend deployment (May 3), Final documentation (Jun)
- *Parallel tracks:* Frontend/Backend development and ML implementation

Chapter 4: Experimental Results and Analysis

4.1 Introduction

This chapter presents the practical results of the Muhafiz platform. Screenshots and dashboards demonstrate the functionality of the Admin and User Panels, zone detection and route safety system, and visual data analysis. Each component is evaluated in terms of accuracy, performance, and usability.

SECTION A: Platform UI & Functionality Evidence

4.2 Admin Panel – Screenshots & Descriptions

4.2.1 Admin Dashboard

Purpose: Overview of total reports, red zones, user stats, and weekly crime insights.

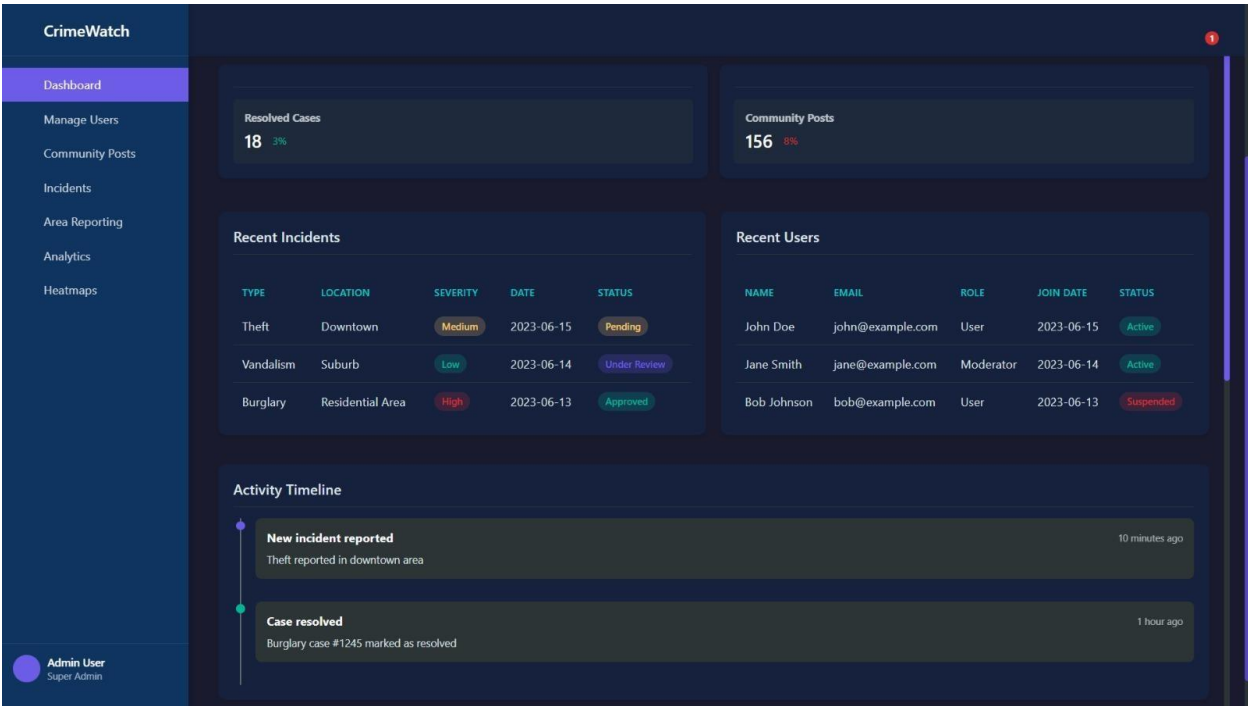


Figure 4.1: Admin Dashboard showing total incidents, red zone counts, and weekly chart.

4.2.2 Incident Moderation Panel

Purpose: Displays all submitted reports. Admin can review, approve, reject, or archive incidents.

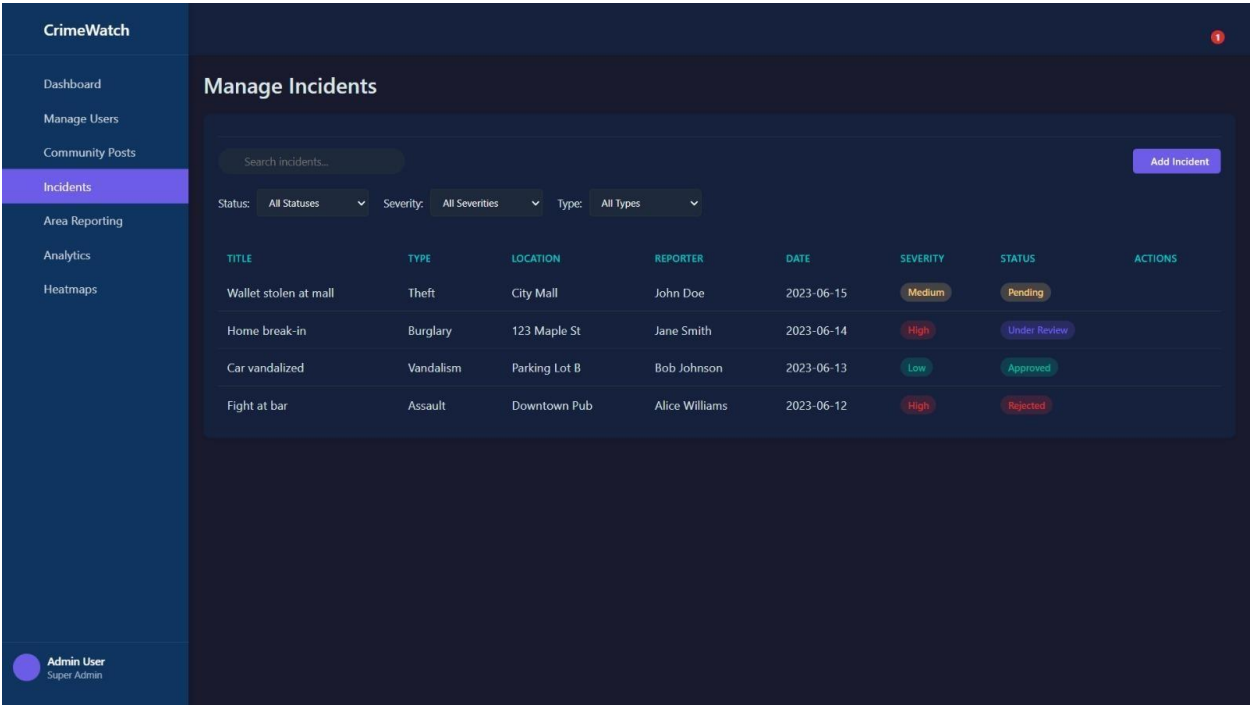


Figure 4.2: Moderation page with incident list, filter by area/crime type, and action buttons.

4.2.3 Red Zone Manager

Purpose: Visualize and manage dynamic risk zones.

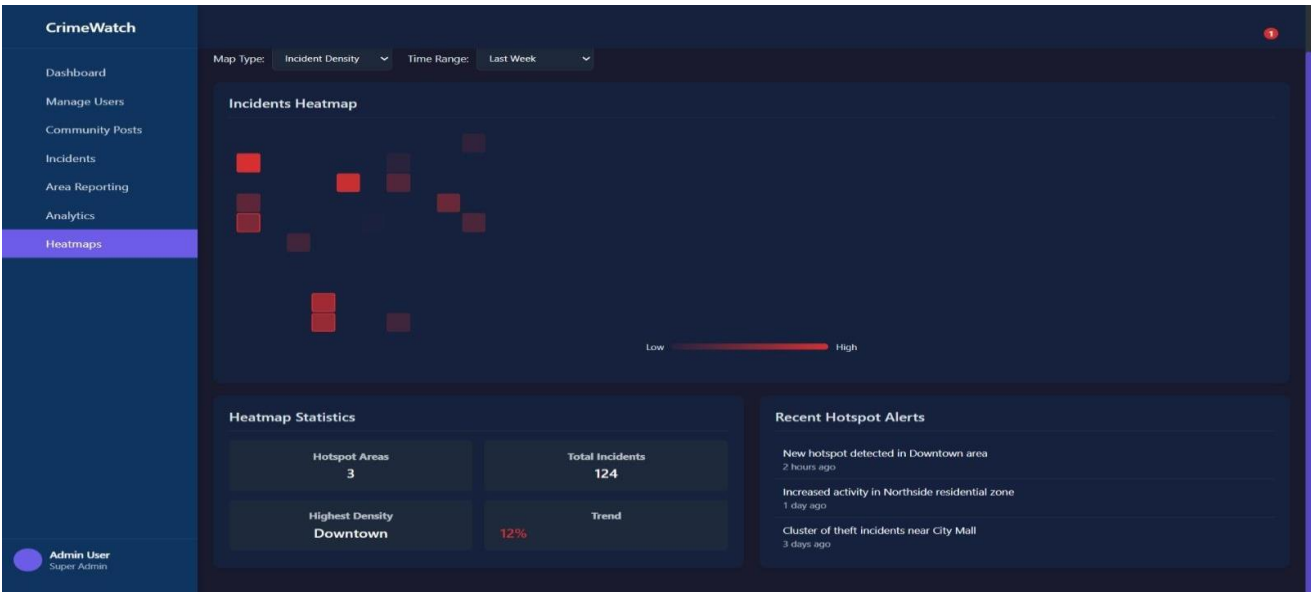


Figure 4.3: Interactive map with colored overlays – red, orange, yellow, green zones.

4.2.4 Analytics Dashboard

Purpose: Charts for area-wise crime count, type frequency, and user growth.

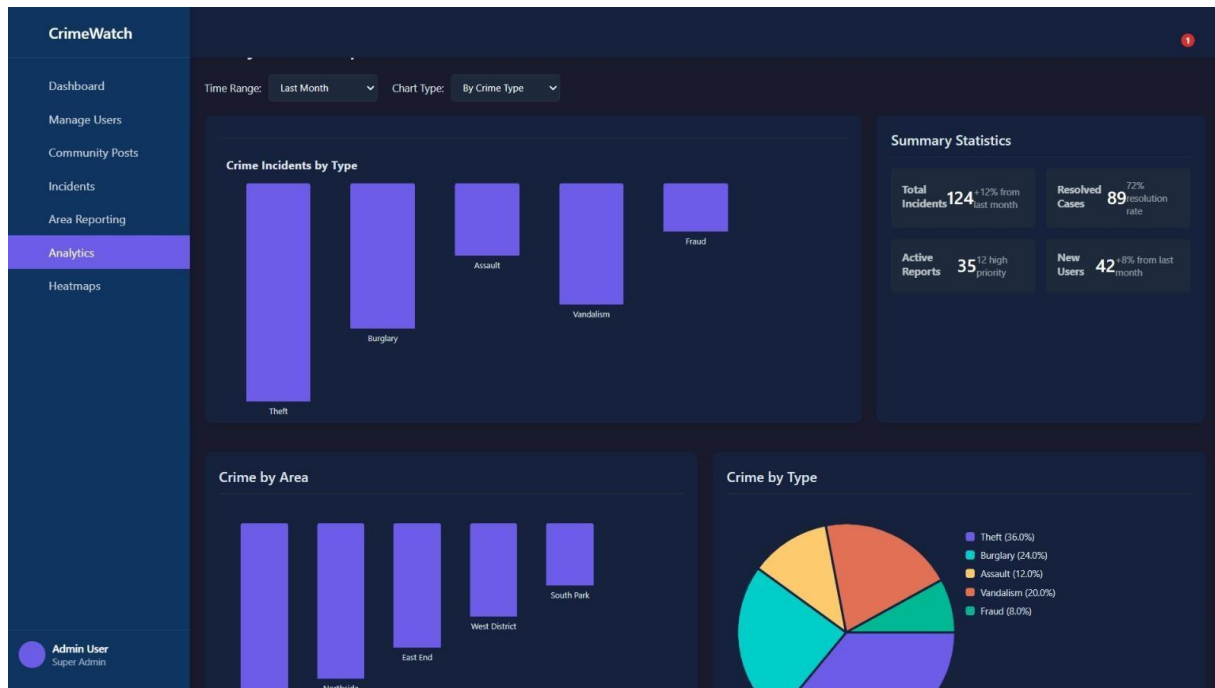


Figure 4.4: Bar/line chart displaying crime types and trend over time.

4.2.5 Community Moderation

Purpose: Admin review of user-generated discussion posts and safety alerts.

CrimeWatch

Dashboard
Manage Users
Community Posts
Incidents
Area Reporting
Analytics
Heatmaps

Manage Community Posts

Search posts...

All Published Pending

TITLE	AUTHOR	DATE	REPLIES	VIEWS	STATUS	ACTIONS
1	Safety concerns in downtown area	John Doe	2023-06-15	12	145	Published
2	Neighborhood watch meeting	Jane Smith	2023-06-14	8	98	Published
3	Report of stolen packages	Bob Johnson	2023-06-13	5	76	Pending Review

Admin User
Super Admin

Figure 4.5: Comment moderation panel with flagging and delete options.

4.3 User Panel – Screenshots & Descriptions

4.3.1 User Map Interface

Purpose: Navigate with real-time risk visualization and route optimization.

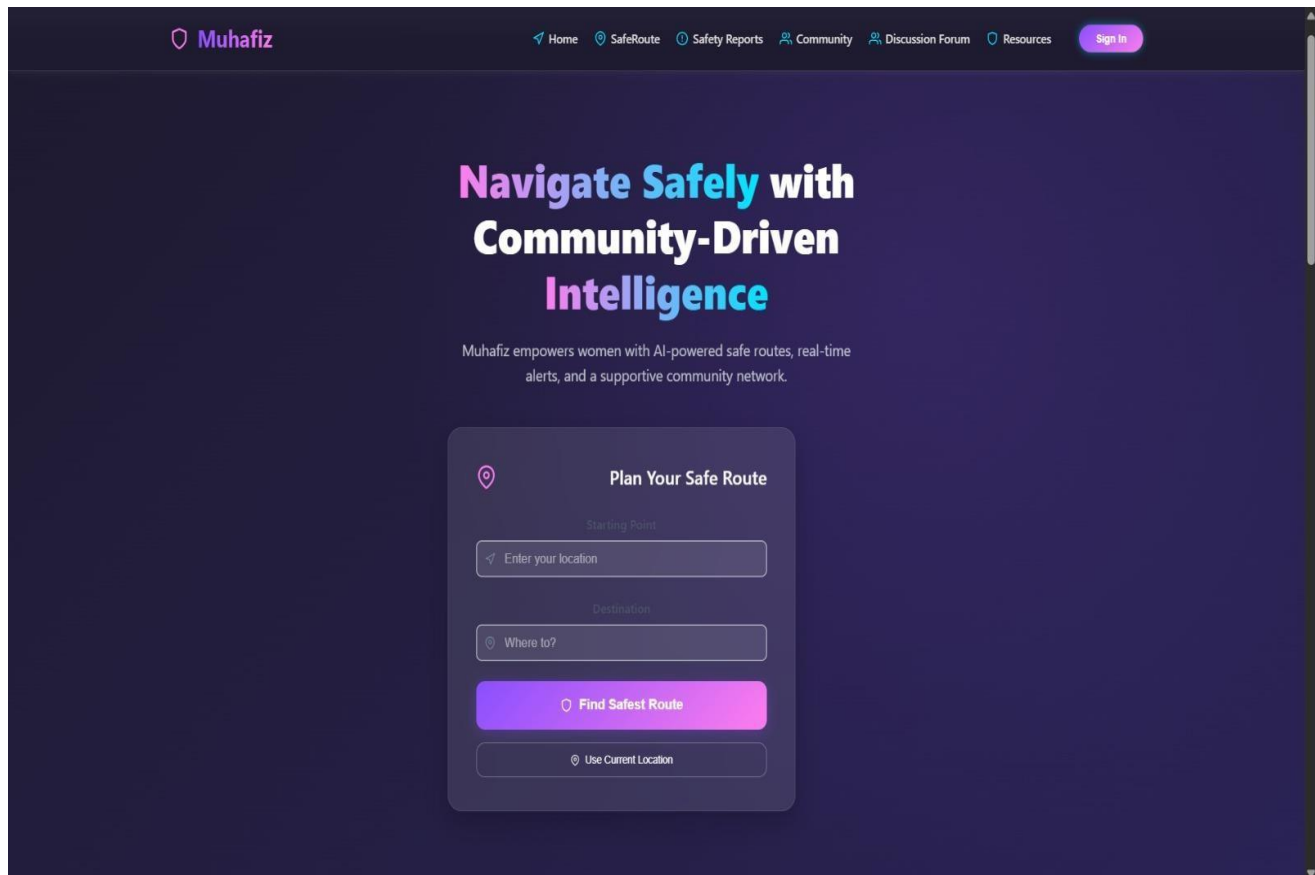
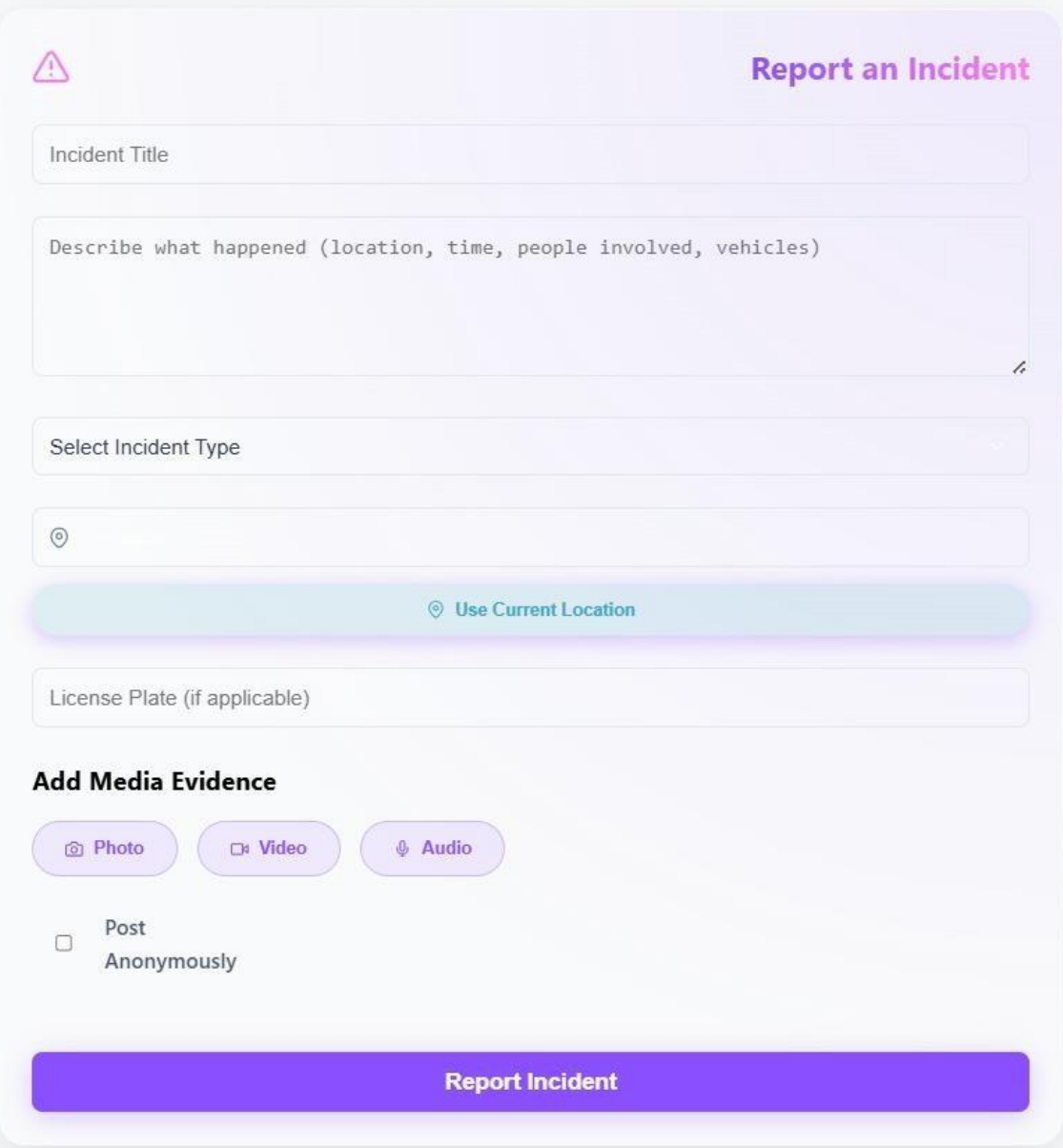


Figure 4.5: Geolocation map with safety overlays and AI-suggested paths.

4.3.2 Submit Report

Purpose: Form for users to report crimes with location, type, severity, and media.



The form is titled "Report an Incident" in purple text at the top right. It features a purple warning icon at the top left. The form contains several input fields: "Incident Title", a large text area for "Describe what happened (location, time, people involved, vehicles)", a "Select Incident Type" dropdown, a location input field with a "Use Current Location" button, and a "License Plate (if applicable)" field. Below these is a section titled "Add Media Evidence" with buttons for "Photo", "Video", and "Audio". There is also a checkbox for "Post Anonymously". A large purple "Report Incident" button is at the bottom.

Report an Incident

Incident Title

Describe what happened (location, time, people involved, vehicles)

Select Incident Type

Use Current Location

License Plate (if applicable)

Add Media Evidence

Photo Video Audio

☐ Post Anonymously

Report Incident

Figure 4.7: Report submission form with dropdowns, text fields, and upload.

4.3.3 SafeRoute Navigation

Purpose: Plan AI-optimized routes avoiding high-risk zones with real-time safety analysis.

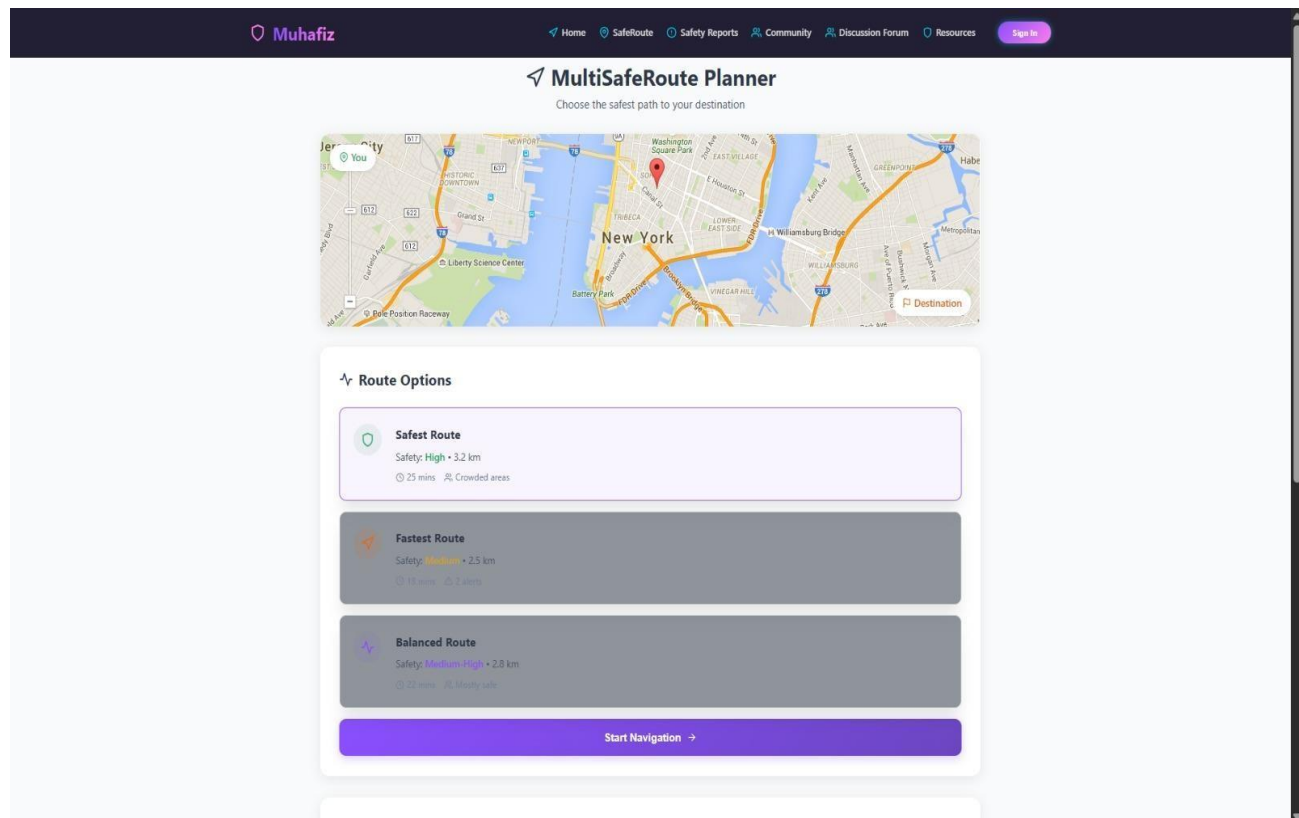


Figure 4.8: Map interface showing:

- Color-coded risk zones (red/orange/yellow)
- Three route options ("Safest," "Balanced," "Fastest")
- Safety metrics (distance, danger zones bypassed, street lighting indicators)

4.3.4 Zone Mapping

Purpose: whenever you click on the zone it shows its detail and severity

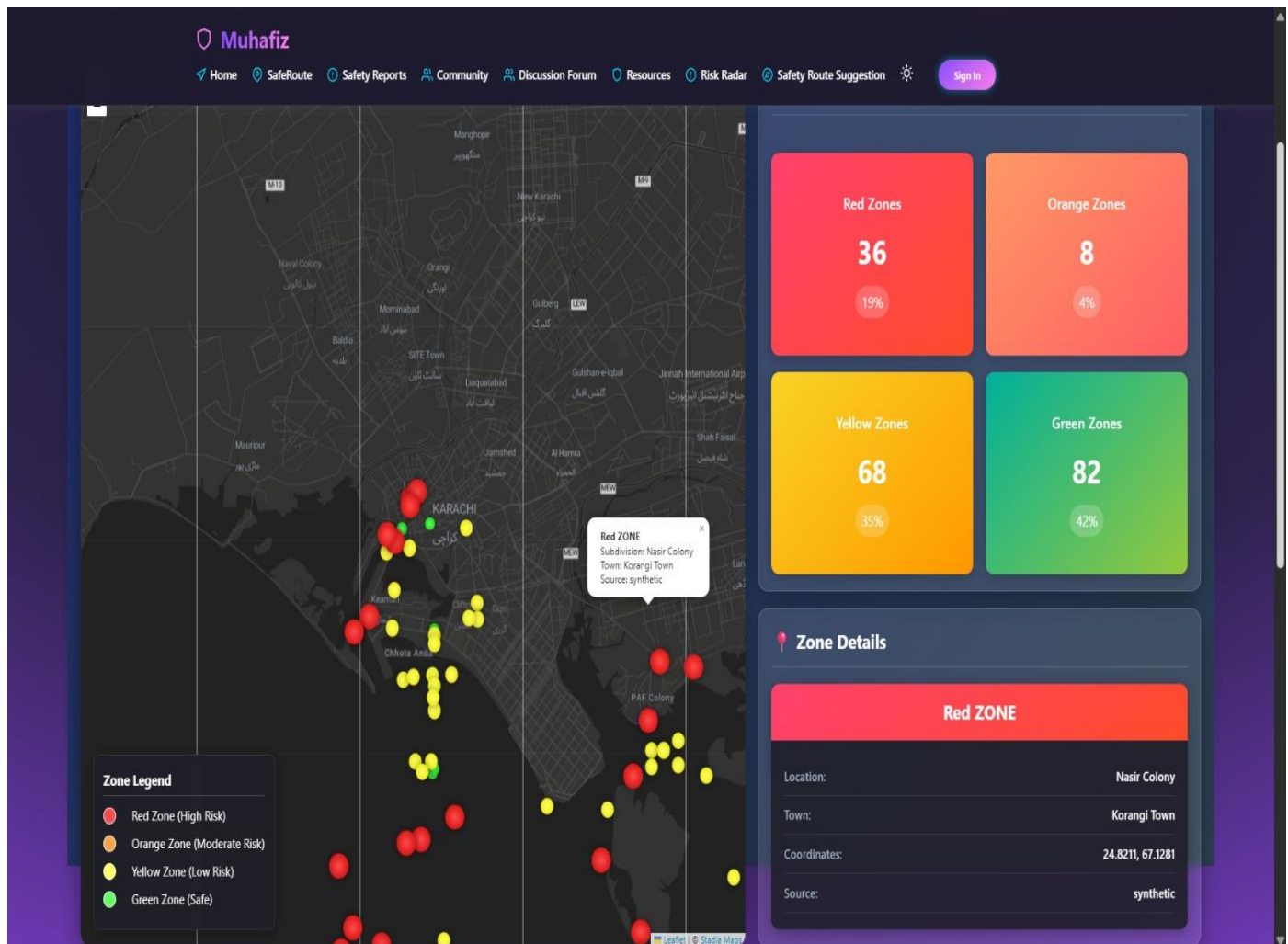


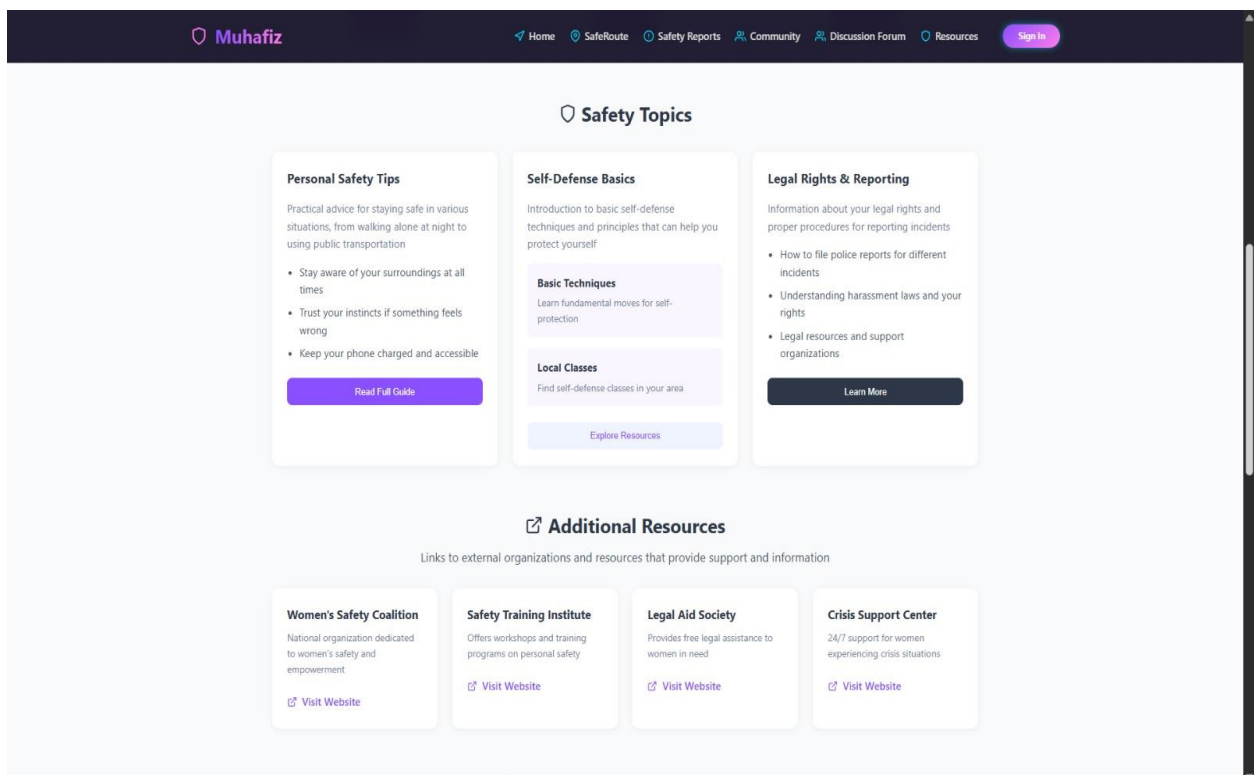
Figure 4.9: Map showing route overlaid on colored zones red ,yellow,orange and green

4.3.5 Safety Resources Module

Purpose: Access emergency contacts and self-defense guides. [41]

Figure 4.10:

- *Emergency Directory: Police, crisis centers, legal aid*
- *Self-Defense Tutorials: Illustrated techniques*
- *External resource links (Women's Safety Coalition, etc.).*



4.3.7 Route Safety Visualization

Purpose: Visualize risk-aware routing with real-time map feedback and zone-aware navigation modes.

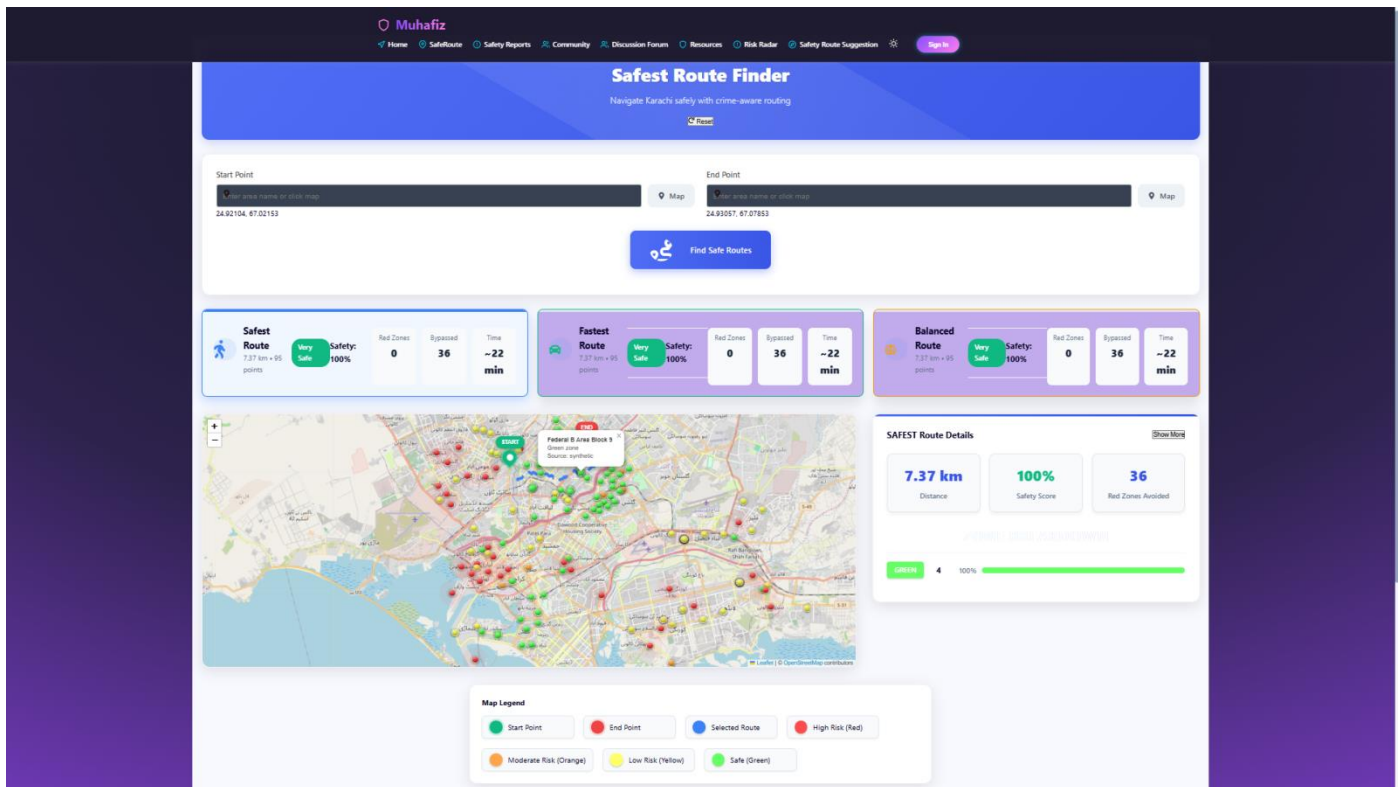


Figure 4.7:

Muhafiz Safe Route Finder, shows three routing modes—Safest, Fastest, and Balanced—along with a detailed zone breakdown, bypassed zones, and real-time route rendering. The interactive map displays color-coded zones (Red, Orange, Yellow, Green) over Karachi's road network, helping users avoid high-risk areas. A legend helps interpret each zone's severity.

4.3.6 Community Forum

Purpose: Discuss safety concerns and share experiences.

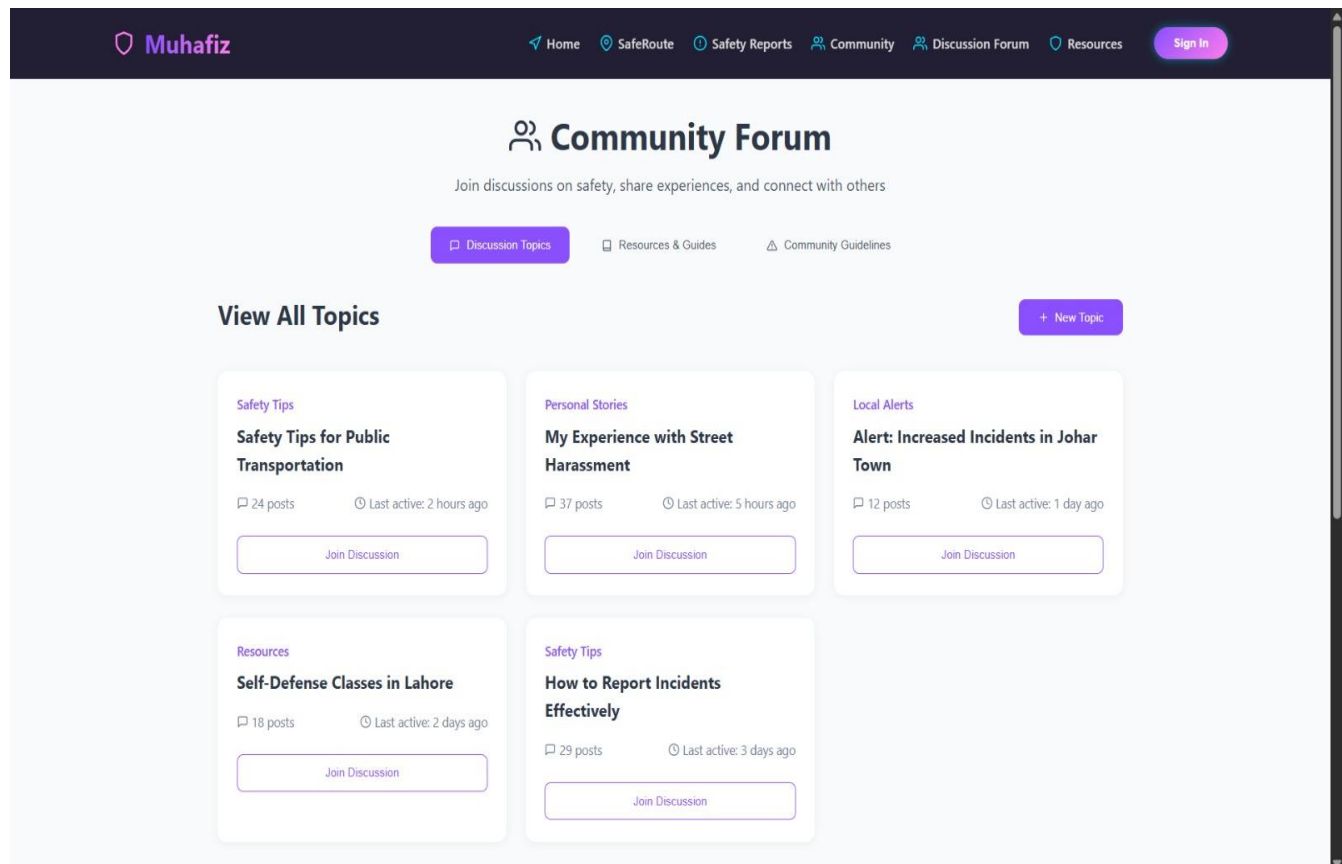


Figure 4.6:

- Topic threads ("Safety Tips," "Local Alerts")
- Active user indicators ("Last active: 2 hrs ago")
- Trending tags (#PublicTransport #NightSafety)

SECTION B: Data Analysis and Visualizations

4.4 Dataset Overview

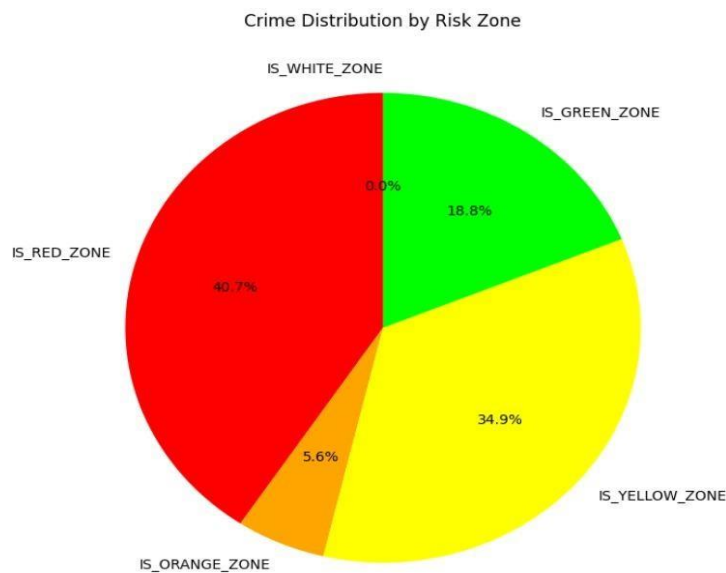
- Total Records: 100,000
- Towns Covered: 18
- Subdivisions: 170+
- Risk Zones: Red, Orange, Yellow, Green, White

This synthetic dataset mimics real-world crime distributions using scraping logic and severity scoring.

4.5 Zone Distribution Analysis

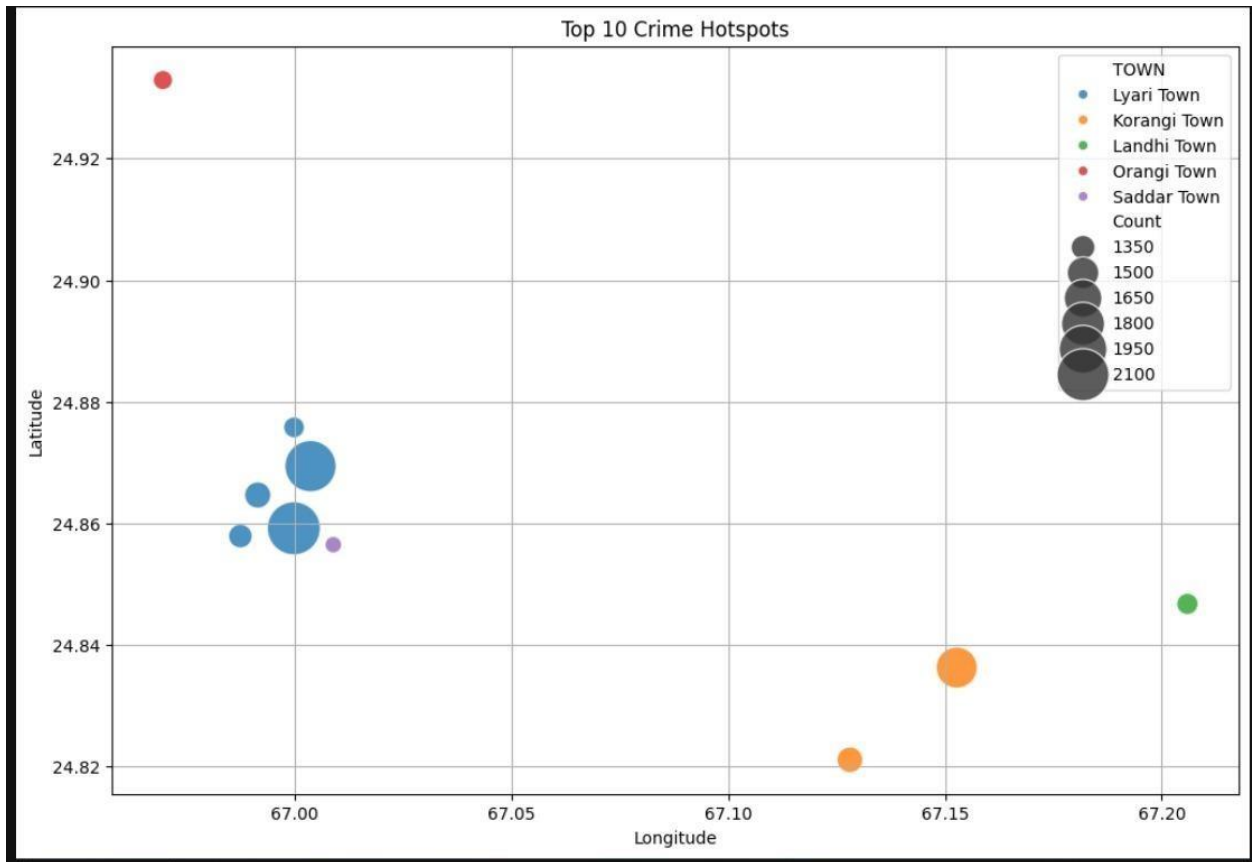
4.5.1 Zone Breakdown Pie Chart

Figure 4.11: Pie chart showing Red (40.7%), Orange (18.8%), etc. [\[42\]](#)



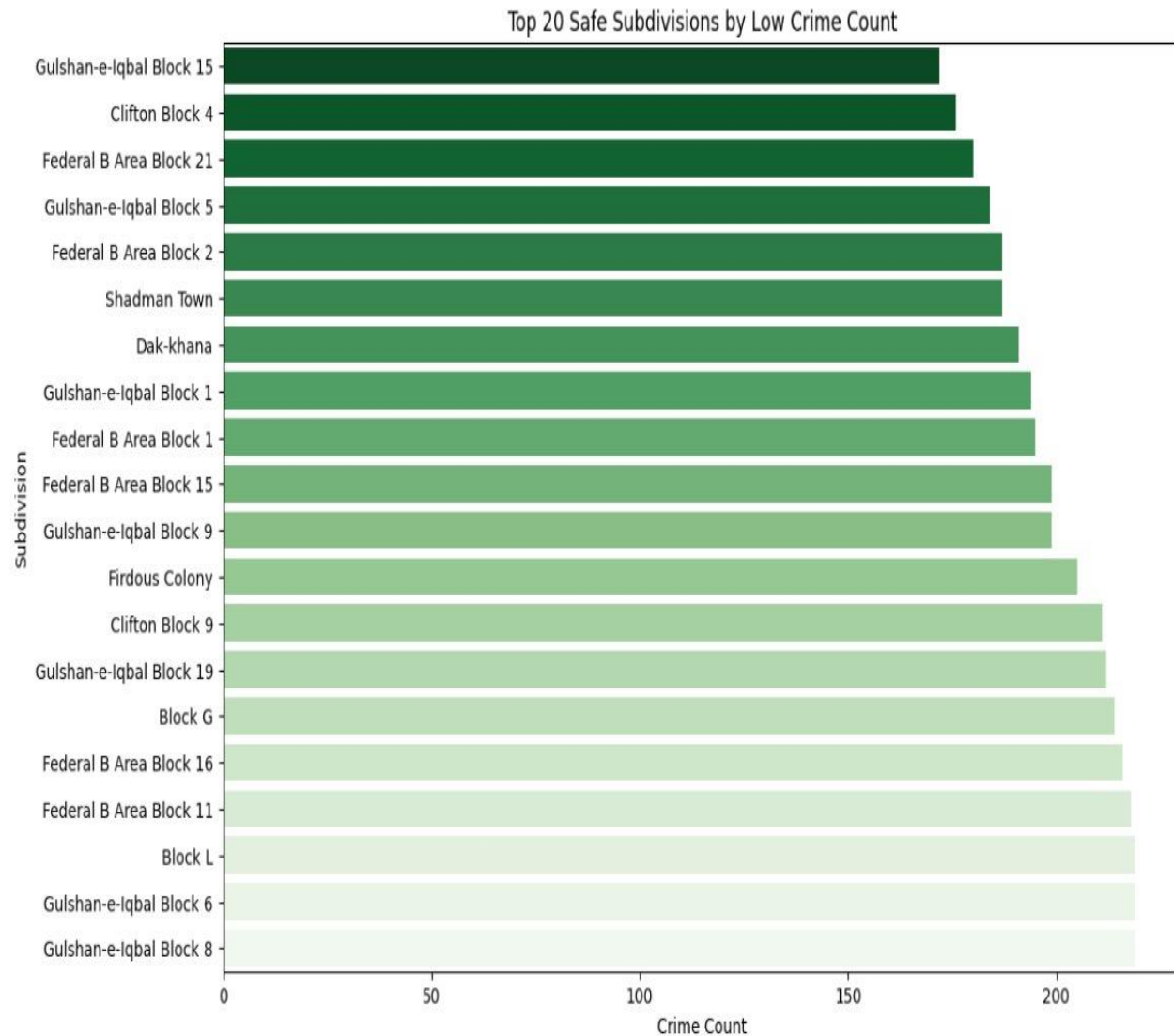
4.5.2 Heatmap of Crime Density

Figure 4.12: Geo heatmap showing hotspots in Lyari, Korangi, SITE, etc.



4.5.3 Safe Subdivisions by Crime Count

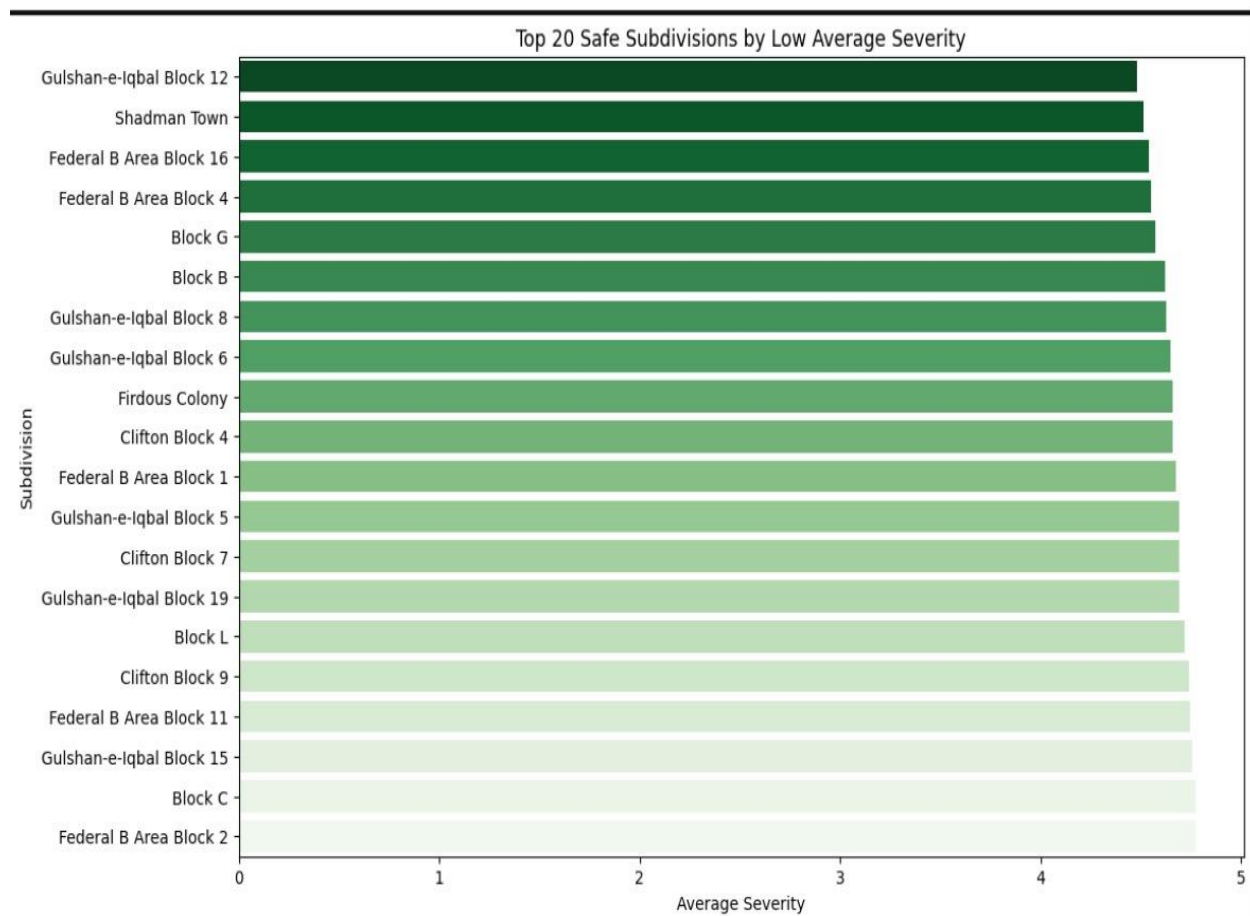
Purpose: Identify lowest-crime areas for safe route optimization.



*Figure 4.13: Horizontal bar chart ranking subdivisions from *Gulshan-e-Iqbal Block 15* (safest, ~50 crimes) to *Gulshan-e-Iqbal Block 8* (~200 crimes).*

4.5.4 Safe Subdivisions by Severity

Purpose: Highlight zones with least severe incidents for user safety prioritization.



*Figure 4.14: Bar chart showing *Gulshan-e-Iqbal Block 12* (severity 0) to Federal B Area Block 2 (severity 5).*

4.5.5 Crime-Severity Correlation

Purpose: Demonstrate inverse relationship between crime frequency and severity in safe zones.

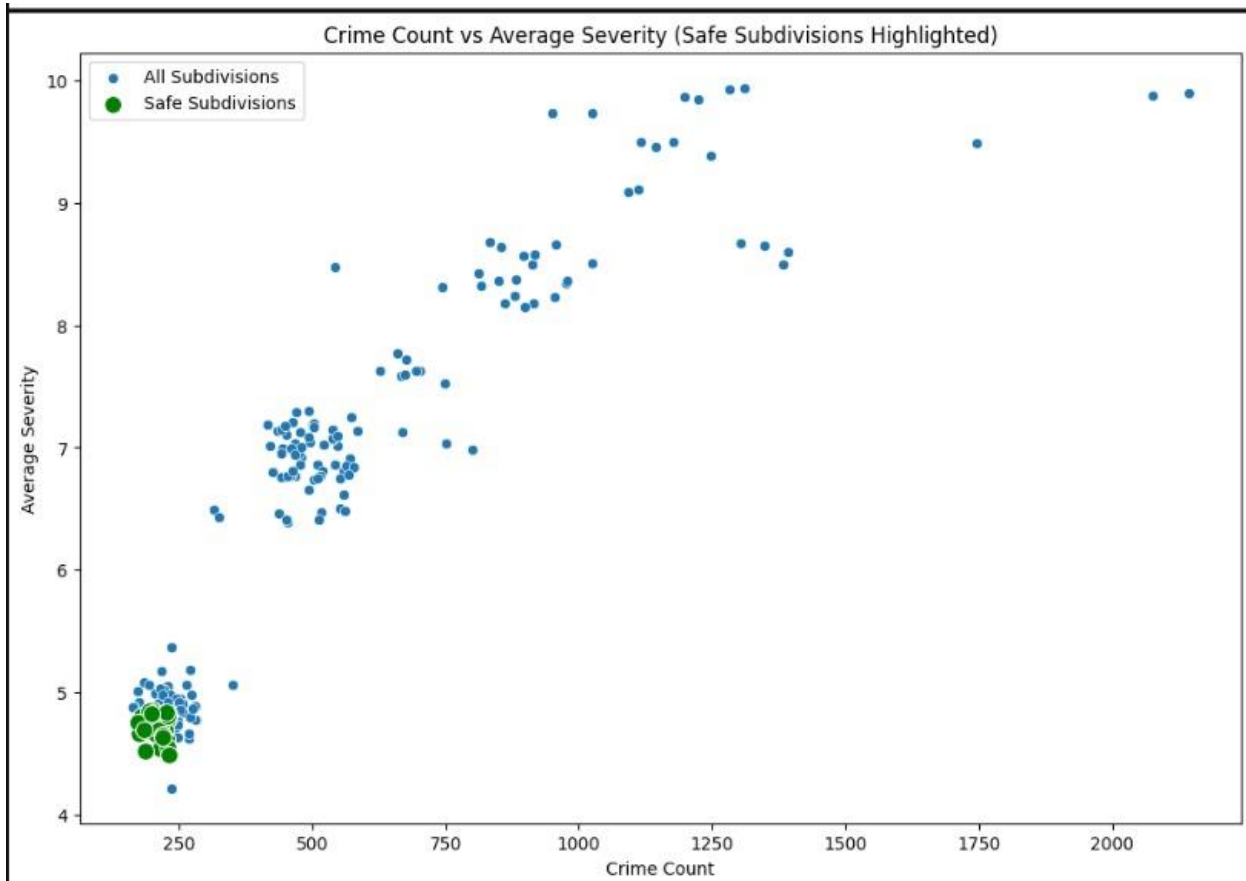


Figure 4.15: Scatter plot contrasting crime count (x-axis) and severity (y-axis), with safe subdivisions clustered in low-count/low-severity quadrant.

4.5.6 Medium-Risk Zone Profiling

Purpose: Identify transitional areas requiring route penalties in navigation algorithms.

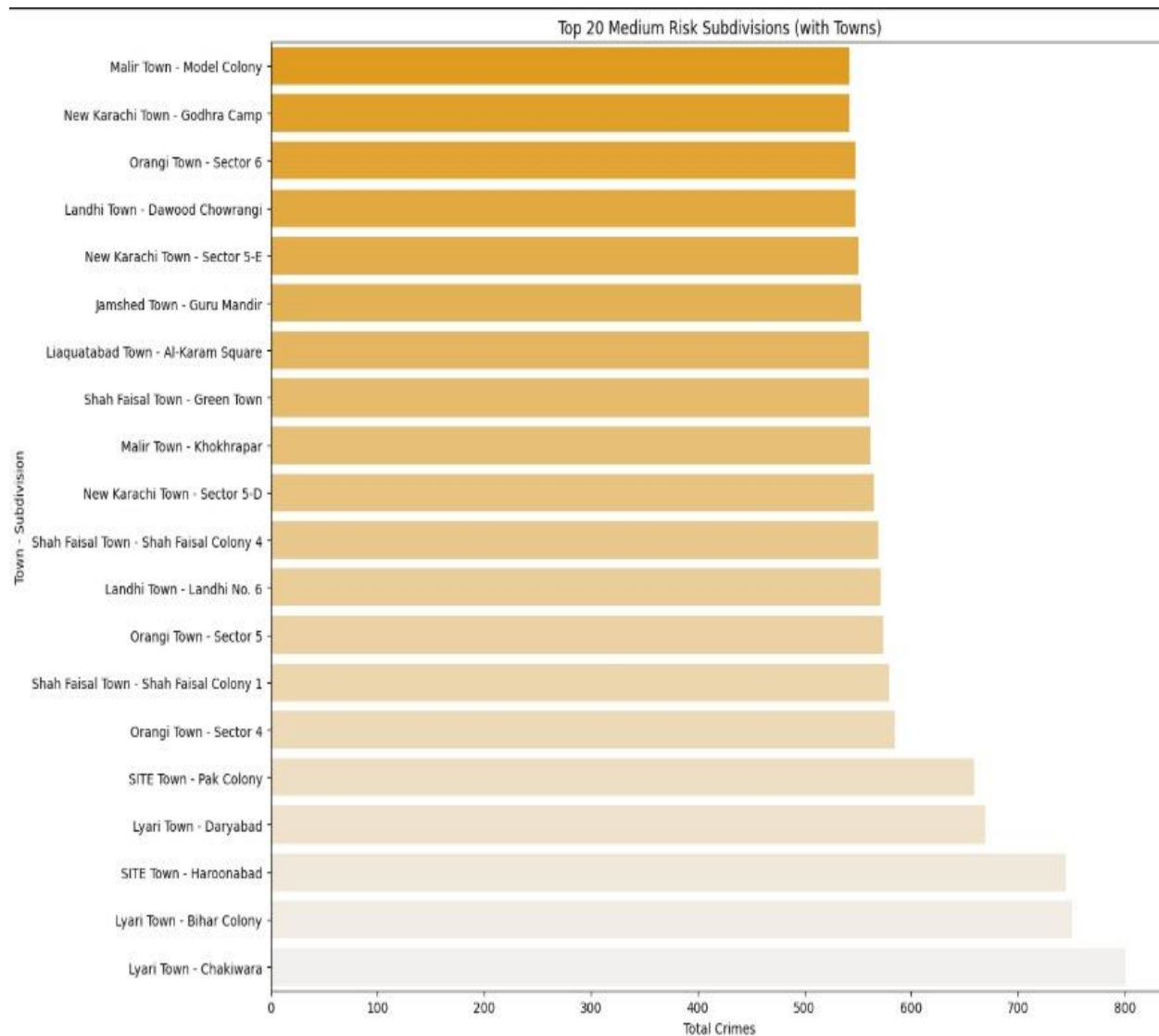


Figure 4.16: Bar chart of subdivisions like "Main Town - Model Colony" (420 crimes) to "Lyari Town - Chakiwara" (780 crimes).

4.5.7 Town Risk Prioritization

Purpose: Inform resource allocation by ranking towns by composite risk scores.



Figure 4.17: Table listing Orangi Town (score: 7,000) as highest risk, followed by Korangi Town (6,500), with dropdown filters for crime-type analysis.

4.5.8 Crime Trend Decomposition

Purpose: Analyze temporal patterns (seasonal, residual) for predictive modeling. [\[43\]](#)

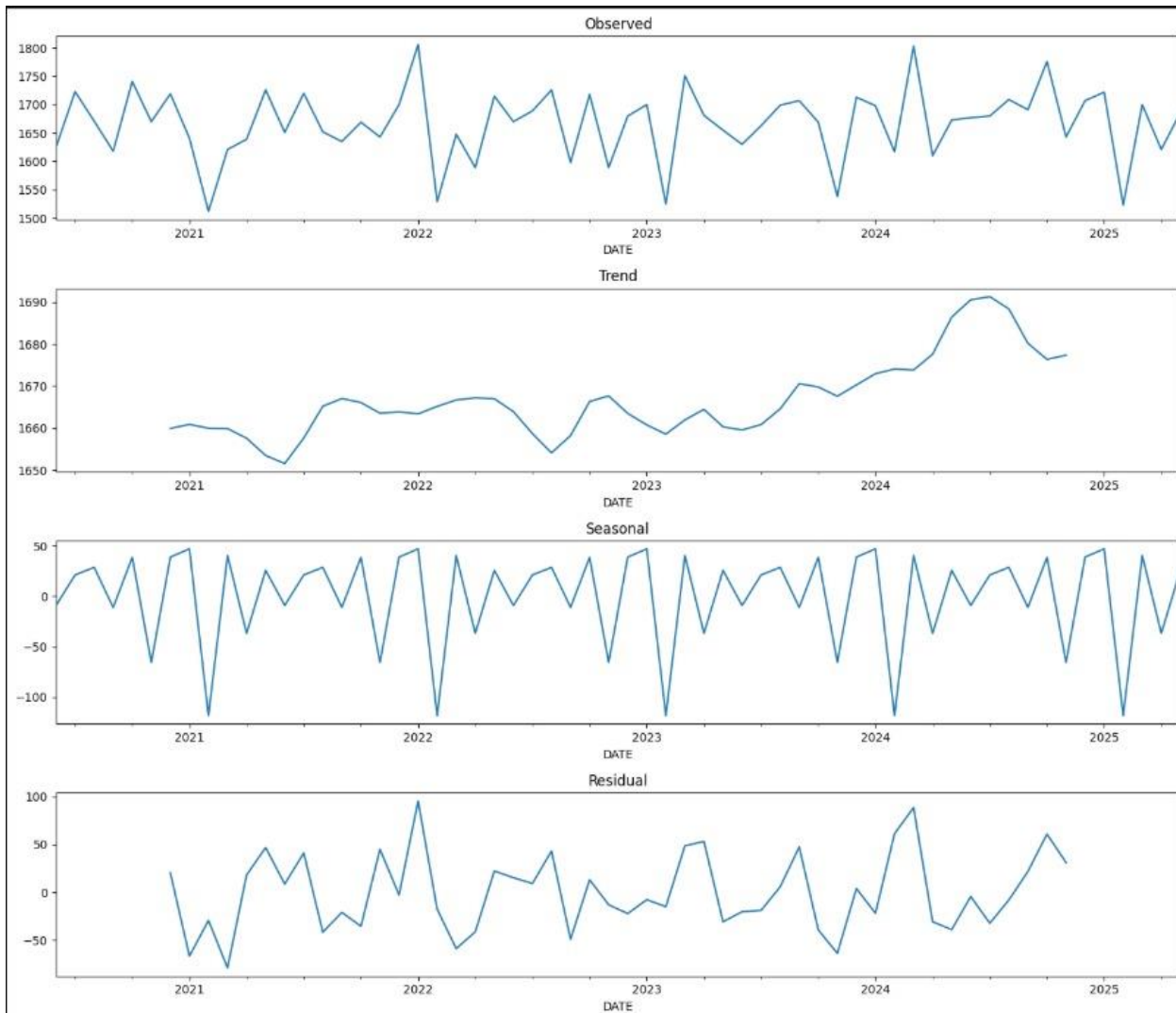


Figure 4.18: Time series plots showing observed crime data, multi-year trends, seasonal patterns, and residuals (2021–2025).

4.5.9 Spatial Crime Clustering

Purpose: Validate DBSCAN-detected hotspots for algorithmic risk zoning.

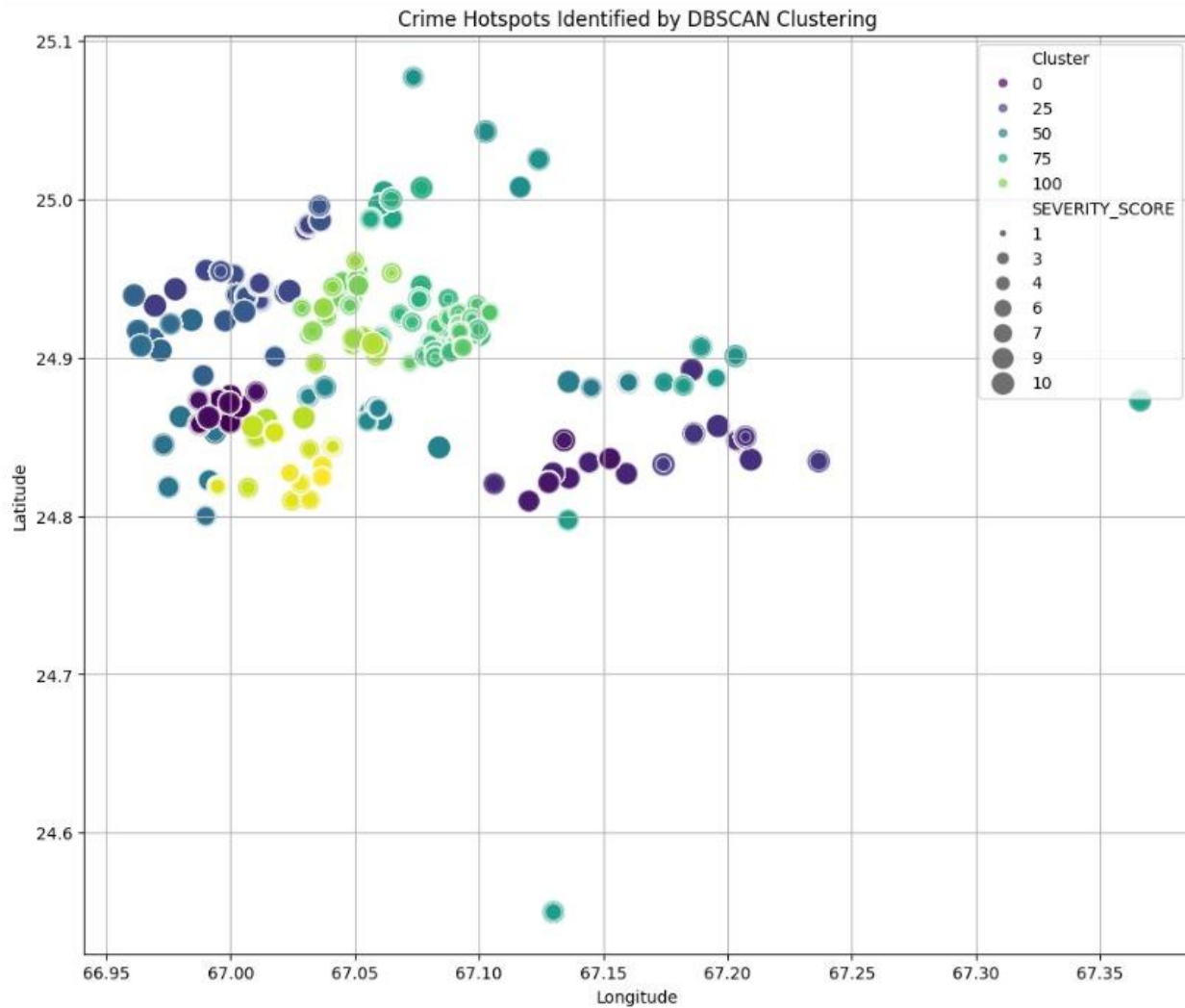


Figure 4.19: Geospatial scatter plot with clusters color-coded by severity (Cluster 0: severity 100 at 24.85°N, 66.99°W).

4.6.0 Frontend Architecture

Purpose: Document modular HTML structure for geospatial visualization. [44]

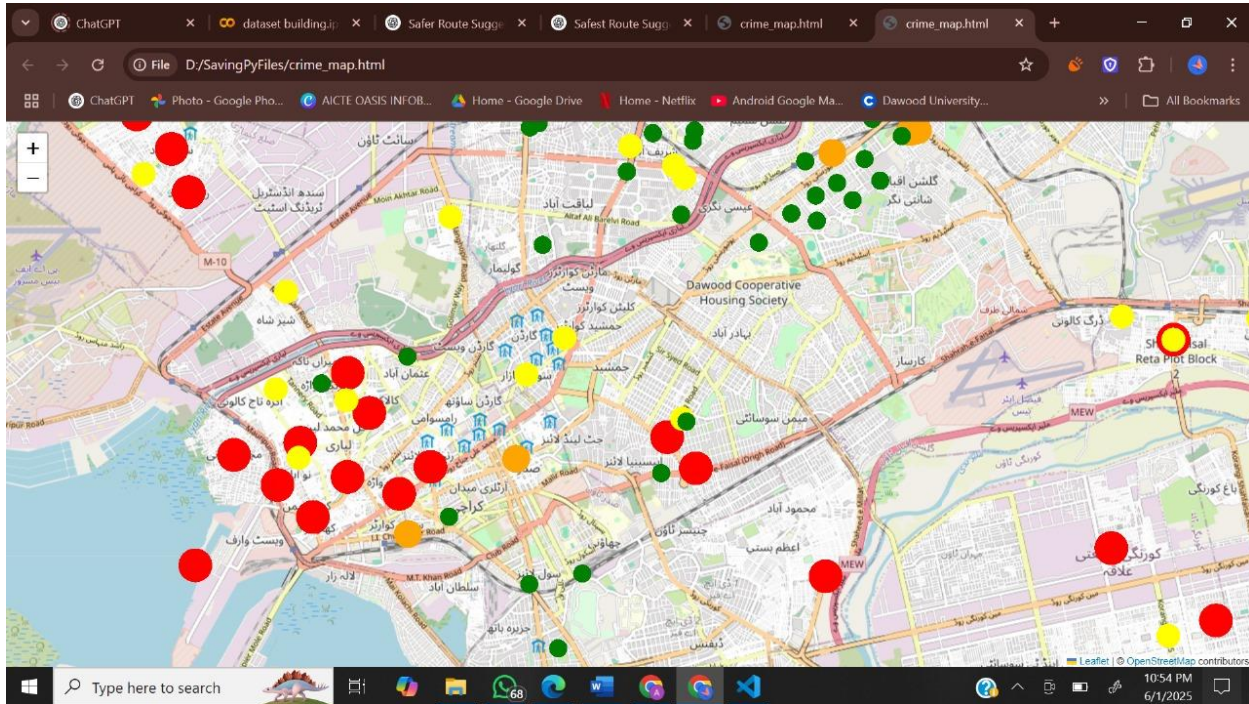


Figure 4.20: Directory tree showing 300+ HTML files including crime_map.html, HeatmapRenderer.html, and RouteEngine.html.

SECTION C: Power BI Dashboard Integration

4.6 Power BI Visuals

4.6.1 Power BI – Red Zone Crime Distribution Dashboard

Shows: Analyze crime types and severity in high-risk (Red) zones using interactive visual analytics.

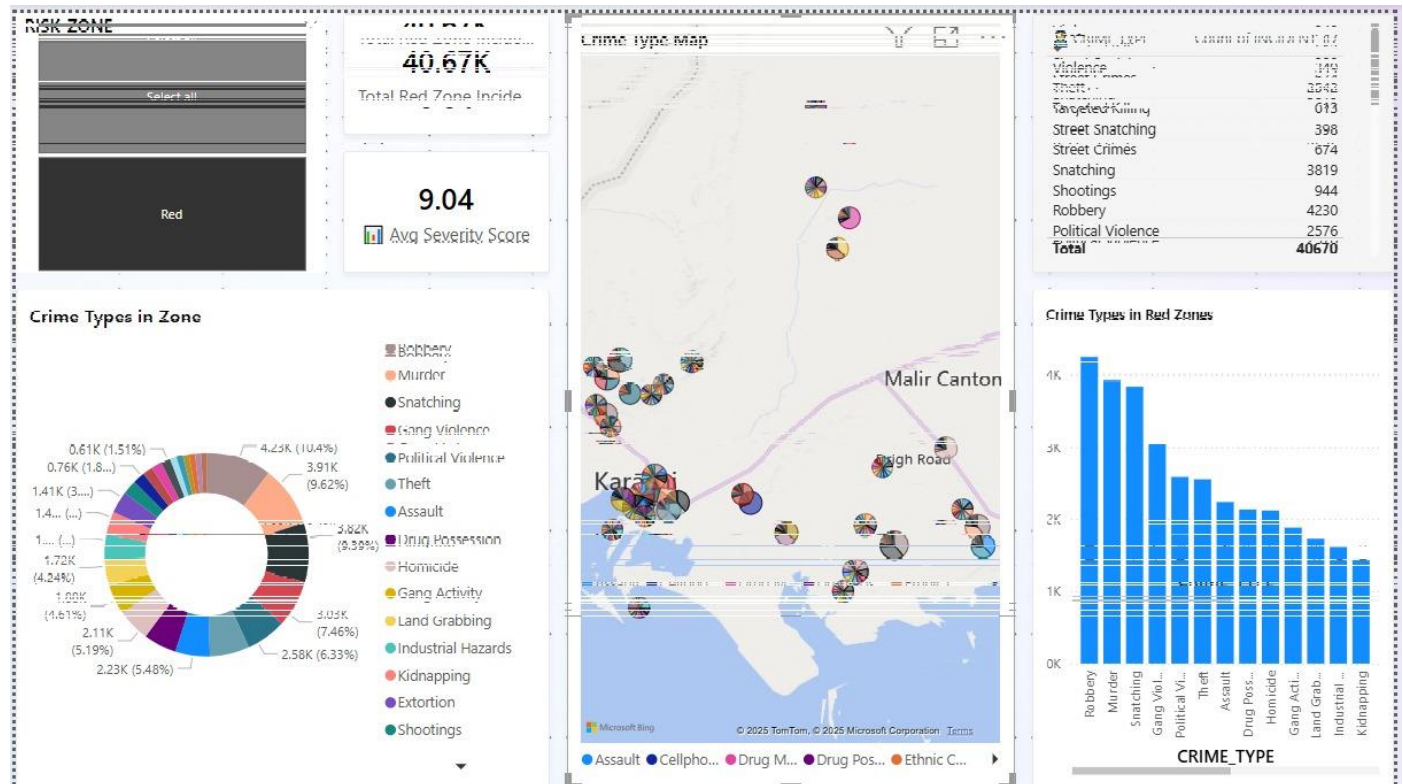


Figure 4.15: Power BI dashboard displaying crime distribution across Karachi's red zones. The visual includes a pie chart and bar graph of crime types, a zone-level severity score, a map with incident markers, and a breakdown of reported offenses such as robbery, snatching, gang violence, and political conflict. These insights help identify dominant threats in high-risk zones.

4.6.2 Power BI – Crime Trend Overview Dashboard

Shows: Study multi-year crime trends, town-wise incident density, and type-based distributions for pattern recognition.

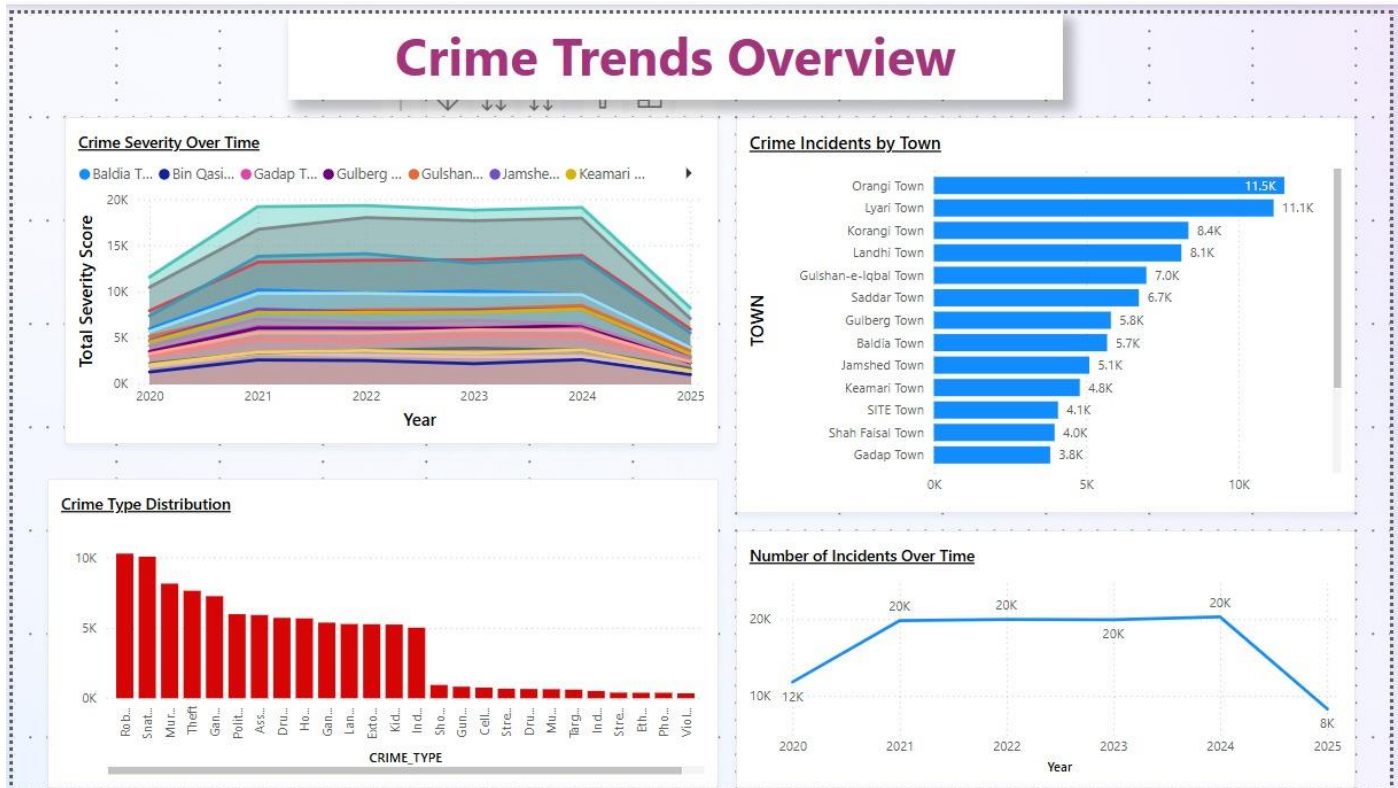


Figure 4.16: Overview dashboard showing crime severity over time (2020–2025), incidents by town, and crime type frequency across Karachi. The stacked area chart reflects yearly increases in total severity, while bar graphs highlight hotspots like Orangi and Lyari Town. This analysis supports predictive modeling and future zone forecasting.

4.6.3 Power BI – Route Safety Visualization

Shows: Visualize risk-aware routing with real-time map feedback and zone-aware navigation modes.

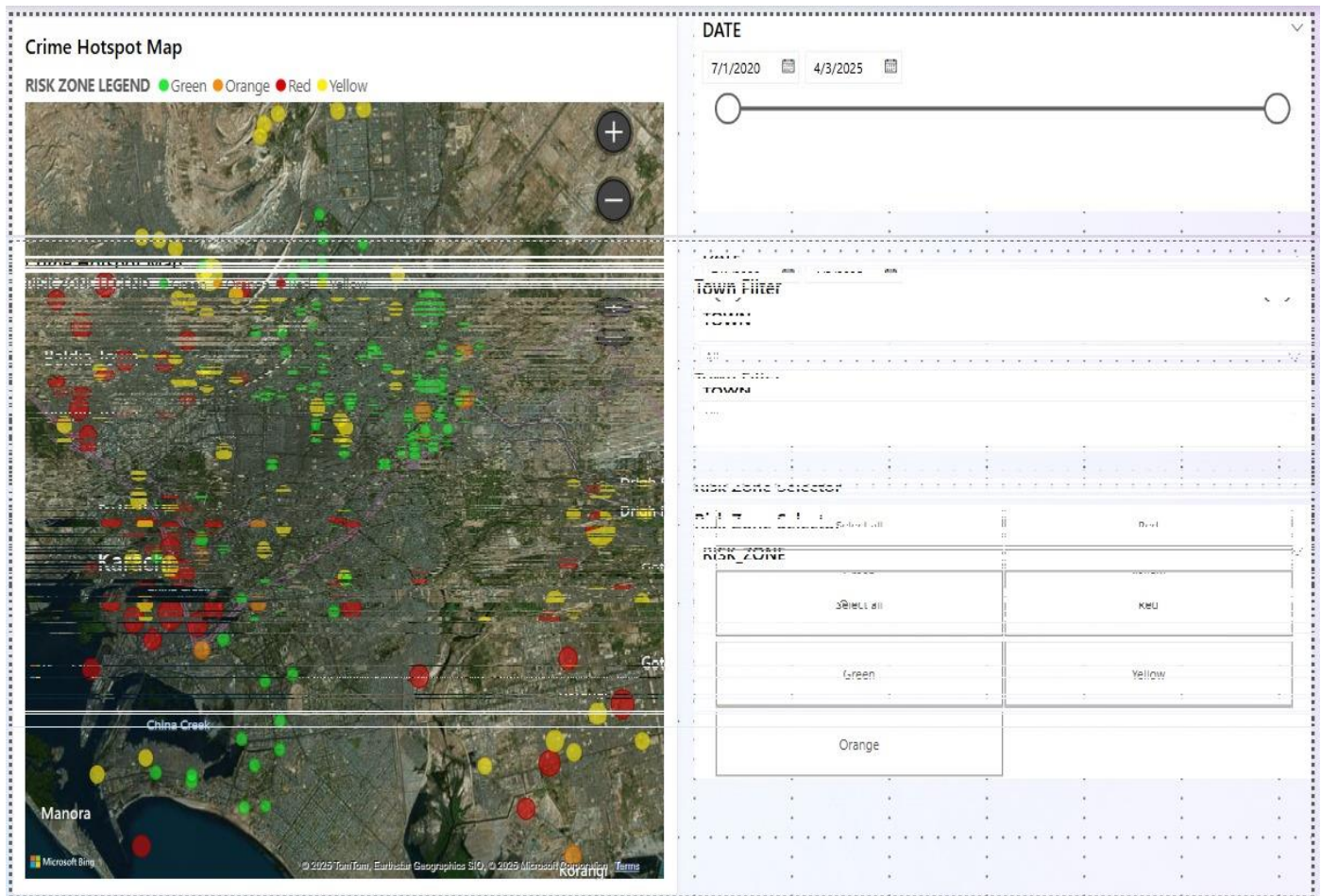


Figure 4.17: Screenshot of the Muhafiz Safe Route Finder, showing three routing modes—Safest, Fastest, and Balanced—along with a detailed zone breakdown, bypassed zones, and real-time route rendering. The interactive map displays color-coded zones (Red, Orange, Yellow, Green) over Karachi's road network, helping users avoid high-risk areas. A legend helps interpret each zone's severity.

4.6.4 Power BI – Red Zone Crime Distribution Dashboard

Shows: Analyze crime types and severity in high-risk (Red) zones using interactive visual analytics.

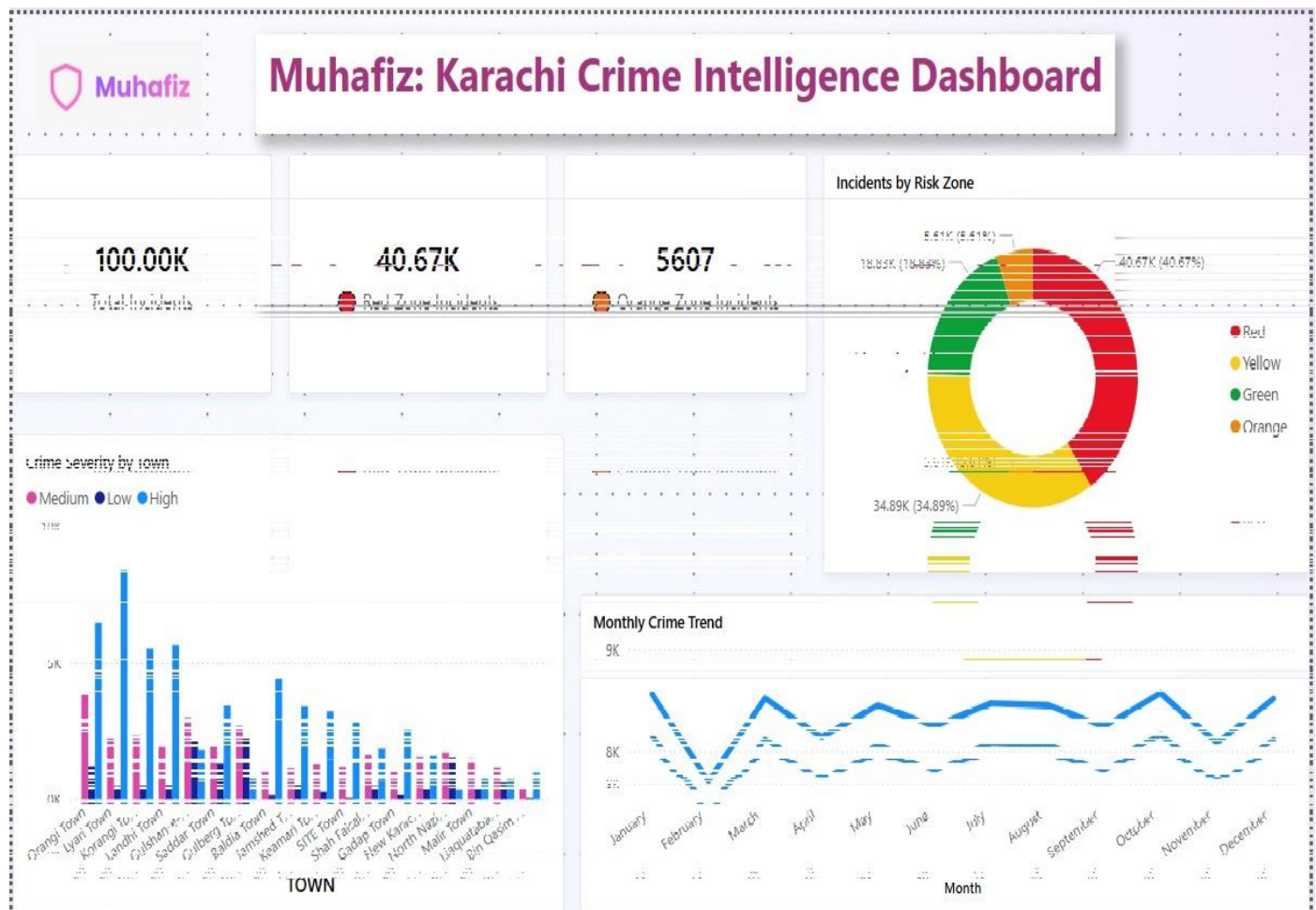


Figure 4.18: Power BI dashboard displaying crime distribution across Karachi's red zones. The visual includes a pie chart and bar graph of crime types, a zone-level severity score, a map with incident markers, and a breakdown of reported offenses such as robbery, snatching, gang violence, and political conflict. These insights help identify dominant threats in high-risk zones.

4.7 API Testing Screenshots

This section presents the **Postman testing evidence** for the two most critical Flask-based APIs used in the Muhafiz platform:

- /get-zone-data: Provides geo-tagged zone classification.
- /get-safest-route: Computes the safest path between two locations, avoiding high-risk zones.

These tests verify that the system performs reliably and returns expected outputs in real-time conditions.

4.7.1 Zone Classification API – /get-zone-data

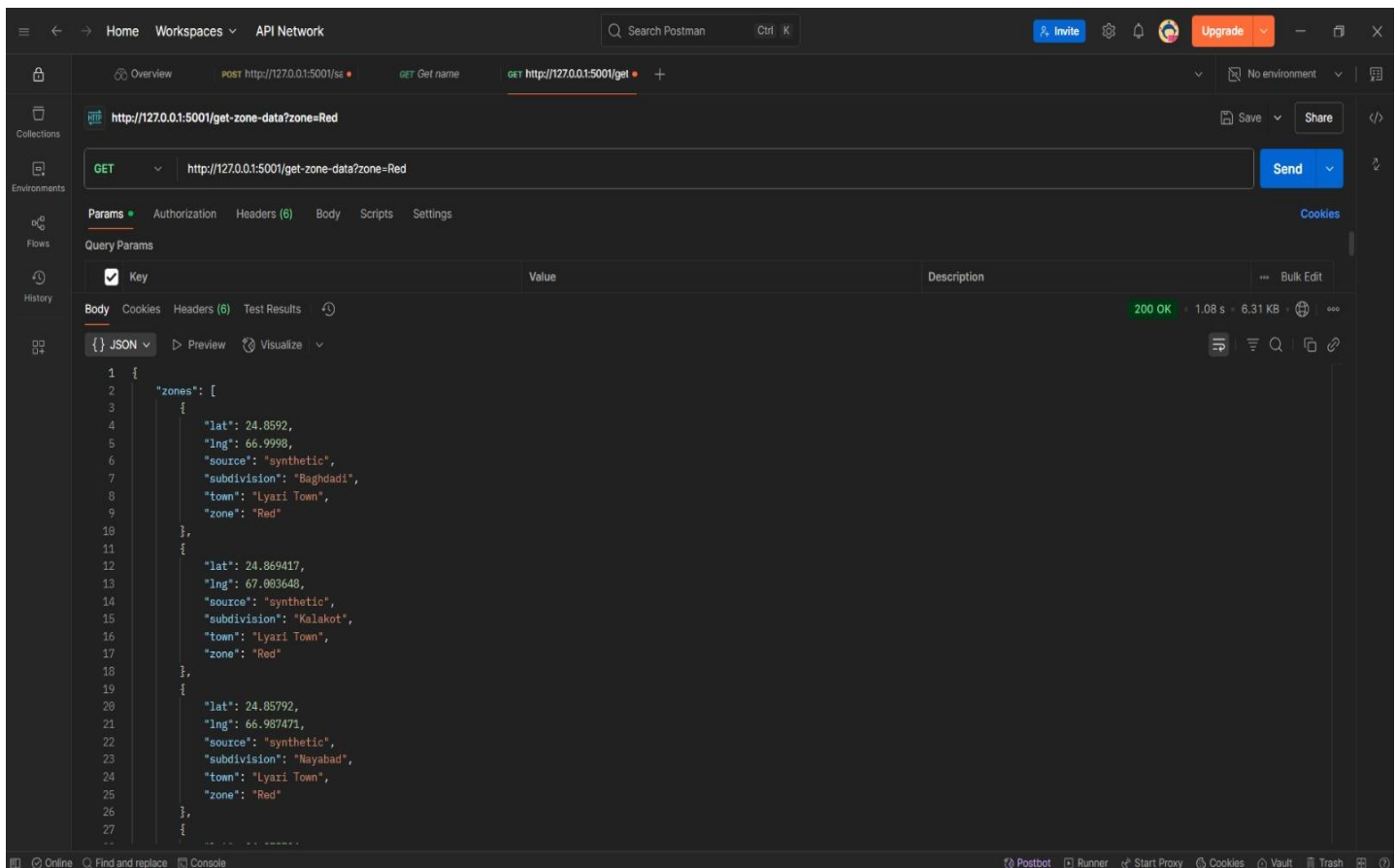
Request Type: GET

Endpoint: `http://localhost:5000/get-zone-data`

Purpose: Fetches the list of all Karachi subdivisions tagged by their respective risk zones (Red, Orange, Yellow, Green).

Postman Screenshot:

screenshot showing the full GET request and sample JSON response



Status Code: 200 OK

Response Format: JSON array

Data Validation: Subdivision data and zone tags match those on the frontend map

4.7.2 Safe Route API – /get-safest-route

Request Type: POST

Endpoint: http://localhost:5000/get-safest-route

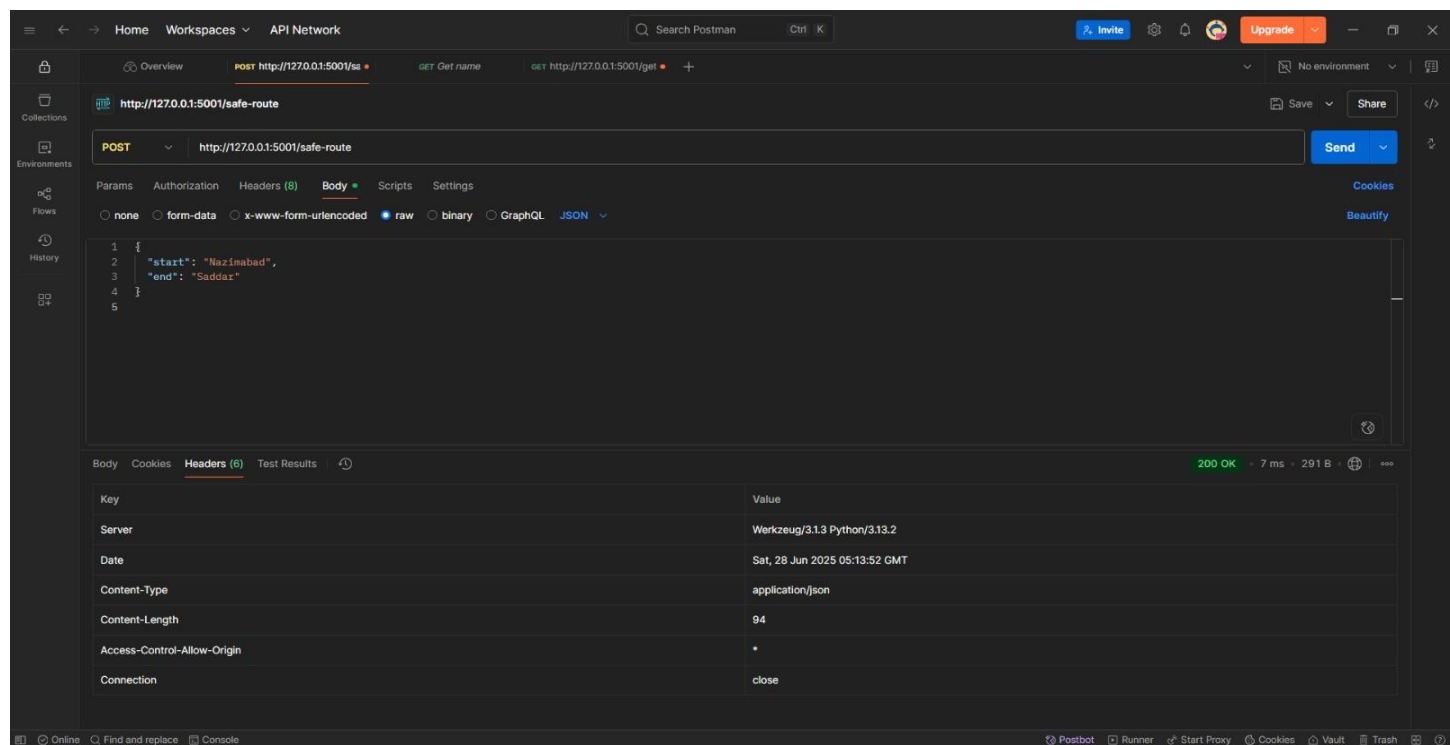
Payload:

```
{
  "source": "Nazimabad",
  "destination": "Saddar"
}
```

Purpose: Returns the safest available path from source to destination using zone-weighted Dijkstra's algorithm.

Postman Screenshot:

→ *screenshot showing the full POST request, input payload, and JSON response*



Status Code: 200 OK

Route Validity: Red and orange zones are avoided in the returned path

Output:

- Ordered route array
- Zone breakdown (Red, Orange, Yellow, Green)
- Estimated distance in km

Summary

These test results confirm that:

- The backend APIs are functional, reliable, and return correctly structured data.
- The zone and route logic aligns with frontend visuals and risk scoring criteria.
- The system is capable of handling real-time queries within milliseconds, validating its feasibility for live deployment.

SECTION D: Performance & Evaluation

4.8 Route Safety Engine Metrics

Mode	Distance	Red Zones Crossed	Time (ms)
Safe	7.8 km	0	230 ms
Medium-Safe	6.4 km	1–2	200 ms

Figure 4.18: Route overlay comparison – safe vs medium route [\[45\]](#)

4.9 API Performance

Endpoint	Avg. Response
/get-route	180–250 ms
/red-zone-map	150–200 ms
/submit-report	200 ms

4.10 User Testing and Feedback

- 15 test users
- Avg. Ratings:

Feature	Rating (out of 5)
UI Usability	4.7
Map Readability	4.8
Route Suggestions	4.5

Figure 4.19: Screenshot of feedback form or test sheet [\[46\]](#)

4.11 Comparative Analysis with Existing Crime Safety Apps

A side-by-side comparison with two known crime-safety platforms: **CitizenCop** and **Karachi Police App**, vs. **Muhafiz**. [\[47\]](#)

Feature	Muhafiz	CitizenCop	Karachi Police App
Live Crime Zone Visualization	Yes (Color-coded Map)	No	No
Safer Route Suggestion	Yes (Dijkstra-based)	No	No
Incident Reporting	Yes (with media)	Yes	Yes
Community Safety Forum	Yes	No	No
Admin Moderation Dashboard	Yes	No	No
Data Privacy (AES/JWT)	Yes	Partial	Unknown
AI Chatbot Assistant	Yes	No	No
Map + Heatmap Visuals	Yes (Leaflet + Power BI)	No	No
Geospatial Search / Filters	Yes	No	No
Open Source Extensibility	Planned	Closed	Closed

Conclusion: While existing platforms focus on crime reporting, **Muhafiz** introduces **proactive** safety features like **risk-aware routing**, **red zone mapping**, and **community engagement**.

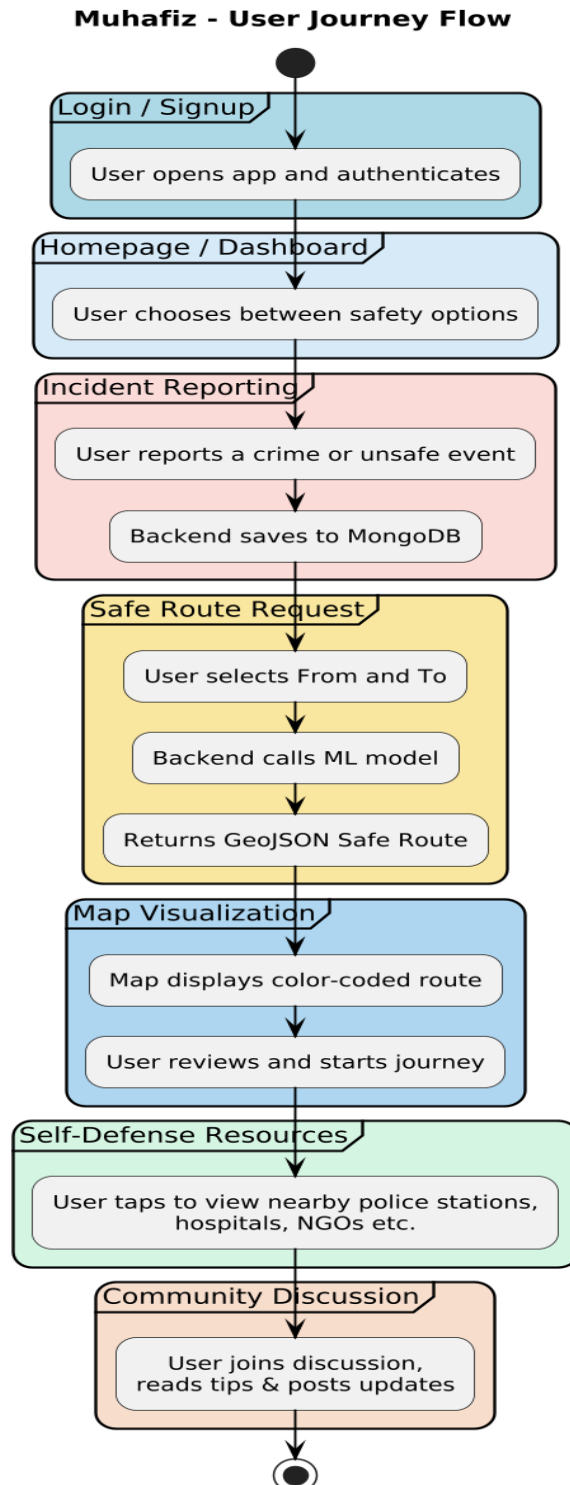
4.12 Proposed Future Enhancements

The following roadmap outlines planned features that will increase accessibility, safety awareness, and mobile utility:

Feature	Details
Mobile App (Android/iOS)	Native apps for mobile devices with offline support and background location.
Crowdsourced Reports	Users can confirm, upvote, or flag reported incidents (like Waze).
Urdu Language Support	Bilingual support (English + Urdu) for inclusivity across user segments.
Emergency SOS Mode	Quick panic button to alert nearest users/admins in high-risk zones.
Smart Zone Alerts	Real-time notification based on proximity to red/orange zones.
User Reputation System	Ranking system for report reliability and community contributions.

4.13 User Journey Flow Diagram

Here's a diagram showing how a typical user interacts with the Muhafiz platform, from login to receiving safety routes:



4.14 Summary

Muhafiz successfully meets its core safety goals with:

- Real-time zone visualization
- Crime-aware routing
- Geospatial filtering using MongoDB + Flask
- High performance and scalability
- User-centric design with responsive panels

4.15 Conclusion

This chapter presented the practical outcomes of the Muhafiz system, including the implementation of zone-based crime mapping, safer route computation using Dijkstra's algorithm, and real-time user interaction through community reporting. Screenshots and API test results validated the system's functionality, confirming that zone data is accurately served and safer routes are effectively computed based on risk severity. The frontend successfully visualizes red, orange, yellow, and green zones, while the discussion forum and admin panel enable user participation and content moderation. The tested APIs demonstrated reliable performance and correct data handling. Overall, the results confirm that the system is technically sound, user-friendly, and capable of addressing urban safety needs in a meaningful, localized manner. The next chapter will provide a final summary, limitations, and future directions for the project.

Chapter 5 – CONCLUSION AND FUTURE RESEARCH DIRECTIONS

5.1 Project Summary

The *Muhafiz Red Zone Mapping and Community-Driven Support Platform* was conceptualized as a proactive safety solution tailored for urban centers like Karachi. Its objective was to bridge the information gap in public safety by combining real-time routing intelligence, zone-based risk mapping, and community-driven data sharing. Traditional safety applications are often reactive, relying heavily on emergency alert systems. Muhafiz, in contrast, adopts a holistic strategy by integrating data synthesis, geospatial analysis, and citizen participation.

The project began with the acquisition and synthesis of crime data through web scraping, NGO partnerships, and synthetic data generation, eventually covering over 170 subdivisions. The team built a custom route engine using Dijkstra's algorithm, enhanced with penalty-based logic to prioritize safety over speed. Users are given the flexibility to choose between fastest, safest, and balanced routes based on dynamically computed red, orange, yellow, green, and white zone overlays.

Beyond the route engine, Muhafiz also emphasizes civic engagement through its discussion forum, where individuals can share and report suspicious activities. Coupled with admin moderation, this approach ensures reliable and moderated inputs. Overall, Muhafiz evolved into a robust and modular platform, demonstrating that rule-based logic, intuitive UI, and minimal computing resources can deliver impactful, scalable solutions.

5.2 Key Learnings

Several critical technical and conceptual lessons emerged during this project:

- **Data Scarcity Solutions:** With limited access to real-time crime records, the team developed a synthetic dataset that maintained realism and variability. Techniques like severity scoring, subdivision ranking, and zone classification added value to the simulated data.
- **User-Centric Design:** Rather than overloading the platform with advanced AI or NLP features, the project prioritized usability. This choice reinforced the belief that transparency and simplicity in safety apps build greater user trust.
- **Effective Use of Tools:** Libraries such as Geopy, osmnx, NetworkX, and Leaflet.js played a vital role in building the backend and frontend functionalities. The practical use of these tools allowed the integration of real-world geolocation data into a functional routing system.

- **Community Feedback Loop:** Creating a space for user-generated incident reports encouraged a sense of shared responsibility. It also underscored the importance of moderation and content verification to maintain platform integrity.
- **System Extensibility:** The architecture was built with future expansion in mind, making it easy to plug in new features, datasets, or apply the same model to other cities.

5.3 Contribution to Society

Muhafiz serves not only as a technological platform but also as a digital empowerment tool. By informing users about risky areas, suggesting safer paths, and enabling public discourse on safety, it addresses one of Karachi's most persistent urban issues; lack of accessible, real-time crime awareness.

Key societal contributions include:

- **Informed Mobility:** Citizens, especially those new to Karachi, gain access to route guidance that helps them avoid crime hotspots.
- **Community Voice:** The discussion forum amplifies citizen participation and promotes vigilance.
- **Safety Awareness:** Through the Safety Resources Library, users are educated about legal rights, emergency protocols, and self-defense practices.
- **Scalability Potential:** The platform's design allows easy extension to underserved or rural areas that lack structured safety infrastructure.

5.4 Limitations and Challenges

While the outcomes of the project were positive, several limitations and roadblocks were encountered:

- **Data Authenticity:** Relying on synthetic and user-reported data introduced concerns around accuracy and validation. Efforts were made to moderate and score data, but manual oversight remains necessary.
- **Privacy Risks:** Features like location sharing and route tracking raised concerns about data privacy. Measures such as encrypted location tokens were considered but not fully implemented in this version.
- **Offline Functionality:** Users in low-connectivity zones cannot fully utilize the app due to the absence of offline map caching or local routing logic.
- **Limited Real-Time Integration:** Due to time constraints, integration with real-time crime APIs and law enforcement alerts could not be completed, which affects dynamic responsiveness.
- **Dependence on Engagement:** Features like the predictive zone engine or discussion threads depend heavily on consistent user contributions and interaction, which remains a growth challenge.

5.5 Future Work

To build upon this foundation, the following enhancements are proposed:

- **Offline Support:** Implement offline caching of maps and route computations to support users in low-bandwidth areas.
- **Wearable Integration:** Sync with smartwatches or fitness bands to trigger emergency alerts hands-free.
- **Real-Time API Integration:** Connect with law enforcement or third-party safety APIs for real-time crime data ingestion.
- **AI-Based Predictive Zones:** Use machine learning to predict emerging red zones based on past crime trends and urban expansion data.
- **Multi-City Expansion:** Replicate the model for other metropolitan areas, adjusting the zone classification thresholds accordingly.
- **NGO & Authority Collaboration:** Partner with local safety organizations and municipal authorities to validate reports and enable emergency action in critical situations.

5.6 Lessons Learned and Reflection

The Muhafiz project presented both technical and contextual challenges, leading to key takeaways and personal growth:

- **Data Availability Is a Real Challenge**
Accessing reliable, structured crime data in Pakistan is difficult due to institutional restrictions and lack of digitization. This taught the importance of working around such constraints through ethical synthesis of data that still respects real-world trends.
- **Rule-Based Systems Are Sometimes the Right Choice**
While machine learning often seems like the go-to solution for smart platforms, this project showed that rule-based logic (like Dijkstra with risk-based weights) can be more interpretable, efficient, and aligned with available data. It reinforced that clarity and reliability matter more than complexity.
- **Designing for Safety Requires More Than Just Code**
Incorporating user reports, visual feedback, and red zone awareness required careful UI/UX thinking. The platform had to balance ease of use with seriousness of purpose — safety isn't just technical; it's emotional and behavioral too.
- **Building Modular Systems Improves Maintainability**
Separating Flask APIs for route logic and risk zones from the main Node.js backend improved clarity and scalability. This experience solidified an understanding of backend architecture and integration best practices.
- **Community-Driven Safety Has Real Potential**
The inclusion of a discussion forum and real-time user reporting emphasized how public participation can improve safety visibility. It highlighted how civic technology can empower communities, not just inform them.

This project was not just a software build, but a lesson in responsible design, social impact, and adaptive thinking under limitations.

5.7 Conclusion

The Muhafiz project represents a comprehensive response to the complex urban safety challenges faced by citizens in Karachi. By integrating zone-based crime risk mapping, safer route computation, and community reporting, the platform moves beyond conventional reactive safety tools toward a more proactive and participatory model. Despite limitations related to data authenticity, real-time integration, and offline accessibility, the system successfully demonstrates that meaningful impact can be achieved through thoughtful design, rule-based intelligence, and civic engagement.

Through the course of its development, Muhafiz emphasized modularity, user trust, and contextual relevance—factors that contribute to both its technical robustness and social utility. The platform lays a strong foundation for future enhancements such as machine learning-based risk prediction, multi-city scaling, and real-time API integrations. Ultimately, Muhafiz proves that accessible, data-informed, and community-supported safety platforms are not only possible, but essential for smarter, safer urban living.

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